FACE ATLAS: transdiagnostic dimensional map of severe mental illness — a missingness-aware Bayesian measurement model and its archetypes across bipolar disorder, schizophrenia and depression

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Abstract

Psychiatric diagnoses aggregate biologically heterogeneous patients, but a dimensional alternative is credible only if the *measurement* is done well—comparable across cohorts, faithful to mixed data types, honest about missingness and uncertainty, and held fixed when it is later validated. We present a methodology that meets these requirements end-to-end and apply it to the harmonized baseline assessment of $N = 9{,}013$ patients with bipolar disorder, schizophrenia or major depression from the three FACE cohorts, with diagnosis withheld from model fitting throughout. The pipeline has four stages. (i) One global, missingness-aware Bayesian sparse bifactor / exploratory structural-equation model with mixed likelihoods and a Gaussian-copula continuous block estimates an eight-dimensional map—a general functional-burden factor (G) and seven specific dimensions (cognition, immunometabolic load, sleep, mania/activation, suicidality, developmental risk and substance use)—from each patient's observed cells only, without imputation. (ii) On the fixed map, an uncertainty-aware structure test shows the patient space is a graded **continuum, not discrete biotypes** (apparent clustering is reproduced by a single-Gaussian null), summarized by a stable five-archetype simplex that describes a patient's clinical–biological state $\approx 9.7\times$ more tightly than the DSM-5 diagnostic label. (iii) Scoring two yearly follow-ups under the frozen model quantifies temporal coherence: the backbone axes keep their meaning, and a between/within/measurement variance decomposition separates durable traits from moving states. (iv) An errors-in-variables Bayesian generalized linear model tests incremental prognostic validity beyond diagnosis, severity and baseline outcome, with a representation benchmark against the raw indicators. Reporting all eight factors uniformly and separating measurement from interpretation, we find that one property of the map recurs across all four stages: a single **immunometabolic axis distinct from the rest of the clinical picture** (correlation with $G \approx 0.10$ versus 0.39/0.42 for cognition/sleep, robust to medication, adiposity and site), which anchors its own archetype corner, is the most temporally durable axis (test–retest intraclass correlation 0.91), and is the worst-prognosis pole for two-year functioning (functional-remission gradient 22%–63% across corners; held-out predictive-density gain +62.8 but only +0.010 individual-level discrimination, co-informative with rather than superior to DSM-5: incremental density +17.3 for the map versus +29.0 for diagnosis). The contribution is the measurement-model pipeline—missingness-native (no imputation), mixed-likelihood, and uncertainty-propagating end-to-end, with the map fixed before any validation stage. The immunometabolic dissociation is what that discipline *reveals*: a property a coarser, point-estimate or listwise-complete measurement would blur, and the existence proof that the pipeline recovers structure rather than imposing it. Placing routinely

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collected biology *inside* the transdiagnostic factor space—as a co-equal latent axis rather than a correlate of symptom dimensions—is, to our knowledge, the first such map across bipolar disorder, schizophrenia and major depression. All findings are internal-validity results on observational data; individual-level clinical utility would require external validation and incident-event outcomes.

Keywords: transdiagnostic psychiatry; dimensional psychopathology; bifactor model; metabolic syndrome; inflammation; bipolar disorder; schizophrenia; major depression; archetypal analysis; precision psychiatry.

Table 1: **Sample characteristics by cohort (FACE baseline, N = 9,013).** Aggregate-only summary; sex coded female. Retention rows are the percentage of the baseline cohort re-assessed at the 12- and 24-month visits.

Characteristic	Bipolar	Schizophrenia	Depression	Overall
N	6,252	2,209	552	9,013
Age, years — mean (SD)	39.3 (13.4)	31.1 (9.6)	51.0 (13.3)	38.0 (13.4)
Age, years — median [IQR]	38 [28–49]	29 [23–37]	52 [42–60]	36 [27–48]
Female — n (%)	3,969 (63.5)	600 (27.2)	323 (58.5)	4,892 (54.3)
Education, years — mean (SD)	14.2 (2.8)	12.3 (2.6)	13.2 (3.2)	14.0 (2.8)
Recruitment sites — n	15	12	13	21
Education missing — n (%)	1,803 (28.8)	1,646 (74.5)	228 (41.3)	3,677 (40.8)
DSM-5 subtype — n (% of cohort)				
Bipolar I	2,635 (42.1)	—	—	
Bipolar II	2,956 (47.3)	—	—	
Bipolar, unspecified	661 (10.6)	—	—	
Schizophrenia	—	1,692 (76.6)	—	
Schizoaffective disorder	—	476 (21.5)	—	
Schizophreniform disorder	—	41 (1.9)	—	
Major depressive disorder	—	—	552 (100.0)	
Retained at 12 months — %	49.2	43.6	42.2	47.4
Retained at 24 months — %	35.6	26.9	24.5	32.8

1 Introduction

Psychiatric diagnoses are clinically indispensable but biologically nonspecific. The categories of the DSM aggregate patients who differ widely in their biology, course and treatment response, and they are undercut by pervasive comorbidity, within-category heterogeneity and diagnostic instability over time [1, 2]. Two influential programmes argue in response that psychopathology is better described by continuous, *transdiagnostic* dimensions than by categorical boundaries: the U.S. National Institute of Mental Health's Research Domain Criteria (RDoC) [1] and the Hierarchical Taxonomy of Psychopathology (HiTOP) [2, 3]. A recurring motif within this work is a general factor that spans disorders—the "p factor"—most defensibly read as a dimension of life impairment rather than a latent liability to specific symptoms [4, 5], though the meaning of such general factors remains contested [6, 7].

This dimensional structure has, however, been mapped almost entirely from a single *kind* of measurement—symptom report [2, 4] or brain imaging [8–10]. The routinely collected *biology* that drives the excess physical-health mortality of severe mental illness—its joint metabolic and inflammatory load—has not been placed *inside* the transdiagnostic factor space across disorders, even though an independent literature suggests it forms a partly separable immunometabolic axis that tracks atypical, energy-related symptoms more than overall severity [11–13], and that metabolic burden follows medication, adiposity and chronicity more than symptom load [14–16].

Table 2: **The eight-factor transdiagnostic map (uniform reporting).** Anchoring indicators, the direct general-factor loading, the correlated-G entanglement, adjudication verdict and cross-cohort measurement invariance for each factor, reported on the same footing. $|\lambda_G|$ is the mean absolute *direct* bifactor loading on G (how each domain relates to general burden under the orthogonal identification); corr G is the correlation from the freely-correlated-G model, reported for the well-covered continuous axes (the correlated arm does not cleanly separate the thin and explicit axes, for which only the direct loading is given). Substance is pinned orthogonal to the correlated block (its cross-factor correlations are non-identifiable). Depression/anxiety instruments are cross-loading windows on G (loadings 0.66–0.80), not a separate factor; the map is otherwise simple-structure apart from three earned cross-loadings into cognition (childhood adversity and poor sleep, $|\lambda| \leq 0.09$, 95% CI excluding zero).

Factor	Anchoring indicators	$\lvert\lambda_G\rvert$	corr G	Verdict	Invariance
G (burden)	FAST, EGF, EQ-5D (no symptoms)	—	—	severity factor	partial (SZ)
Cognition	TMT, WAIS coding, fluency	0.20	0.39	confirmed	invariant
Immunometabolic	BMI, lipids, glycaemia, BP, CRP, leukocytes	0.06	0.10	confirmed (single biology axis)	invariant
Sleep	PSQI total + subscales	0.20	0.42	confirmed	invariant
Mania	YMRS, Altman	0.13	—	confirmed	Altman floors (DR)
Suicidality	ISF ideation / attempt (binary)	0.11	—	confirmed (mixed)	3-cohort
Developmental risk	CTQ, perinatal, family history	0.07	—	confirmed (proxy)	invariant
Substance	alcohol/cannabis SUD, nicotine	0.07	—	confirmed (orthogonal)	BP–SZ (2-cohort)

Whether this load is separable from a transdiagnostic severity axis, and by how much, has not to our knowledge been measured directly.

We address this not with a single analysis but with a **four-stage measurement-and-validation pipeline**, and we treat the pipeline itself—not any one finding—as the contribution. Its methodological novelty is not a new estimator but a *discipline* assembled from three commitments that are rarely held jointly: the map is estimated from each patient's observed cells only, with *no imputation*, under mixed likelihoods and a Gaussian-copula continuous block; per-patient *uncertainty is propagated end-to-end*, so every downstream test (structure, temporal coherence, prognosis) operates on posterior coordinates rather than point scores; and the map is *fixed before any validation stage*, with diagnosis held out, so each result is a property of the measurement rather than an artifact fitted to the outcome. The immunometabolic dissociation that recurs across all four stages is the existence proof that this discipline reveals structure a looser measurement would blur. **(i) Measurement:** a global, missingness-aware Bayesian model places routine clinical, cognitive and biological data in one coordinate system with diagnosis held out, and reports all eight resulting factors uniformly. **(ii) Structure:** on the fixed map, an uncertainty-aware structure test asks whether the patient space is a graded continuum or a set of discrete kinds, and summarizes it with archetypes rather than imposed clusters. **(iii) Temporal coherence:** the map is scored forward onto follow-up under the frozen model. **(iv) Prognosis:** its incremental value beyond diagnosis and severity is tested against a hard autoregression bar and benchmarked against the raw indicators. Two commitments run throughout: we

Table 3: **The five archetypes.** Corners of the clinical–biological continuum—directions of maximal phenotype on the convex hull of the cloud, not discrete biotypes (the space carries no density gap; single-Gaussian falsification null $z = 1.13$, n.s.). $A = 5$ is the largest cross-seed-stable corner count (minimum Tucker congruence 0.979, explained variance 0.601, clean stability cliff to 0.436 at $A = 6$). Sizes are dominant ($\arg\max$) memberships over $N = 9{,}013$ patients; the inferential object is the continuous simplex weight vector, not the hard label.

Archetype	Defining profile (latent z-scale)	Clinical reading	N (dominant)
A0	↑sleep +2.8, ↑mania +1.7, ↓severity	*activation/sleep* (BP-enriched)	1,448
A1	↑severity +1.9, ↓immunometabolic −1.9, ↓developmental	*severe, clean-biology*	1,584
A2	↑immunometabolic +3.5, ↑severity +2.4, ↑suicidality	*immunometabolic burden* (the biology corner)	1,426
A3	↑developmental +2.8, ↑suicidality +1.4, ↓immunometabolic −2.1	*trauma/suicidality*	2,004
A4	↓sleep −2.7, ↓severity −2.0, ↓mania	*low-burden/well*	2,551

report the eight factors *uniformly* (loadings, general-factor loadings, factor correlations) and keep *measurement separate from interpretation*—the clinically salient properties of the map are read off neutral structural summaries, not built into them. The structural question is itself contested: normative-modelling studies report individual heterogeneity so extensive that "the average patient" is barely informative [8, 9], whereas the B-SNIP programme reports replicable, brain-based psychosis "biotypes" that cut across DSM boundaries [17–19]—a tension whose resolution may depend on which measurement space is interrogated.

We use latent modelling as a measurement tool, not as a claim that latent factors are more real than the items that define them. Routine clinical and biological data are high-dimensional, redundant and pervasively incomplete; a latent model fit to each patient's observed cells pools many noisy, overlapping indicators into a few axes, places heterogeneous data types (laboratory values, ordinal scales, counts) on one comparable scale, and gives every patient a position with quantified uncertainty rather than discarding incomplete records. Whether that representation *adds* anything over the raw indicators is then an empirical question, which we test directly: against the full raw indicator set under a matched classifier, the eight-dimension map is a sufficient representation for predicting deterioration and near-sufficient for recovery.

Using the harmonized baseline of the three FACE cohorts—bipolar disorder, schizophrenia and major depression, $N = 9{,}013$—we build *FACE-ATLAS*: one global, missingness-aware Bayesian bifactor / exploratory structural-equation model with mixed likelihoods, fit to each patient's observed cells only, that places clinical, cognitive, behavioural and laboratory measurements in one coordinate system while holding diagnosis out as metadata. We then score the fitted map forward onto two years of follow-up to test whether it persists and whether it predicts outcome. The pipeline yields an eight-dimension map—a general functional-burden factor and seven specific axes; a graded continuum rather than discrete biotypes *in this measurement space*, read through five archetypal extremes; a geometry that persists, with durable biology and moving symptoms; and a real but modest, *group-level* prognostic signal for functioning that is co-informative with diagnosis rather than a replacement for it. Running through all four stages is one recurring empirical property—a single immunometabolic axis that is distinct from the rest of the clinical picture, the most temporally durable dimension, and the worst-prognosis pole for functioning.

Methodologically, FACE-ATLAS is a latent-variable dimensional measurement model that combines mixed and Gaussian-copula observation layers with an observed-cell likelihood for structured missingness [20–22]: routine biological load is placed inside a transdiagnostic psychiatric factor space, its separation from functional burden is measured rather than assumed, and its temporal and prognostic implications are tested without converting a continuum into biotypes.

2 Methods

The complete development, with proofs, is in the Supplement (A–D). Throughout, $i = 1, \ldots, N$ indexes patients, $j = 1, \ldots, J$ indicators, $k = 1, \ldots, K$ the $K = 7$ specific factors and G the general factor; O_i is patient i's set of observed indicators and $C_i \subseteq O_i$ its observed continuous cells.

2.1 Cohorts, harmonization and the no-imputation data layer

Data were the baseline (visit V0) of three FACE cohorts (Fondation FondaMental expert centres): bipolar disorder, schizophrenia and major depression (N = 9,013; BP 6,252, SZ 2,209, DR 552; 21 recruitment sites). A harmonized data dictionary maps each modelled variable to its per-cohort source column, a harmonization rule and per-variable plausibility ("sanity") bounds; out-of-range values are set to missing (never clipped, never imputed). Deterministic skip-logic decoding sets structural zeros only where a gate is explicitly negative (for example, attempt count = 0 when ever-attempted = 0); this is decoding, not imputation. Each variable carries a likelihood family and a missingness type. Identifiers (patient, cohort, DSM-5 subtype, visit, site) are never modelled as indicators; cohort and subtype enter only as covariates, invariance-grouping variables or validation labels. Of the 201 usable harmonized common variables, 143 form the modelled-indicator pool (the comparator for the raw-indicator prognostic benchmark); the remaining 58 are identifiers, covariates or validation labels and are never factor indicators. The fitted factor map carries 109 of these indicators—88 in the Gaussian-copula continuous block and 21 in the explicit non-Gaussian block—across its eight axes.

No a-priori power calculation was performed: this is a secondary analysis of the complete baseline assessment of the three FACE cohorts, in which all N = 9,013 available patients are modelled (balancing by subsampling is rejected in favour of cohort weighting). The sample is sized for the study's *group-level* estimands—the factor structure, the biology–burden correlation ϕ_{Gd}, and incremental held-out predictive density—which it resolves well beyond chance (for example, all 37 immunometabolic loadings and the ϕ_{Gd} ordering have 95% intervals excluding the relevant null); it is correspondingly *not* powered for individual-level binary discrimination, so the small functional-remission AUC gain (+0.010) reflects this scope rather than an underpowered analysis.

2.2 Candidate ontology and soft priors

Ten candidate constructs defined a *soft-prior* ontology rather than a fixed output. Mapping the candidates against the common-variable coverage fixes the eligible set and is itself a primary result: a construct with no common indicator is reported *not testable*, never back-filled with an invented proxy. Each (indicator, factor) cell receives a prior *role*—primary, plausible-cross, unlikely or forbidden—whose default is shrinkage rather than exclusion, so the model can confirm, split, merge, downgrade, reject or declare not-testable any candidate. Formally, the loading in cell (j,k) carries the role-dependent prior

$$\lambda_{jk} \mid \rho(j,k) \;\sim\; \begin{cases} \mathcal{N}_{+}(0.70,\, 0.25^2) & \textsc{primary (home cell)}, \\ \text{regularized horseshoe (5)} & \textsc{plausible cross}, \\ \delta_0 & \textsc{unlikely (reported map)}, \\ \delta_0 & \textsc{forbidden}, \end{cases} \tag{1}$$

with $\mathcal{N}_{+}$ the positive truncation that sign-anchors each home loading and δ_0 a point mass at zero; the factor correlations take an LKJ(η=2) prior and the remaining priors are weakly informative (Supplement A). Off-home loadings therefore fall into two operative tiers. The curated PLAUSIBLE CROSS cells are freed under the regularized horseshoe of (5); the remaining ≈ 980 UNLIKELY cells are fixed at exactly zero in the reported map (`soft_unlikely=False`),

$$\lambda_{jk}^{\text{unlikely}} = 0 \qquad \text{(hard zero, reported map)},$$

rather than given a tight prior. Hard-zeroing these ≈ 980 cells is the principled choice for identifiability: freeing them instead as $\mathcal{N}(0, 0.05^2)$ floods every factor column with weak cross-loadings and dilutes the thin specific factors (substance use, mania/activation), degrading mixing. We additionally report the soft variant, $\lambda_{jk}^{\text{unlikely}} \sim \mathcal{N}(0, 0.05^2)$, as a documented *sensitivity* arm, which is congruent with the hard-zero map on the well-anchored backbone (Supplement A).

2.3 The measurement model

The measurement model rests on a small set of core equations. We posit that a few latent factors generate the many measured quantities, so each patient is summarized by a low-dimensional coordinate. Patient i carries a factor vector of one general functional-burden axis G and $K = 7$ specific axes, given a standardized multivariate-normal prior,

$$f_i = (G_i, D_{i1}, \ldots, D_{iK})^{\top} \;\sim\; \mathcal{N}_{K+1}(\mathbf{0},\, \Phi)\,, \qquad \operatorname{diag}(\Phi) = \mathbf{1}. \tag{2}$$

The unit diagonal of the factor correlation matrix Φ fixes the latent scale; G is the burden axis every indicator may share, while $D_{i1}, \ldots, D_{iK}$ are the seven domain-specific axes (cognition, immunometabolic, sleep, mania/activation, suicidality, developmental risk and substance use), so the map is eight-dimensional: one general axis plus seven specifics. The immunometabolic axis is a single biology factor carrying the cardiometabolic and inflammatory markers together (body-mass index, glycated haemoglobin, lipids, C-reactive protein), with bmi$\to$immunometabolic ≈ 0.95 and crp$\to$immunometabolic ≈ 0.37. Equation (2) is the formal content of the idea that a patient is *one general burden axis plus specific axes.*

Each indicator j is a linear reflection of those factors through a single predictor,

$$\eta_{ij} = \alpha_j + \lambda_{jG}\, G_i + \sum_{k=1}^{K} \lambda_{jk}\, D_{ik} + \beta_j^{\top} c_i, \tag{3}$$

with item intercept α_j, a general loading λ_{jG} onto G, specific loadings λ_{jk} (row j of the loading matrix $\Lambda = [\,\lambda_G \mid \Lambda_D\,]$), and item-level covariates c_i that shift an indicator's mean but are never themselves factor indicators. Equation (3) *is* the bifactor exploratory structural-equation model [20, 21]: bifactor because every indicator may load on G *and* on its home specific factor; exploratory, but *targeted*, because only the curated PLAUSIBLE CROSS cells of (1) are admitted under the sparsity-inducing horseshoe prior (5), while the great majority of off-home cells (the ≈ 980 UNLIKELY cells) are fixed at exactly zero rather than left open under a diffuse prior. A validator that instead frees *every* off-home cell under the same horseshoe shows the zeros are data-consistent rather than assumed: $\approx 83\%$ of the freed cells shrink to ≈ 0, recovering the same anchored map (detailed below). The predictor feeds a per-type observation density—Gaussian or log-Gaussian for continuous and laboratory measures, ordered-logit for ordinal, Bernoulli for binary, negative-binomial for counts—and continuous indicators additionally enter through a rank-based *Gaussian-copula* marginal $z_{ij} = \Phi_{\mathcal{N}}^{-1}(\hat{F}_j(X_{ij}))$ ($\hat{F}_j$ the frozen empirical distribution function of indicator j), so that forcing heterogeneous variables onto a shared scale never manufactures covariance and the latent dependence the claims live in is left unchanged.

The loading as a measurement slope. Equations (2)–(3) separate two intrinsically different objects. The factor f_i is a per-patient latent *position*—a random coordinate with prior $\mathcal{N}(\mathbf{0}, \Phi)$—whereas a loading is a fixed *slope*: the rate at which a measurement changes as that position moves, shared by every patient. On the latent linear-predictor scale this slope is exact and patient-independent, $\partial\eta_{ij}/\partial f_{ik} = \lambda_{jk}$, so the loading matrix is nothing but the Jacobian of the measurement with respect to the latent, $\Lambda = [\,\partial\eta_{ij}/\partial f_{ik}\,]_{j,k}$. On the response scale, for the single-index likelihoods (the continuous, log-, Bernoulli and count blocks) with conditional mean $\mathbb{E}[X_{ij} \mid f_i] = h_j(\eta_{ij})$ and inverse link $h_j = g_j^{-1}$, the chain rule gives the marginal effect of a factor on an indicator and its population average,

$$\frac{\partial\,\mathbb{E}[X_{ij} \mid f_i]}{\partial f_{ik}} = h_j'(\eta_{ij})\,\lambda_{jk}, \qquad s_{jk} \equiv \mathbb{E}\Big[\frac{\partial\,\mathbb{E}[X_{ij} \mid f_i]}{\partial f_{ik}}\Big] = \kappa_j\,\lambda_{jk}, \qquad \kappa_j = \mathbb{E}\Big[h_j'(\eta_{ij})\Big] \geq 0, \quad (4)$$

so the *expected derivative* of an indicator with respect to a factor is exactly its loading, rescaled by a non-negative, link-dependent constant κ_j (identity link $h_j' \equiv 1$, so $s_{jk} = \lambda_{jk}$; logit $h_j'(\eta) = \sigma(\eta)\{1 - \sigma(\eta)\}$; log link $h_j'(\eta) = e^{\eta}$; the ordinal and copula blocks carry the same slope λ_{jk} on their latent scale). The two objects are complementary, not interchangeable: the factors say *where* a patient lies in the coordinate system, while the loadings say *how fast* each instrument reads out as one moves along an axis—a position and the slopes that read it out are different kinds of quantity. This is also why neither is fixed by the likelihood alone: rotating the factors trades position against slope while leaving the data untouched. The Gaussian likelihood is invariant under any invertible factor rotation M—replacing (Λ, Φ, f) by $(\Lambda M,\ M^{-1}\Phi M^{-\top},\ M^{-1}f)$ leaves $\Sigma = \Lambda\Phi\Lambda^{\top} + \Psi$ unchanged—so the model is not identified from the likelihood alone (Prop. 2). Identification comes from the soft-zero loading priors, which are *not* rotation-invariant: the posterior concentrates at the orientation where small loadings align with their soft zeros and home loadings are positive and substantial. A prior-free (flat-prior) refit that recovers the same loadings to Tucker congruence $\varphi = 1.00$ therefore shows the map is recovered from the data, not imposed (Cor. 1).

The off-home cross-loadings are governed by a designed sparsity mechanism that keeps the map mostly simple-structure while letting a few evidence-backed cross-loadings emerge. On each off-home specific↔specific cell we place a regularized ("Finnish") horseshoe prior [23, 24],

$$\lambda_j = z_j\,\tau\,\tilde{\eta}_j, \qquad \tilde{\eta}_j^2 = \frac{c^2\eta_j^2}{c^2 + \tau^2\eta_j^2}, \qquad z_j \sim \mathcal{N}(0,1), \quad \eta_j \sim \mathrm{HalfCauchy}(1), \quad \tau \sim \mathrm{HalfNormal}(\tau_0), \tag{5}$$

with $\tau_0 = 0.05$ and slab $c = 0.30$, fit non-centred. The three pieces each do one job: the *global* shrinkage τ pulls the whole set toward zero (the prior's default belief is *no* cross-loadings); the heavy-tailed *local* scale η_j lets a single cross-loading the data insist on escape that shrinkage; and the *slab* c caps the escape ($\tilde{\eta}_j \to c/\tau$ for large η_j) so a cross-loading stays small (it is not a home loading). The prior is therefore *default-off, evidence-on, magnitude-capped.* This is exactly what protects the thin specific factors—substance and mania/activation are each defined by very few indicators—from dilution: spurious cells from other factors are crushed to zero by the global shrinkage, so a thin column stays defined only by its sign-anchored home indicators (home loadings are not horseshoed), while a clinically real weak cross-loading can still surface with a credible interval. Substance is additionally pinned orthogonal to the correlated block, its cross-factor correlations being non-identifiable. A continuous sparse-ESEM validation arm that frees every off-home cell confirms the structure is *recovered from the data, not imposed*: roughly

83% of the freed cells shrink to ≈ 0, and the credible remainder is folded back into the full mixed map, which retains three small cross-loadings—childhood adversity (CTQ-37, -0.094), sleep latency (PSQI-latency, $+0.057$) and daytime dysfunction (PSQI-daytime, -0.070) onto cognition, each with a 95% interval excluding zero. Given total freedom the model reproduces known clinical cross-talk and nothing spurious, which is positive evidence the map is well specified (Supplement A).

Robustness of the bifactor specification. Bifactor models are known to fit flexibly and, under some fit indices, to be preferred for reasons that are partly artifactual [25, 26]. We guard against this in three ways, and the immunometabolic dissociation survives all three. First, the finding does not depend on a general factor at all: the correlated-G arm of (10)—no forced orthogonal G—reproduces the same axis ordering, with the immunometabolic axis the least entangled with severity. The dissociation is therefore not a bifactor artifact. Second, simple structure is chosen by the data, not imposed by the prior: in the all-cells-freed sparse-ESEM validation arm above, $\approx 83\%$ of the freed off-home cells shrink to near zero, recovering the same anchored map. Third, the reported map fixes the ≈ 980 implausible cross-loadings at exactly zero (Eq. 1), which prevents the general factor from absorbing thin specific factors—the failure mode behind most bifactor over-extraction.

Given the latent scores the indicators are conditionally independent (local independence: once the factors are fixed, only item-specific noise remains, a diagonal residual covariance Ψ). Because the factors are then the only source of shared covariance, the complete-data density factorizes over items and the unobserved f_i is integrated out patient by patient,

$$L(\Theta) = \prod_{i=1}^{N} \int \Big[\textstyle\prod_{j \in O_i} p(X_{ij} \mid f_i, \Theta) \Big] \mathcal{N}(f_i; \mathbf{0}, \Phi)\, \mathrm{d}f_i, \tag{6}$$

the first-principles likelihood—a product over independent patients, a product over items (local independence), and the integral that removes the latent (the sum rule). Equations (2)–(3) also generate a structured covariance among the indicators,

$$\Sigma = \Lambda\, \Phi\, \Lambda^\top + \Psi = \underbrace{\lambda_G \lambda_G^\top}_{\text{general: rank one}} + \underbrace{\Lambda_D \Phi_D \Lambda_D^\top}_{\text{specific factors}} + \underbrace{\Psi}_{\text{private noise}}, \tag{7}$$

where $\Lambda = [\, \lambda_G \mid \Lambda_D \,]$ collects the loadings and Φ_D is the specific-factor correlation block; the cross-terms between G and the specifics vanish under the bifactor orthogonality $\mathbb{E}[G_i D_i^\top] = \mathbf{0}$ (Prop. 3). Equation (7) is the generative claim in one line—G leaves a rank-one footprint that touches every item, the specific factors contribute a block, and Ψ absorbs what no factor explains—and it is also why imputation is forbidden: a filled-in cell would inject covariance that (7) would misread as a factor.

Accordingly the model is fit to each patient's *observed cells only.* With F_j the density of indicator j, the observed-data log-likelihood is

$$\log p(X_{\text{obs}} \mid \Theta) = \sum_{i=1}^{N} \sum_{j \in O_i} \log F_j(X_{ij} \mid \eta_{ij}, \theta_j)\,. \tag{8}$$

A missing cell contributes no term and no completed $N \times J$ matrix is ever formed; deterministic skip-logic decodes structural zeros as *observed* values, not imputed ones. For the Gaussian block this is exact full-information maximum likelihood: marginals of a normal are read off by deleting coordinates, so the observed continuous sub-vector is

$$X_{i,C_i} \sim \mathcal{N}(\mu_{i,C_i}, \Sigma_{C_i}), \qquad \Sigma_{C_i} = \Lambda_{C_i} \Phi \Lambda_{C_i}^\top + \Psi_{C_i}, \tag{9}$$

keeping only the observed rows of Λ and the observed diagonal of Ψ; summing $\log \mathcal{N}$ over each patient's observed sub-vector is precisely the full-information maximum-likelihood (FIML) objective for incomplete data, so *marginalizing* the missing cells is the marginal property itself, not an imputation step (Prop. 1). Equation (8) is the no-imputation invariant on which every downstream analysis rests.

Two factor covariances are estimated. The primary *bifactor* fit holds G orthogonal to the specific axes (Φ block-diagonal); this is the natural null for the biology question. A *correlated-G* arm then unzips that zero block,

$$\Phi = \begin{pmatrix} 1 & \phi_{Gd}^\top \\ \phi_{Gd} & \Phi_D \end{pmatrix}, \qquad \phi_{Gd} = \mathrm{Corr}(G, D) \in [-1, 1]^K, \tag{10}$$

adding to the model-implied covariance the cross-terms the orthogonality had removed,

$$\Sigma = \lambda_G \lambda_G^\top + \underbrace{\lambda_G \phi_{Gd}^\top \Lambda_D^\top + \Lambda_D \phi_{Gd} \lambda_G^\top}_{\text{freed by correlating } G \text{ with the specifics}} + \Lambda_D \Phi_D \Lambda_D^\top + \Psi. \tag{11}$$

The arm thus turns "biology is separable from severity" from a built-in constraint into the *measured* vector ϕ_{Gd}: a domain is severity-entangled when its entry is large and separable when it is near zero (immunometabolic ≈ 0.10, versus cognition 0.39 and sleep 0.42 in the same freely-correlated model—so the separation is a cross-domain ordering the model is free to violate, not a consequence of the bifactor zero block). Reporting the bifactor general loadings $|\lambda_{jG}|$ and the correlated-G correlations ϕ_{Gd} together—they agree on the ordering—is what makes the separation a measurable estimand rather than an artefact of the orthogonality constraint (Figure 7 in the Results; Supplement A). The biology–G result is reported unadjusted and under a covariate-adjustment ladder (age, sex, education, site, antipsychotic exposure, adiposity) implemented by residualizing each indicator before the factor model.

2.4 Estimation, confirmation, invariance and scoring

The model is fit globally to the full sample ($N = 9{,}013$) under per-patient cohort weights $w_i = N/(3\, n_{c(i)})$, with $n_{c(i)}$ the size of patient i's cohort, so that each of the three cohorts contributes equal total weight $N/3$ while every patient is retained and $\sum_i w_i = N$. Balancing by subsampling would discard ~92% of the bipolar sample; weighting equalizes cohort influence without that loss. This is a composite-likelihood weighting: the point estimate is the balanced transdiagnostic estimand, and because the resulting posterior standard deviations are order-correct rather than fully calibrated we validate them by cross-seed congruence. The map's insensitivity to the weighting is quantified by the robustness battery below, one arm of which refits under inverse-cohort weighting against the unweighted full-sample fit (minimum Tucker $\varphi = 0.96$). The global posterior was reached by continuation: a well-conditioned sub-model was fit and then deformed toward the full target one source of difficulty at a time, each converged posterior warm-starting the next, and *only the global fit is interpreted.*

The Gaussian block is marginalized analytically: evaluating (9) naively costs $\mathcal{O}(|C_i|^3)$ per patient, and the Woodbury identity reduces it to the $(K{+}1)$-dimensional factor space,

$$\Sigma_{C_i}^{-1} = \Psi_{C_i}^{-1} - \Psi_{C_i}^{-1} \Lambda_{C_i} \big(\Phi^{-1} + \Lambda_{C_i}^\top \Psi_{C_i}^{-1} \Lambda_{C_i}\big)^{-1} \Lambda_{C_i}^\top \Psi_{C_i}^{-1}, \tag{12}$$

whose inner $(K{+}1) \times (K{+}1)$ solve is independent of $|C_i|$ and is shared by all patients with the same missingness pattern (computed once per pattern, with the matrix-determinant lemma for the log-determinant); this removes the per-patient latent funnel and makes full-sample estimation tractable on a workstation (Lemma 1). The reported map is fit on the full sample ($N = 9{,}013$), not on a balanced subsample: the Gaussian-copula continuous block is estimated at full N by marginalizing the per-patient latents in closed form via (12), and the explicit non-Gaussian block carries per-patient latents and is fit cohort-weighted at full N in the production run. Fitting the full sample rather than a balanced $N = 2{,}000$ approximation is a deliberate cost—the full-N mixed fit is the computational long pole of the pipeline—accepted so that the map reflects every patient's observed data rather than a down-sampled stand-in. Balanced $N = 2{,}000$ configurations appear only as a diagnostic scale and in the no-anthropometrics sensitivity arm, where they are labelled as such. Sampling used the No-U-Turn sampler [27] in PyMC [28] with the NumPyro/JAX backend [29]; a stage advanced only on standard convergence criteria (acceptable $\hat{R}$, adequate effective sample size, zero divergences, no Heywood cases, acceptable posterior-predictive behaviour), with a cross-seed stability guard for the joint mixed fit. Convergence is assessed against a two-tier gate. The continuous backbone—which carries the loadings and factor correlations—certifies at the strict bar ($\hat{R} \leq 1.01$, bulk-ESS ≥ 400, zero divergences). The full mixed map, which adds per-patient discrete latents for the explicit non-Gaussian indicators (suicidality, substance counts), is reported at the largest sample that mixes ($\hat{R} \leq 1.03$, within the < 1.05 convergence range of [30]); reproducibility for this block is established by cross-seed stability (Tucker congruence $\varphi = 0.99$) rather than by a single chain's $\hat{R}$ alone. Identification does not rest on information criteria, which favour bifactor over correlated-factor structures for reasons of functional form that are independent of which model generated the data [25, 26, 31]. We rely instead on three prior-free checks performed in-engine. First, an independent flat-prior refit (identification constraints only, fresh seed, no warm start) reproduced the loadings and factor correlations to three decimals (Tucker congruence $\varphi = 1.00$), so the structure is driven by the data rather than by the soft loading priors. Second, the correlated-G arm (Eq. 10) tests the G–specific orthogonality directly rather than imposing it. Third, posterior-predictive checks gave a Bayesian standardized root-mean-square residual ≈ 0.07 for the continuous block and reproduced 21/22 non-Gaussian indicators within the 90% predictive interval. As a secondary check WAIC and PSIS-LOO [30, 32] also ranked the bifactor above unidimensional and correlated-factor alternatives, but for the reason above we treat this as confirmatory only. Measurement invariance across BP/SZ/DR was tested by multi-group loadings and Bayesian differential item functioning, and robustness by leave-one-cohort-out congruence, subsampling, site cluster-bootstrap and inverse-cohort weighting.

Each patient's coordinates are the expected a posteriori (EAP) factor scores from their observed cells; for an all-continuous pattern,

$$\hat{f}_i = \mathbb{E}[f_i \mid X_{i,C_i}] = \Phi\Lambda_{C_i}^{\top}\Sigma_{C_i}^{-1}(X_{i,C_i} - \mu_{i,C_i}), \qquad S_i = \Phi - \Phi\Lambda_{C_i}^{\top}\Sigma_{C_i}^{-1}\Lambda_{C_i}\Phi, \tag{13}$$

obtained by MCMC when non-Gaussian cells are present. The posterior mean $x_i \equiv \hat{f}_i \in \mathbb{R}^8$ and the posterior covariance S_i are the objects every downstream analysis consumes: a patient with one observed indicator on an axis is *not* treated as equally characterized as one with six, so each downstream step propagates S_i rather than treating coordinates as equally measured.

2.5 A fast variational re-estimation

As an efficiency-oriented alternative to NUTS, we re-estimate the same measurement model—the bifactor/ESEM atlas of Eqs. (2)–(8), identical ontology and mixed likelihood—as a generalized linear latent-variable model [33] fit by stochastic variational inference [34, 35] in PyTorch [36]. Each patient's coordinate receives a variational posterior $q_i(f_i) = \mathcal{N}(\mu_i, \Sigma_i)$, and the model is fit by minimizing the negative evidence lower bound (observed cells only; the soft loading prior enters as a penalty; reparameterized gradients [37]), with the global parameters as maximum-a-posteriori estimates. This trades NUTS's exact posterior sampling for optimization within an approximate family: it is $\sim$100$\times$ faster ($\approx$4.5 minutes at full N on a CPU, versus hours) and reproduces the NUTS loading map and patient coordinates closely, at the cost of attenuated inter-factor correlations under a mean-field posterior—so it serves as a fast operational and generative engine while NUTS remains the authority for correlation magnitudes and credible intervals. The full model, inference and head-to-head comparison are in Supplement G.

2.6 Pre-specification and the downstream estimands

The analysis followed a pre-specified staged plan: the candidate construct ontology and the biology–burden correlation ϕ_{Gd}—the primary estimand—were fixed a priori (the soft-prior matrix, Supplement A), with the stratification, temporal and prognostic arms secondary. As a secondary analysis of an existing cohort the study was not prospectively registered; we therefore report all pre-planned arms, label post hoc analyses as such, and interpret the resulting multiplicity with caution. Each downstream analysis then treats the measurement model as *fixed*—reusing or re-scoring coordinates without re-discovering the map—and propagates each coordinate's posterior covariance S_i from (13) rather than treating the scores as exact.

2.7 Stratification: structure testing and archetypes

On the coordinates $\{x_i, S_i\}_{i=1}^N$ structure was tested *before* any clustering by a multi-statistic battery (Hopkins, Hartigan's dip, the gap statistic, the silhouette and HDBSCAN), each computed over the posterior draws so that measurement noise cannot manufacture structure, and anchored by a *single-Gaussian falsification null*: synthetic data drawn from one Gaussian $\mathcal{N}(\bar{x}, \widehat{\mathrm{Cov}}(x))$ with the empirical mean and covariance of the coordinates—structureless by construction—are run through the identical pipeline, and the real cloud's best partition (silhouette 0.140) is compared with the null's (0.137 ± 0.002),

$$z_{\mathrm{sil}} = \frac{s^\star_{\mathrm{real}} - \mathbb{E}[s^\star_{\mathrm{null}}]}{\mathrm{sd}[s^\star_{\mathrm{null}}]} = 1.13 \text{ (n.s.)}, \tag{14}$$

so the real partition separates patients no better than a structureless blob: the verdict is a graded *continuum*, not discrete biotypes. The continuum is summarized by archetypal analysis [38], which writes each patient as a convex blend of A extreme phenotypes that are themselves convex combinations of patients,

$$\min_{W,B} \sum_{i=1}^{N} \Big\| x_i - \sum_{a=1}^{A} w_{ia}\, z_a \Big\|^2 \quad \text{s.t.} \quad z_a = \sum_{j=1}^{N} b_{ja}\, x_j, \quad w_{ia}, b_{ja} \geq 0, \quad \textstyle\sum_a w_{ia} = \sum_j b_{ja} = 1, \tag{15}$$

i.e. $X \approx WZ$ with $Z = BX$ and W, B row-stochastic, which forces the archetypes z_a onto the convex hull of the cloud; the simplex weight $w_i \in \Delta^{A-1}$ is a patient's soft membership and is propagated by refitting across posterior draws. The number of corners ($A = 5$) is the largest with cross-seed Tucker congruence ≥ 0.8 (a clean stability cliff at $A = 6$, $0.979 \to 0.436$), set by stability rather than parsimony. A complementary measurement-error mixture treats x_i as a noisy reading of a latent position and deconvolves the *known* S_i (extreme deconvolution [39]),

$$x_i \ \sim \ \sum_{r=1}^{R} \pi_r\, \mathcal{N}(m_r,\ \Sigma_r + S_i), \qquad \pi \sim \mathrm{Dirichlet}\big(\tfrac{\alpha}{R}\mathbf{1}\big)\,, \tag{16}$$

so imprecise coordinates spread their mass across regions and prior-dominated axes self-down-weight; the posterior responsibilities are the soft tessellations. Because the mixture BIC is essentially flat in the number of regions R, no R is privileged: a nested family is exported and the operative granularity deferred to incremental predictive validity downstream. The representation is read against the held-out labels by the variance a partition explains per axis, $\eta^2_d = 1 - \sum_k \sum_{i \in \mathcal{C}_k} (x_{id} - \bar{x}_{kd})^2 / \sum_i (x_{id} - \bar{x}_d)^2$, and by the adjusted Rand index against diagnosis: the $A = 5$ archetypes explain mean $\eta^2 = 0.256$ of the eight-axis coordinate variance against 0.026 for the DSM-5 grouping ($\approx 9.7\times$) yet are nearly orthogonal to it (ARI 0.021).

2.8 Temporal coherence: invariance and trait/state

Follow-up visits were scored on the *fixed* $V0$ model—the frozen $(\hat{\Lambda}, \hat{\Phi}, \hat{\Psi})$, the frozen rank-copula maps $\hat{F}_j$ of (A36) and the frozen covariate residualization—via the EAP map (13); re-scoring $V0$

through this path reproduces the baseline coordinates at Pearson $r = 0.99$, so $V0$, $V1$ and $V2$ share one scale. Longitudinal measurement invariance was tested by the Tucker congruence of the per-visit loadings against $V0$ (all four backbone axes $\varphi \geq 0.95$). Trait was separated from state by a measurement-error random-intercept model that plugs the *known* baseline measurement variance s_i^2 (the relevant diagonal entry of S_i): writing x_{it} for patient i's coordinate at visit t,

$$x_{it} = \underbrace{\gamma_t}_{\text{visit fixed effect}} + \underbrace{u_i}_{\text{trait}} + \underbrace{e_{it}}_{\text{state}} + \underbrace{m_{it}}_{\text{measurement}}, \qquad u_i \sim \mathcal{N}(0, \tau^2),\ e_{it} \sim \mathcal{N}(0, \omega^2),\ m_{it} \sim \mathcal{N}(0, s_i^2), \tag{17}$$

the trait fraction being the intraclass correlation

$$\text{ICC} = \frac{\tau^2}{\tau^2 + \omega^2}, \tag{18}$$

reported as a posterior with a 94% credible interval. Plugging s_i^2 prevents a low-reliability axis from masquerading as state, and the visit fixed effects γ_t prevent a shared population trajectory—the shift $\gamma_{V2} - \gamma_{V0}$—from being miscounted as individual instability. The immunometabolic biology axis is the single most durable (ICC = 0.91), with cognition trait beside it (0.70) and the symptom axes state or mixed, while the population improves on symptoms (severity shift −0.46, suicidality −0.84) and holds on biology (+0.04).

2.9 Prognosis: errors-in-variables incremental validity

Prognosis tested whether a baseline coordinate predicts a two-year outcome *incrementally* beyond diagnosis, current severity and the baseline value of the outcome. Because coordinates are estimated rather than observed, a naive regression on $\hat{f}_i$ attenuates, so we fit an errors-in-variables Bayesian generalized linear model that carries each coordinate's posterior standard deviation s_{id} as a measurement model on a latent regressor θ_i,

$$\theta_{id} \sim \mathcal{N}(\hat{f}_{id}, s_{id}^2), \qquad g(\mathbb{E}[y_i]) = \beta_0 + \beta_W^\top W_i + \beta_\theta^\top \theta_i, \tag{19}$$

for outcome y_i (two-year functioning or severity) and link g, over a fixed ladder of designs W: a reference (age, sex, baseline outcome, baseline severity), then +DSM-5 subtype, +map (continuous coordinates, archetypes or a tessellation), or +both. Incremental value is the held-out expected log predictive density by Pareto-smoothed importance-sampling leave-one-out [30],

$$\Delta\text{ELPD} = \widehat{\text{elpd}}_{\text{loo}}(\text{model}) - \widehat{\text{elpd}}_{\text{loo}}(\text{reference}), \tag{20}$$

declared *predictive* only when it exceeds twice its standard error; for the binary endpoints—two-year functional remission (reaching GAF ≥ 71 from baseline impairment) and deterioration (a GAF drop ≥ 10), each pooled across the 12- and 24-month visits—we also report the cross-validated change in AUC. Robustness used inverse-probability-of-attrition weighting, restriction to the reliable axes and a permutation null, and a separate representation benchmark compared the map against all 143 raw indicators under a matched gradient-boosting classifier (Supplement D).

3 Results

3.1 Sample and observed-data modelling population

The analysis sample was the harmonized baseline (visit V0) of the three FACE cohorts: $N = 9{,}013$ patients (bipolar disorder [BP] 6,252; schizophrenia [SZ] 2,209; major depression [DR] 552) recruited across 21 expert-centre sites (Table 1). We deliberately fit the measurement model on the *full* sample without completeness selection. Of the 201 usable harmonized common variables (the full role breakdown is in the Methods), the 143-variable modelled-indicator pool has mean cell missingness 39.8%, preserved as missing and never imputed; admission to the map is *coverage-agnostic* above a two-cohort floor—the observed-cell likelihood lets a two-cohort marker serve as an indicator where it is measured, so about one-third of the fitted indicators are present in only two cohorts. Supplement A.1 (Table A6) classifies all 225 harmonized variables by role and the selection rule. Diagnosis (cohort and DSM-5 subtype) was held out of the model and used only as a validation or metadata label. As detailed in the Methods, the FACE-ATLAS map is one global, cohort-reweighted, full-sample Gaussian-copula fit; because the same structure is recovered under a plain Gaussian likelihood, the central findings do not depend on that modelling choice—a robustness result, not an assumption (Supplement A). Follow-up visits at 12 and 24 months (retained: 47.4% and 32.8% of V0; Extended Data Fig. E3) were used only to test temporal coherence and prognosis, never to define the map. The full derivation of the model—the observed-data likelihood, the rotation and identification argument, the covariance identity and the equivalence with full-information maximum likelihood—is given in Supplement A. Figure 1 presents the atlas at a glance; the study design and the three core modelling principles are detailed in Extended Data Fig. E1.

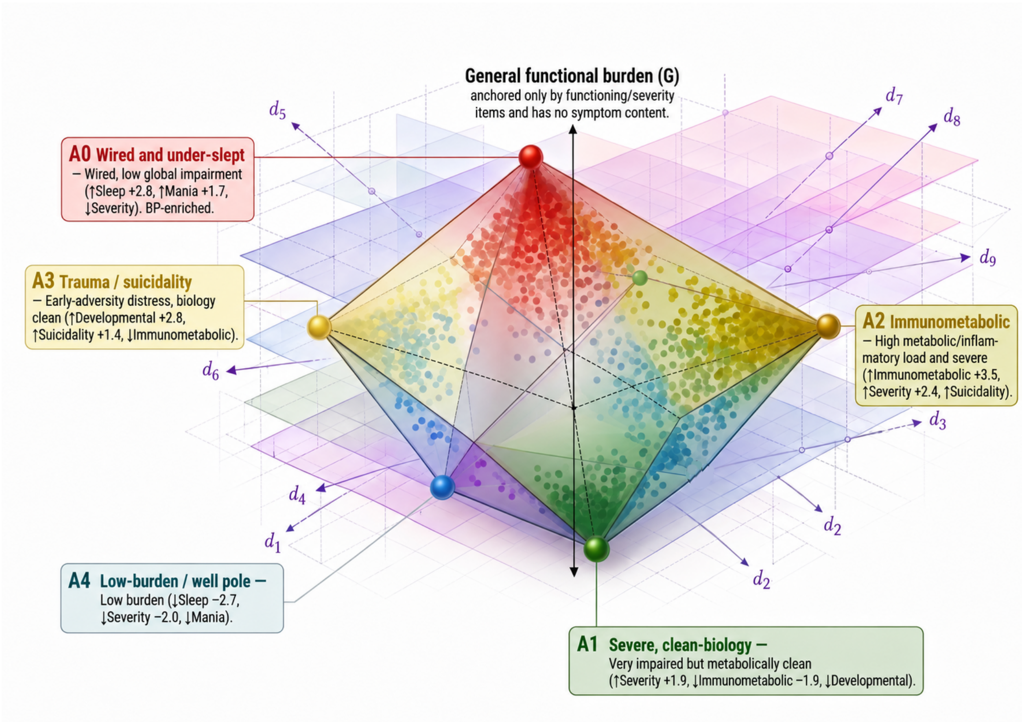

Figure 1: **Illustration of the FACE ATLAS: latent dimensions of severe mental illness and corresponding archetypes.** The global, missingness-aware Bayesian bifactor / exploratory structural-equation model, fit to the FACE baseline (N = 9,013) with diagnosis held out, yields an eight-dimension transdiagnostic latent space—a general functional-burden factor (G) and seven specific dimensions. The patient cloud is a graded continuum (best silhouette 0.140, indistinguishable from a single-Gaussian null 0.137 ± 0.002), read through five archetypal extremes whose convex blends describe every patient: an *activation / sleep* corner (A0), a *severe, clean-biology* corner (A1), an *immunometabolic* corner (A2; the biology pole), a *trauma / suicidality* corner (A3), and a *low-burden* pole (A4). Immunometabolic load is largely dissociated from general burden; it is the most durable axis over two years; the archetype phenotype improves held-out two-year functional prediction (ΔELPD $+62.8 \pm 11.2$); and two-year functional remission ranges 22%–63% across corners, the immunometabolic corner worst. A1 and A2 carry comparable severity but opposite biology.

3.2 The fitted object: a posterior coordinate for each patient

The primary output of FACE-ATLAS is not a cluster label but a *posterior coordinate distribution*. For patient i the model returns an eight-dimensional coordinate

$$f_i = \big(G_i,\ D_{i,\text{cog}},\ D_{i,\text{immet}},\ D_{i,\text{sleep}},\ D_{i,\text{mania}},\ D_{i,\text{suic}},\ D_{i,\text{dev}},\ D_{i,\text{subst}}\big) \in \mathbb{R}^8, \tag{21}$$

one general functional-burden coordinate G_i and seven specific coordinates (cognition, immunometabolic, sleep, mania, suicidality, developmental risk, substance use)—the seven specific axes are established as distinct, data-supported dimensions in §3.5 (Table 2). What the model actually delivers is the posterior of this coordinate given the patient's *observed* indicators x_{i,O_i} alone—no imputation, no completed data matrix (Methods):

$$p(f_i \mid x_{i,O_i}) \ \approx\ \mathcal{N}\Big(\hat{f}_i,\ S_i\Big), \tag{22}$$

whose mean $\hat{f}_i$ is the patient's *position* and whose covariance S_i is the *uncertainty* of that position—a location with error bars, not a category. Two patients can share an estimated immunometabolic score yet differ in its reliability (a complete laboratory panel versus a single observed marker), and every downstream analysis in this paper consumes the full $(\hat{f}_i, S_i)$ rather than treating coordinates as exact, so a sparsely measured patient is never mistaken for a precisely located one. The unit of analysis is thus a position on eight continuous axes with quantified uncertainty, not a biotype label.

3.3 How sharply is a patient located? Information accumulation and a minimal-indicator rule

The covariance S_i in (22) is not a fixed property of the model but a closed-form function of *which* indicators a patient has. Because the indicators are conditionally independent given the factors, each observed cell contributes additively to the posterior *precision*: for the continuous (Gaussian-copula) block the posterior covariance after observing a set of indicators is

$$S^{-1} = \Phi^{-1} + \sum_{j \text{ observed}} \frac{1}{\sigma_j^2}\,\lambda_j\lambda_j^\top, \qquad \hat{f} = S \sum_{j \text{ observed}} \frac{\lambda_j\, x_{ij}}{\sigma_j^2}, \tag{23}$$

the information form of the EAP score map (Methods, Eq. 13), with $\sigma_j^2 = 1 - \lambda_j^\top \Phi \lambda_j$ the indicator's residual variance. Two consequences follow. First, **the position uncertainty can only shrink as instruments arrive**: the summed term in (23) is positive semidefinite, so every observed cell tightens the ellipsoid, and a patient with no observed indicators on an axis is returned the prior rather than a false-precise point. Second, **the localization is anisotropic**: the rate of collapse on each axis is set by the loading-weighted information it receives, so a heavily instrumented axis (immunometabolic, 37 markers; cognition; general burden) reaches a posterior SD around 0.2–0.3 from a handful of items, while a thin axis (substance and mania, two to four indicators) stays wide even at full coverage. This is the quantitative content of "a sparsely measured patient is never mistaken for a precisely located one," and Figure 2 follows one real, de-identified patient (BP-62162, bipolar disorder) through the whole pipeline as a single illustrative worked example. The patient's posterior places them at the extreme of the immunometabolic axis (+3.57 SD, posterior SD 0.22) from only six observed metabolic items, while cognition and suicidality are prior-dominated (zero items observed, posterior SD $\approx$ 0.99; Fig. 2A). The archetype layer summarises this position as an 86% weight on the immunometabolic corner A2 (Fig. 2B). Locating the patient in the archetype-by-cohort prognostic atlas (§3.5–§3.9 below) returns an interval-valued forecast rather than a point estimate—27% predicted functional remission within bipolar disorder, the lowest-remitting archetype in that cohort against 73% at the low-burden pole (Fig. 2C)—and the model further identifies the single most informative

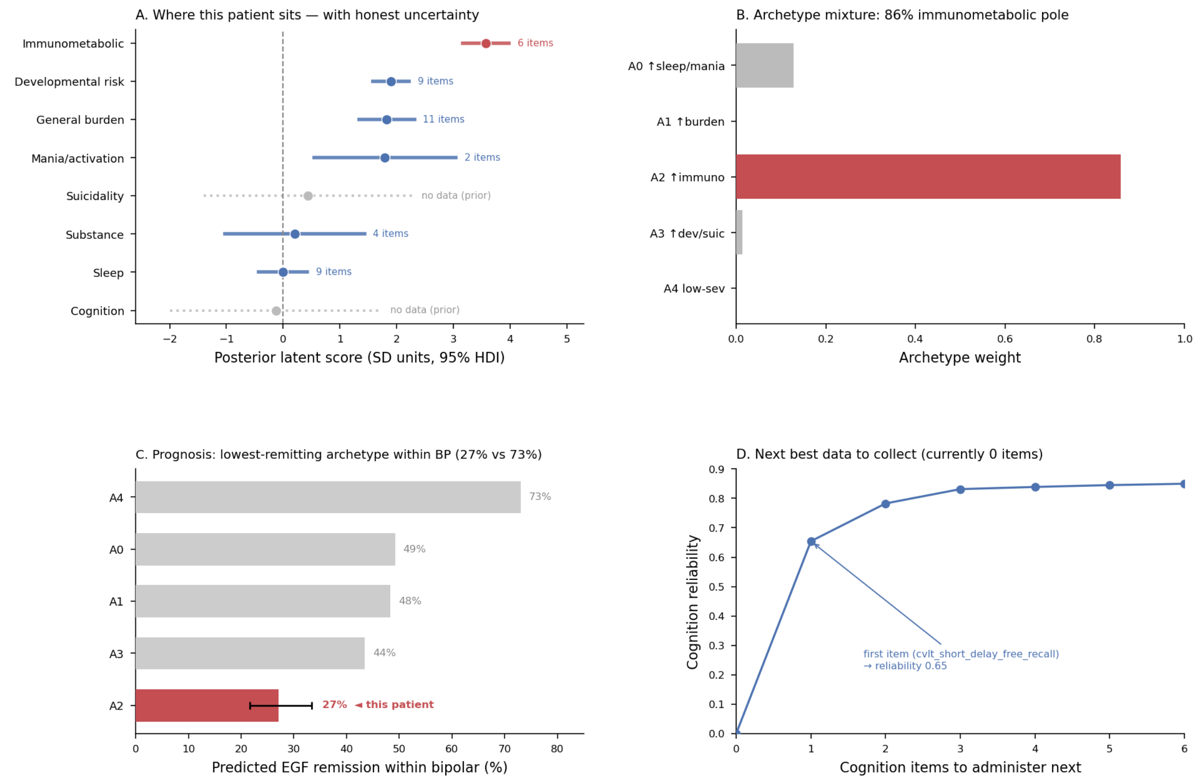

Figure 2: **A worked patient: coordinates → archetype → prognosis → next best measurement.** One real, de-identified patient (BP-62162, bipolar disorder) followed end to end through the pipeline; illustrative of the instrument on a single case, not a population-level result. **(A)** Posterior latent score on all eight axes with 95% highest-density intervals; the patient sits at the extreme of the immunometabolic axis (+3.57 SD from six observed items) while cognition and suicidality are prior-dominated (no observed items, shown as wide grey intervals). **(B)** The archetype layer summarises this position as an 86% weight on the immunometabolic corner (A2). **(C)** Locating the patient in the archetype-by-cohort prognostic atlas gives an interval-valued forecast: 27% predicted two-year functional remission within bipolar disorder, the lowest of the five archetypes (versus 73% at the low-burden pole A4). **(D)** Because cognition is currently unmeasured for this patient, the model identifies the single most informative next item: one verbal-memory test raises cognition reliability from 0 to 0.65.

unmeasured indicator: one cognitive item (`cvlt_short_delay_free_recall`) would raise the cognition axis's reliability from 0 to 0.65 (Fig. 2D). Because this is one patient, the vignette is illustrative rather than the evidence base; the population-level reliability curves and cohort-level measurement gaps that generalise it are given next.

A loading-dependent minimal-indicator rule. Equation (23) also answers a practical design question—how many indicators must a factor carry before a patient's coordinate on it is acceptably precise—and the answer is set not by the raw *count* of indicators but by the *information* they carry. For a single standardized factor measured by m conditionally-independent indicators the posterior SD is the accumulated information

$$\mathrm{SD}(f) = \frac{1}{\sqrt{1+I}}, \qquad I = \sum_{j=1}^{m} \frac{\lambda_j^2}{1-\lambda_j^2}, \tag{24}$$

in which each indicator contributes $i(\lambda_j) = \lambda_j^2/(1-\lambda_j^2)$, its signal-to-noise (reliability) ratio. This per-indicator information is steeply nonlinear in the loading (Fig. 3B): $i(0.3) = 0.10$,

$i(0.5) = 0.33$, $i(0.7) = 0.96$ and $i(0.95) = 9.3$, so a single strong indicator can be worth dozens of weak ones. Requiring a posterior SD no larger than a tolerance τ is the information threshold $I \geq 1/\tau^2 - 1$; for a target SD ≤ 0.5 (reliability 0.75) this is $I \geq 3$, met by about 4 indicators at $\lambda = 0.7$, 9 at $\lambda = 0.5$ or 31 at $\lambda = 0.3$ for equal loadings (Fig. 3A)—or by a single very strong anchor on its own. The cut-off is therefore loading-dependent, not a fixed instrument count, with minimal count $m^\star = \lceil(\frac{1}{\tau^2} - 1)(1 - \lambda^2)/\lambda^2\rceil$.

Real factors do not lie on any single equal-loading curve, because their loadings are heterogeneous, and because the eight axes span four measurement families (Gaussian, logistic, graded-response, negative-binomial) whose per-indicator information takes a different closed form in each. Computing the exact Fisher information every indicator contributes at the population mean under its own family—rather than the equal-loading approximation above—and adding indicators in decreasing-information order gives the realised reliability-vs-items curve for each of the eight axes on the same footing (Fig. 3, left). Three to six items recover most of each axis's reliable signal. On the immunometabolic axis, the anthropometric triad—body-mass index, weight and waist circumference—reaches a score reliability of 0.85 in three items and the axis plateaus near 0.88 (Fig. 3, right); the remaining 34 indicators (the full lipid and inflammatory panel, of the axis's 37 total markers) add less than 0.02 each. Two axes are intrinsically short: mania/activation, with only two indicators, cannot exceed a reliability of 0.408, and substance use, with four, cannot exceed 0.429. This is a bank limitation, not a missingness problem—95–96% of patients have at least one item on each axis—and it is itself an informative finding: those two constructs are under-instrumented in the current battery, not under-measured in these patients. The practical lesson for instrument design is therefore not "collect more questionnaires" but "include at least one high-loading marker per intended dimension," and, for mania and substance use specifically, a finer or graded instrument rather than additional weak items.

3.4 A few well-chosen indicators recover most of each dimension

This reliability-vs-items analysis is, in miniature, an adaptive-assessment result: it identifies, per dimension, the minimal set of indicators that recovers most of the reliable signal, so a compact axis-targeted battery can be deployed at a new site with a stated reliability rather than the full instrument set—directly informing which measures a future cohort must collect to place patients on the map, and which are redundant. Because the Fisher information above is evaluated at the population mean, the curves in Fig. 3 are an upper bound on the efficiency a fully adaptive test (one that re-evaluates information at each patient's provisional estimate) would realise; and because the loadings and item parameters are held at their point estimates, the analysis does not itself propagate parameter uncertainty. These mania/activation and substance-use ceilings are a property of the instrument bank, not of the measurement approach (coverage is not the issue, §3.3).

From uncertainty to a shared battery: designing the next cohort's measurement. The same arithmetic that asks "how many items per axis" also answers the consortium-level question: given a fixed measurement budget, which items across *all* axes should a future cohort collect, and where is each existing cohort under-measured? By the matrix-determinant lemma applied to (23), observing a not-yet-measured indicator j reduces the differential entropy (log-volume) of the posterior ellipsoid by $\frac{1}{2}\log(1 + \lambda_j^\top S\lambda_j/\sigma_j^2)$—a closed-form *value of information* that depends on what has already been measured. Greedily adding the single most-informative item across all eight axes (recomputed after each, using the exact per-family Fisher information of §3.3), a shared 27-item cross-axis battery brings every axis online at mean reliability 0.70 (Fig. 4, left); the first item selected on each axis is its clinically canonical instrument (CTQ, PSQI, ISF, BMI, CVLT, FAST, then the short substance and mania scales). The same framework exposes where existing cohorts are currently under-measured (Fig. 4, right): schizophrenia participants were assessed on markedly fewer immunometabolic items than bipolar participants (20.7 vs. 34.9

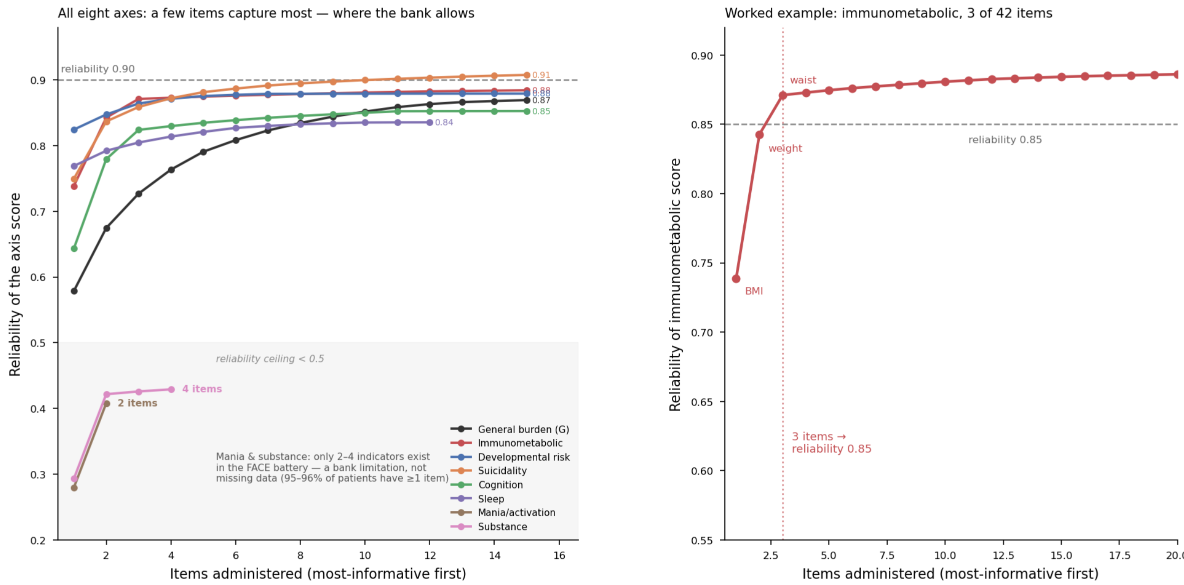

Figure 3: **Adaptive assessment: how many instruments are enough?** Reliability of the axis score against the number of items administered (most-informative first), computed from the exact Fisher information each indicator contributes under its own likelihood family (Gaussian, logistic, graded-response, negative-binomial), so all eight axes are compared on the same footing. **Left**: all eight axes. Six well-instrumented axes climb to reliability 0.85–0.90 within about ten items; mania/activation (two indicators) plateaus at 0.408 and substance use (four indicators) at 0.429—a ceiling set by the size of the instrument bank, not by missingness (95–96% of patients have at least one item on each). **Right**: worked example for the immunometabolic axis—the anthropometric triad (BMI, weight, waist) reaches reliability 0.85 in three items and the full 37-indicator axis plateaus near 0.88.

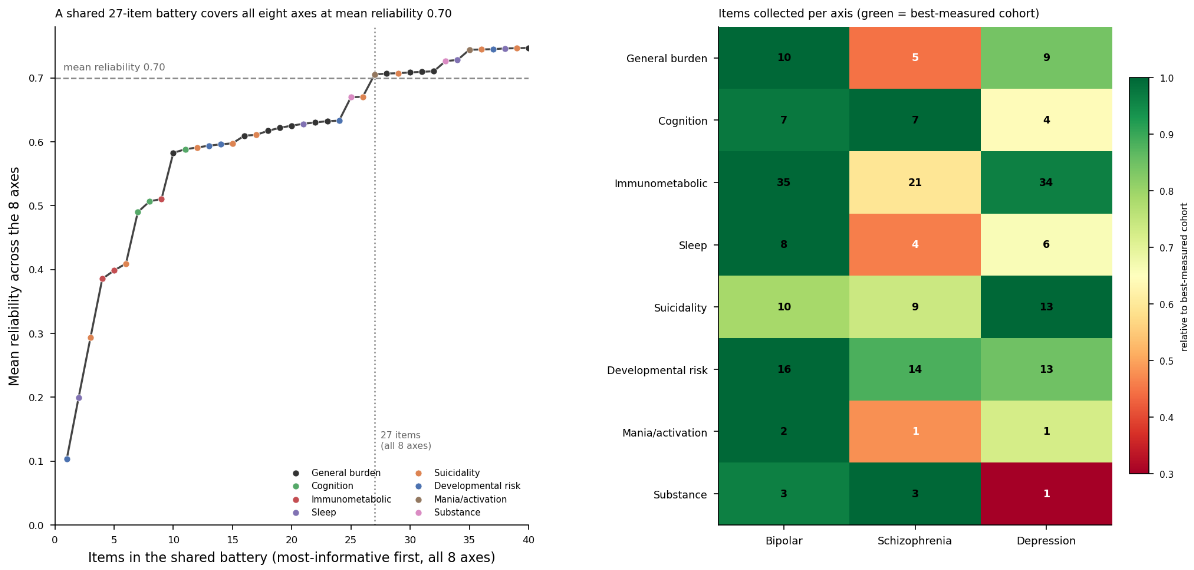

Figure 4: **Value of information: what a future cohort should collect. Left**: greedily adding the single most-informative item across all eight axes (exact Fisher information per likelihood family), a shared 27-item battery brings every axis online at a mean reliability of 0.70; the first item chosen on each axis is its canonical instrument (e.g. CTQ, PSQI, ISF, BMI, CVLT, FAST). **Right**: mean items collected per axis by cohort, shaded relative to the best-measured cohort. Schizophrenia is under-measured relative to bipolar disorder on immunometabolic (20.7 vs. 34.9 items) and sleep (3.9 vs. 8.4) indicators, and depression is thinnest on substance-use items—the highest-value additions for future data collection at each site.

items on average) and fewer sleep items (3.9 vs. 8.4), while depression is thinnest on substance-use items—precisely the axes on which each cohort's under-measurement is most consequential. For a consortium harmonising protocols across disorders, this converts the measurement model into an actionable specification—a common core battery plus disorder-specific top-ups—rather than a post-hoc analysis. The item ranking uses pooled item parameters, so it assumes a common ordering across cohorts; the gaps themselves are empirical. The arithmetic that quantifies a coordinate's uncertainty therefore also prescribes how to reduce it most cheaply—a concrete clinical and consortium-level use of the map beyond description.

3.5 The eight dimensions of the transdiagnostic atlas

From ten candidate constructs, one global model returned a stable eight-dimension structure: a general functional-burden factor (G) plus seven weakly correlated specific axes—cognition, immunometabolic, sleep, mania, suicidality, developmental risk and substance use (Figure 5; Table 2). Inter-factor correlations among the specific axes were weak throughout (mean absolute off-diagonal ≈ 0.08; Figure 6b), confirming genuinely distinct axes rather than one collapsed dimension; the single notable coupling was a modest mania–sleep correlation (≈ 0.23). G was anchored only by functioning and global-severity indicators (FAST loading 0.78, EGF 0.60) and carried no symptom content, supporting its conservative reading as transdiagnostic *functional burden* rather than a symptom-liability "p" factor. Depression and anxiety instruments (MADRS, QIDS, STAI) did not form a separate factor; they loaded 0.61–0.82 on G as cross-loading "windows" onto burden. The biology axis is a single *immunometabolic* factor—cardiometabolic and inflammatory markers (body-mass index, HbA_{1c}, lipids, C-reactive protein) load on one common dimension, anchored by body composition (BMI loading ≈ 0.95, CRP ≈ 0.37; all 37 loadings with credible intervals in Table A7), the data placing cardiometabolic and immune burden on a shared axis rather than two. Body size anchors this axis but does not constitute it: excluding the three anthropometric indicators and refitting, a coherent cardiometabolic–inflammatory axis survived—all 34 remaining markers credible, re-anchored on blood pressure, lipids and inflammation, and highly congruent with the full axis (Tucker $\varphi = 0.90$; Supplement A.7). Of the ten prior candidate factors, anhedonia was rejected (absorbed into G and the depression/anxiety windows), and a further three constructs with no common indicator—impulsivity, negative symptoms and sensory abnormality—were declared *not testable*, each verdict an outcome of the analysis rather than an assumption (Table 2). The structure was not an artefact of the priors: an independent flat-prior refit reproduced the loadings and factor correlations essentially exactly (Tucker congruence $\varphi = 1.00$). Information criteria also ranked the bifactor above unidimensional and correlated-factor alternatives, but we treat this as secondary evidence, since such criteria are known to favour bifactor structures for reasons of model form independent of the data-generating structure [25, 26]. Absolute fit was acceptable across both likelihood blocks (continuous Bayesian standardized root-mean-square residual ≈ 0.07; 21/22 non-Gaussian indicators reproduced within the 90% posterior-predictive interval). *Clinically,* the axes answer two different questions: G answers *how impaired* the patient is, while the seven specific axes answer *what kind* of burden the patient carries.

The map is otherwise simple-structure—each indicator loads on its home axis and G—by design rather than by fiat. A sparsity-inducing regularized (“Finnish”) horseshoe prior is placed on every off-home specific-to-specific loading: a global shrinkage term pulls the whole set toward zero (the prior’s default belief is “no cross-loadings”), heavy local tails let a genuinely supported cross-loading escape, and a slab caps its magnitude. This protects the thin factors (substance, mania, each defined by very few indicators) from dilution while letting small, clinically real cross-talk emerge. Given total freedom—a continuous sparse-ESEM refit that frees every off-home cell—$\approx$ 83% shrank to ≈ 0 (the simple structure is *recovered from the data*, not imposed), and exactly three cross-loadings survived into the full model with 95% credible intervals excluding zero, all onto cognition: childhood adversity (CTQ, -0.094) and two sleep items (PSQI latency $+0.057$, daytime dysfunction -0.070). Given the freedom to load anywhere, the model reproduced known clinical cross-talk—adversity and poor sleep weighing weakly on cognition—and nothing spurious: positive evidence that the map is well-specified, not an artefact of rigid zeros (Supplement A).

3.6 How each domain relates to general burden

Because the bifactor identification fixes G orthogonal to the specifics by construction, a domain’s relationship to general burden is read from its indicators’ *direct loadings on G* (Figure 6). On this metric three axes sit at the periphery, loading only weakly on G: immunometabolic (≈ 0.06), substance (≈ 0.07) and developmental-risk (≈ 0.07)—a near-tie that, by itself, singles out none of them. To convert “biology is separable from severity” from a modelling constraint into a measured quantity, we re-estimated the model allowing G to correlate freely with the specific axes. The resulting vector of general–specific correlations, $\phi_{Gd} = \mathrm{Corr}(G, D)$ (Methods, Eq. 10), is the *primary estimand* of FACE-ATLAS. Under this correlated-G model, immunometabolic load was the *least* entangled with functional burden: a correlation with G of ≈ 0.10, against 0.39 for cognition and 0.42 for sleep (Figure 7). In shared-variance terms the gap is stark—immunometabolic load shares only $\approx 1\%$ of its variance with general burden, against $\approx 15\%$ (cognition) and $\approx 18\%$ (sleep).

What distinguishes immunometabolic load from the other two peripheral axes is that it is decoupled from the *entire* clinical picture, on a freely-estimated basis. Substance is orthogonal *by construction* (pinned, because its cross-factor correlations are non-identifiable), so its independence is imposed rather than measured; developmental-risk, though weak on G, couples to the symptom axes (sleep and suicidality, $\Phi \approx 0.20$) and indexes a historical antecedent rather than a current state. Immunometabolic load alone is weakly coupled both to functional

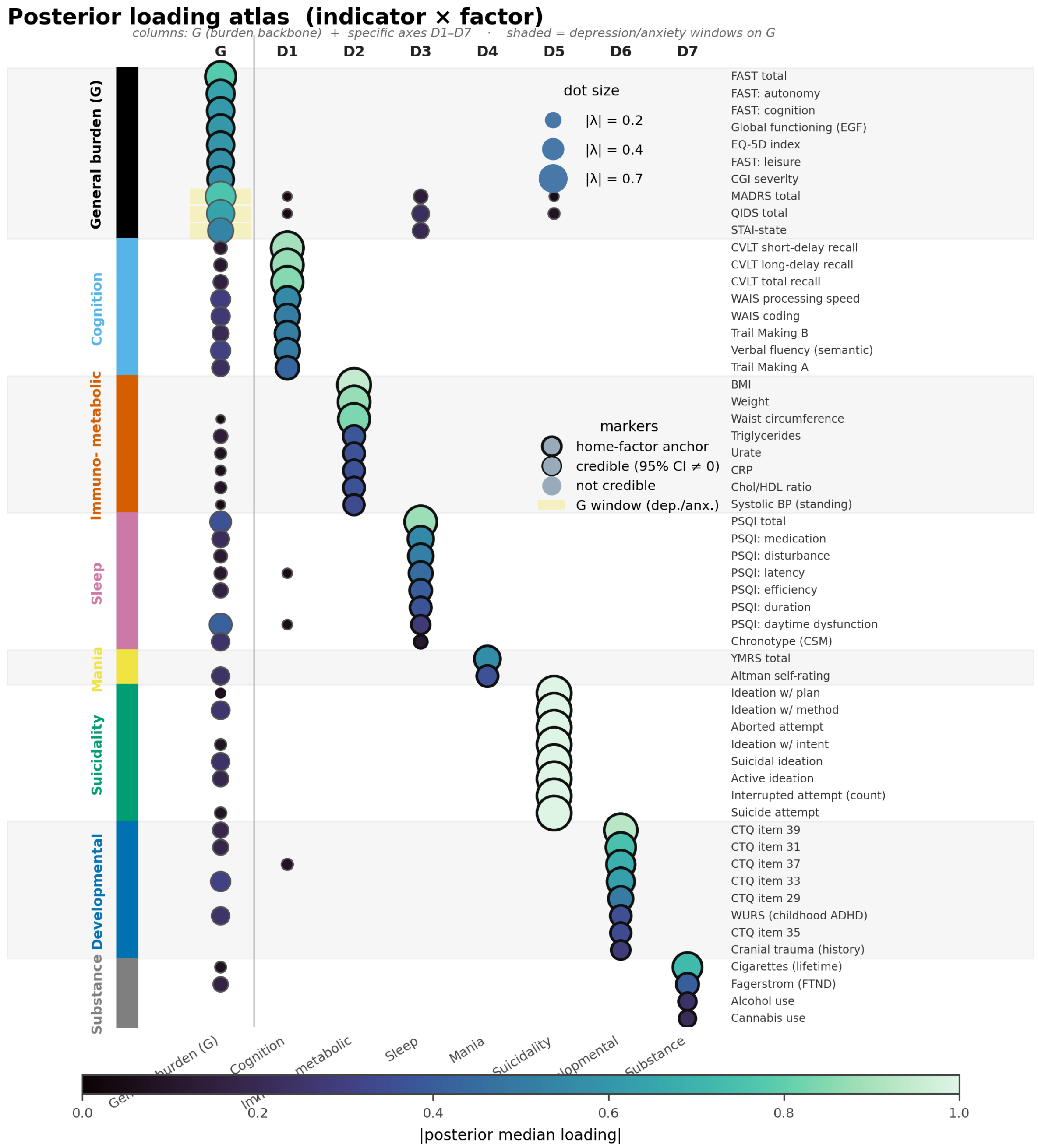

Figure 5: **Posterior loading atlas of the measurement model.** Rows are clinical indicators grouped by their home factor (coloured left strip and alternating bands); columns are the general functional-burden dimension G and the seven specific latent axes (D1–D7). Dot size and colour encode the absolute posterior median loading; outlined dots indicate loadings whose 95% credible interval excludes zero, and a heavier ring marks the home-factor anchor. Specific axes are each anchored by a coherent clinical indicator block, whereas G captures broad functional burden and carries cross-loading "windows" over the depression and anxiety instruments (MADRS, QIDS, STAI; shaded on the G column). The three data-supported off-home cross-loadings—childhood adversity and two sleep items onto cognition—render as outlined dots in the cognition column. The full eight-factor map is shown (continuous backbone plus the mania and substance axes from the joint mixed fit); the complete atlas over all modelled indicators is Extended Data Fig. E2. This structure supports an atlas-like reading of the latent space: specific dimensions are clinically localised axes, while G acts as a transdiagnostic burden backbone. Factor-wise interpretability and the inter-factor correlation matrix are in Figure 6.

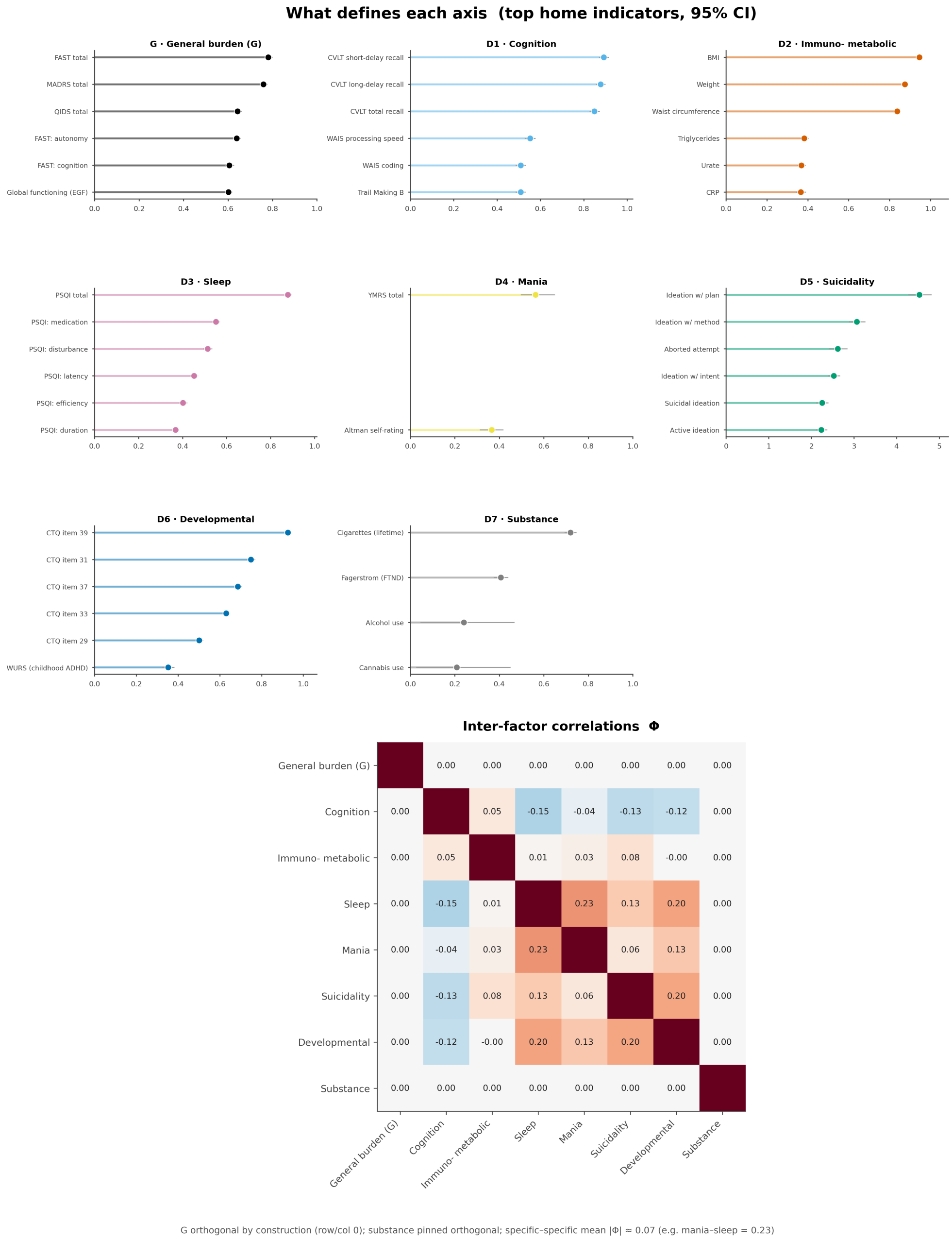

Figure 6: **Factor interpretability and inter-factor correlations.** (**a**, top) Factor-wise lollipops: for each axis (G and D1–D7), its strongest home indicators with 95% credible whiskers—"what defines each axis." (**b**, bottom) The inter-factor correlation matrix Φ is weak throughout (specific–specific mean |off-diagonal| ≈ 0.08); G is orthogonal to the specific axes by construction (top row and column). The one notable specific coupling is mania–sleep (≈ 0.23); substance is pinned orthogonal to the correlated block (its cross-factor correlations are non-identifiable), and the biology×symptom cells all stay at the off-diagonal floor.

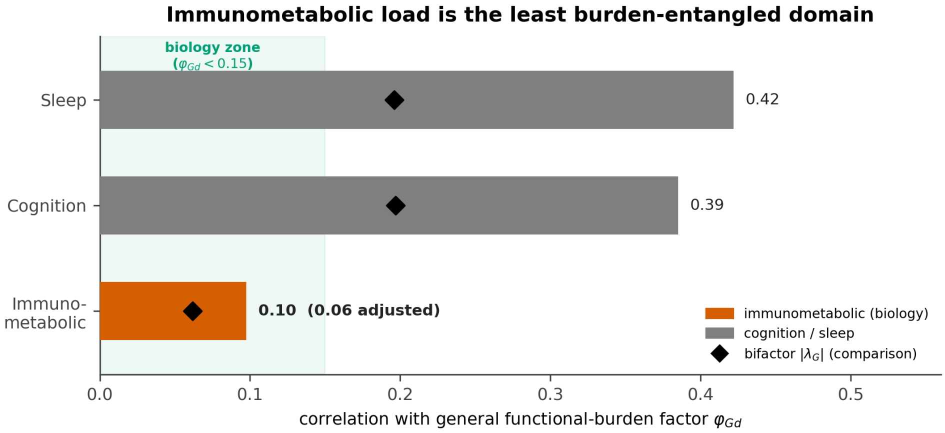

Figure 7: **Immunometabolic load is the least burden-entangled domain.** Correlation of each continuous specific axis with the general functional-burden factor G, $\phi_{Gd} = \mathrm{Corr}(G, D)$, from the correlated-$G$ refit (bars); the bifactor general loading $|\lambda_G|$ (diamonds) is shown for comparison. Immunometabolic load falls in the "biology zone" ($\phi_{Gd} < 0.15$) and stays there under adjustment for medication, adiposity and site (annotated value), whereas cognition and sleep partly track G. (The correlated-G sensitivity model resolves biology into inflammatory and metabolic factors; these are averaged into the single immunometabolic axis of the map, Supplement A.)

burden *and* to the symptom axes—its largest inter-factor correlation is ≈ 0.08, against ≥ 0.16 for every other freely-correlating factor (Figure 6)—and is, uniquely, an earned "island" in the map. (We note one boundary: the correlated-G arm cleanly separates biology from cognition and sleep but not from developmental-risk, so the immunometabolic–developmental distinction rests on the symptom-decoupling in Φ, not on ϕ_{Gd}; Supplement A.) Two patients with the same overall impairment can therefore carry very different immunometabolic loads—a patient's biological burden rides largely on an axis that the global clinical-severity picture does not capture. This dissociation was robust to the obvious confounders. Adjusting each indicator for age, sex, education and recruitment site, and then additionally for antipsychotic exposure, did *not* raise the immunometabolic correlation—if anything it lowered it (to ≈ 0.07), and the immunometabolic axis remained the least entangled at every step (Figure 7; Supplement A). Consistent with a large independent literature (Discussion), we state this as relative, not absolute, independence: immunometabolic load is *largely independent of a general functional-burden axis*, the low correlation quantifying—at the low end of what that literature would predict—a pattern previously described only qualitatively. *Clinically,* this means two patients with comparable functional burden can have sharply different cardiometabolic and inflammatory profiles: severity cannot be used as a proxy for biological load. This is the property a biology-aware stratification can exploit, and it is one the downstream stages carry forward and re-test rather than assume.

3.7 The coordinate cloud is continuous, not discretely clustered

We next asked whether these eight coordinates fall into separable kinds. They do not—a *negative structural result.* On the cloud $\{(\hat{f}_i, S_i)\}$, a structure-discovery battery run over the posterior draws—cluster tendency, modality, silhouette, the gap statistic, a Gaussian-mixture information criterion, density-based clustering and a topological summary—returned a continuum verdict: the silhouette peaked at only 0.146 (below the 0.15 separation floor), the gap-statistic optimum was a single component, density-based clustering assigned 100% of points to noise, and the topological summary was a single connected component. A single-Gaussian *falsification null* made the verdict decisive: the best partition of the real cloud separated patients no better than slicing a structureless Gaussian blob of identical covariance (best silhouette 0.140 for the real data versus 0.137 ± 0.002 for the null, $z = 1.13$, n.s.; the Gaussian-mixture-optimal number of components collapsed to one in all 20 posterior draws; Figure 8a). There are, in this clinical–biological space, no well-separated, reproducible discrete clusters. This is a statement about discrete clustering in this representation (Gaussian-copula ESEM coordinates at $N = 9{,}013$ under this pipeline), not a claim that biological subtypes are absent in other feature spaces or larger samples [8, 40, 41], nor a denial of the clinical heterogeneity we do find.

The continuum is genuine and transdiagnostic, not an artefact. Diagnosis is intermixed throughout the cloud (the soft tessellation's adjusted Rand index is 0.006 against the seven DSM-5 subtypes and 0.011 against cohort, both ≈ 0; the $A = 5$ archetype assignment is likewise near-orthogonal, ARI 0.021, §3.9; Figure 8b); overall burden forms a smooth gradient along the principal axis (Figure 8c); and immunometabolic load varies along a *different* direction (Figure 8d)—the biology–G dissociation, made visible. The verdict is not a missingness artefact (a classifier predicting position from each patient's coverage pattern performed below baseline, accuracy 0.611 versus a 0.683 baseline, lift -0.072) and is a tighter description of the coordinate cloud than diagnosis (free-mixture information criterion 185,600 versus 188,200 for a DSM-5-constrained model, $\approx 3\times$ the explained variance at lower BIC). These same properties explain why clustering the *raw* indicators can appear to yield biotypes where the latent space does not [42]: a raw partition is driven largely by measurement-side structure—structured missingness, a few heavy-tailed laboratory variables, redundant indicator blocks, and site-specific instruments that re-encode diagnosis—which the measurement model neutralizes by construction (observed-cell likelihood, per-indicator and Gaussian-copula likelihoods, shared factors, and diagnosis held out under invariance), so the structure that remains is genuinely transdiagnostic and tested against a structureless null rather than assumed. *Clinically,* the appropriate object is therefore a coordinate with uncertainty (Eq. 22), not a categorical biotype.

3.8 The measurement map is stable to holding out recruitment sites

The map is estimated once on the full sample; a reviewer may reasonably ask whether its structure is an artifact of the particular sites that contributed patients. Because the cohort spans 21 expert centres, we can test this directly: for each of the 15 sites contributing at least 100 patients, we refit the entire measurement model with that site's patients removed (held-out N ranging from 134 to 1,308) and compared the leave-site-out map to the full-data map by Tucker's congruence coefficient per factor. Refitting used the variational path, which reproduces the reference posterior loadings at Tucker 0.957–0.999 (Annex G), making 15 refits computationally tractable.

The immunometabolic axis—the load-bearing structure of this paper—was essentially invariant to which site was removed: loading congruence with the full-data map was $\varphi \geq 0.999$ in every fold (minimum 0.9993, median 0.9998). Its dissociation from general severity, measured at the score level as $|\text{corr}(\text{immunometabolic, general burden})|$, remained between 0.073 and 0.082 across all folds (full-sample score-level value ≈ 0.08), and the immunometabolic axis was the least severity-correlated of the seven specific factors in all 15 of 15 refits. The two single-diagnosis holdout sites (one bipolar-only, one schizophrenia-only) behaved no differently from the multi-diagnosis sites. Across all eight dimensions, every factor cleared its pre-registered congruence bar (Tucker ≥ 0.95 for the four major axes, ≥ 0.85 for the four thin specific factors) in all 15 folds; the single lowest per-factor value across the entire analysis was $\varphi = 0.917$, for a two-to-four-item thin factor at a small held-out site ($n = 237$, Monaco), still comfortably above its bar (Extended Data Fig. E6).

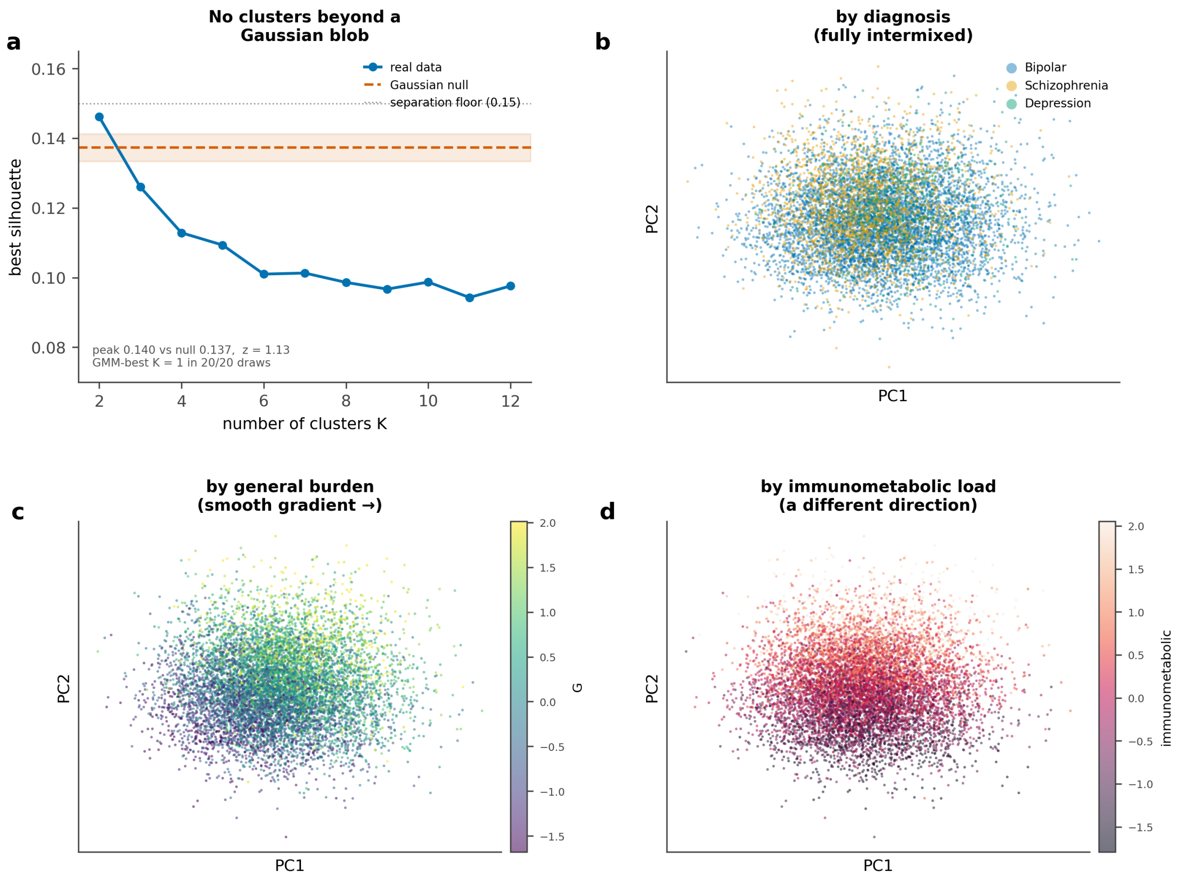

Figure 8: **The patient space is a continuum, not biotypes.** (**a**) A single-Gaussian falsification null: the best silhouette achievable on the real eight-dimensional cloud is no better than on a structureless Gaussian of matched covariance ($z = 1.13$, n.s.); the mixture-optimal number of clusters is one in all posterior draws. (**b–d**) Principal-component embedding of the coordinates (N = 9,013), coloured by diagnosis (fully intermixed), by general burden G (a smooth gradient), and by immunometabolic load (a gradient in a *different* direction).

3.9 Five archetypal extremes define the shape of the continuum

A continuum has no clusters, but it still has a *shape*. Archetypal analysis recovers that shape as a small set of reproducible extreme corners—a *positive geometric result* that coexists with the negative clustering result above. Each patient is written as a convex blend of $A = 5$ extreme phenotypes,

$$\hat{f}_i \;\approx\; \sum_{a=0}^{4} w_{ia}\, z_a, \qquad w_{ia} \geq 0, \qquad \sum_{a=0}^{4} w_{ia} = 1. \tag{25}$$

The five z_a are extreme points on the convex hull of the patient cloud, and the weights w_{ia} are *barycentric coordinates*: they quantify how strongly each patient is pulled toward each extreme. Dominant-archetype labels are used only for summaries; the inferential object is the continuous weight vector w_i, not a class assignment. The number of corners ($A = 5$) was set by cross-seed stability rather than by a parsimony rule—five corners were the largest reproducible solution (cross-seed Tucker congruence 0.979, explained variance 0.60), with a clean stability cliff at six corners (0.979 → 0.436)—and uncertainty was propagated by refitting Eq. (25) across posterior draws (Supplement B).

The five extremes separate biology from symptoms from severity (Figure 9; Table 3): an *activation / sleep* corner (A0; high sleep and mania), a *severe, clean-biology* corner (A1;

Table 4: **Archetype composition by cohort (diagnosis-blind).** Share of each cohort's patients whose dominant (arg max) archetype is each corner. Every corner is populated in all three cohorts and each diagnosis is a mixture of corners; the archetype assignment is essentially orthogonal to DSM-5 (adjusted Rand index 0.021).

Dominant corner	BP	DR	SZ
A4 low-burden / well	0.30	0.11	0.29
A3 trauma / suicidality	0.24	0.22	0.18
A0 activation / sleep	0.19	0.06	0.10
A1 severe, clean-biology	0.14	0.31	0.25
A2 immunometabolic	0.13	0.30	0.19

high overall severity but low immunometabolic load), an *immunometabolic* corner (A2; high immunometabolic load with severity and suicidality)—the biology pole—a *trauma / suicidality* corner (A3; high developmental risk and suicidality), and a *low-burden* pole (A4; everything low). The immunometabolic corner A2 and the severe corner A1 carry comparable general burden but opposite biological profiles—A1 versus A2 is precisely why *severity is not biology*, the central dissociation now rendered as geometry. The symptom space itself resolves into two distinct corners—activation/sleep (A0) and trauma/suicidality (A3)—rather than one combined symptom pole. Patients sit *between* the corners far more than at them: the weight vectors are diffuse (median normalized entropy 0.65), so most patients are convex blends rather than single-corner cases. Critically, because immunometabolic load varies continuously without a density *gap*, it never forms a clean partition boundary at any granularity; it surfaces instead as a *direction of maximal phenotype*—an archetypal corner—which is exactly why the continuous coordinates and the weight vectors w_i, not any hard tessellation, are the primary objects downstream. *Clinically,* the archetypes provide a vocabulary for reading the continuum, not a replacement for it.

Read against the held-out diagnostic labels, the archetype assignment is essentially orthogonal to DSM-5 (adjusted Rand index 0.021) and every corner is populated in all three cohorts—the stratification analogue of diagnosis-as-metadata. Each diagnosis is a *mixture* of corners and each corner draws from every diagnosis (Table 4): the activation/sleep corner is bipolar-enriched (mania is a bipolar feature) and the biology corners (A1/A2) are depression/schizophrenia-enriched, yet none is diagnosis-specific. The simplex nonetheless describes *where a patient sits* on the eight-dimensional map far more tightly than the diagnostic label: averaged over the eight axes, the archetype assignment accounts for $\eta^2 = 0.256$ of the coordinate variance against only 0.026 for the DSM-5 grouping—an $\approx$9.7-fold improvement, at lower BIC. A categorical diagnosis is thus a weak organiser of a patient's clinical–biological state and the archetype mixture an informative one, transdiagnostically; whether the mixture also *predicts* better, beyond being a tighter *description*, is the prognostic question taken up below.

Stated precisely, the archetypes separate on severity *and* profile. Severity (G) is in fact the single strongest separating axis ($\eta^2 = 0.47$)—the simplex spans a clear gradient from the well pole (A4) to the severe poles (A1/A2)—but the corners separate almost as strongly on immunometabolic load (0.39), sleep (0.38) and developmental risk (0.35). This is the value over a one-number severity score: at any given severity the archetype distinguishes patients by biological and symptom *profile*, so A1 and A2—equally severe, biologically opposite—are different corners a severity score cannot tell apart. Choosing soft archetypal corners over hard clustering is itself the methodological point: on a cloud with no density gaps, convex mixtures of extreme phenotypes neither manufacture boundaries the data lack nor blur patients into centroid "averages," and they preserve a patient's position relative to *every* pole rather than collapsing it to a bin.

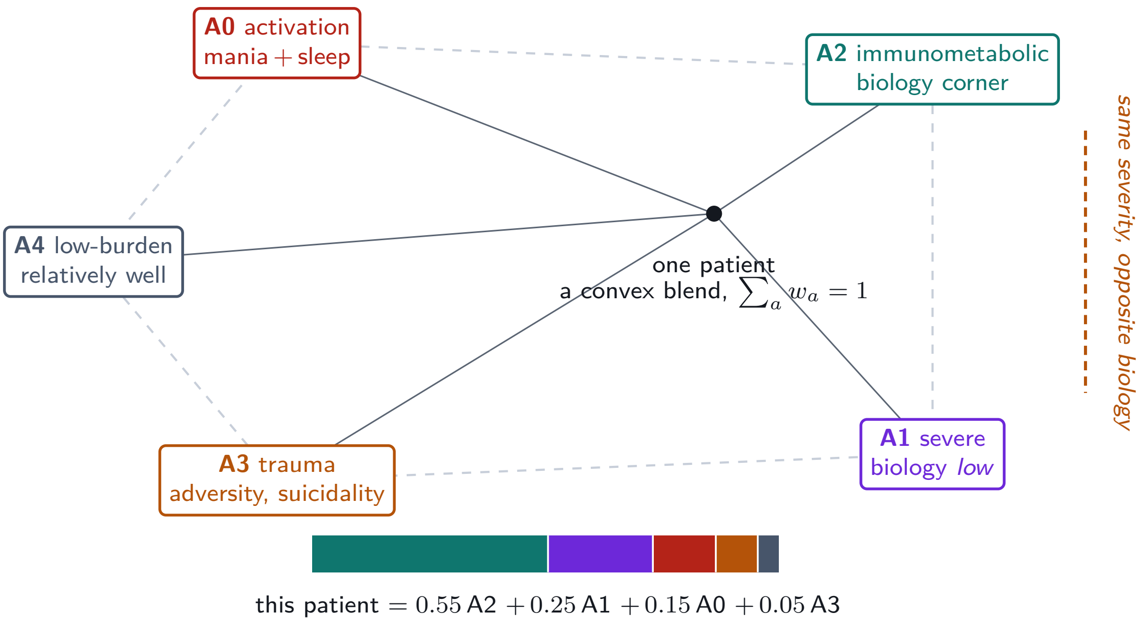

Figure 9: **The archetype simplex: every patient a convex blend of five extremes.** The five archetypal extremes are the corners—**A0** activation / sleep, **A1** severe but clean-biology, **A2** immunometabolic (the biology pole), **A3** trauma / suicidality, and **A4** low-burden—and every patient sits *inside* the simplex as a convex blend whose barycentric weights w_i sum to one (worked example, foot). The central dissociation is geometric: **A1** and **A2** are distinct corners at essentially the *same* general burden but opposite biology—the biology–severity dissociation, rendered as a direction one can point to.

The archetype simplex is a fog, not five clusters. The continuum verdict is equally visible in archetype space. Writing each patient as their membership vector $w_i \in \Delta^4$ over the five corners and measuring how *blended* they are by the entropy $H(w_i) = -\sum_a w_{ia} \log w_{ia}$, the 9,013 patients fill the interior of the simplex rather than piling at its vertices (Fig. 10): the median entropy is 1.24 of a possible $\log 5 = 1.61$ nats, 74% of patients have no majority archetype (largest weight < 0.5), and only 0.4% sit near a single corner (largest weight > 0.8). The archetypes are extremes that *span* a continuous cloud, not labels that *partition* it—the same conclusion the structure tests reach, now read directly off the membership weights.

3.10 The geometry persists over two years (the map persists)

Scored forward onto the 12- and 24-month visits on the *fixed* model (observed cells, uncertainty propagated, never re-estimated; baseline coordinates reproduced at Pearson $r \approx 0.99$ through this path), the measurement remained invariant for all four testable backbone axes (longitudinal Tucker congruence $\varphi = 0.987$–0.996; the immunometabolic axis invariant at $\varphi = 0.987$; Extended Data Fig. E4). Coordinate change can therefore be read as patient change rather than instrument drift. A measurement-error decomposition then separated trait from state (Figure 11). The durable, trait-like axes were the biological and cognitive ones, with immunometabolic load the single most durable axis (test–retest intraclass correlation: immunometabolic 0.91, cognition 0.70); the moving, state-like axes were the symptom ones (substance 0.49, sleep 0.47, suicidality 0.43, developmental risk 0.39). General burden was trait *by rank* (intraclass correlation 0.62): patients largely kept their relative position even as the cohort as a whole improved on severity and symptoms (population shift -0.46 on G and -0.84 on suicidality, while immunometabolic load stayed static, $+0.04$). The archetype geometry persisted—each patient's weight vector w_i

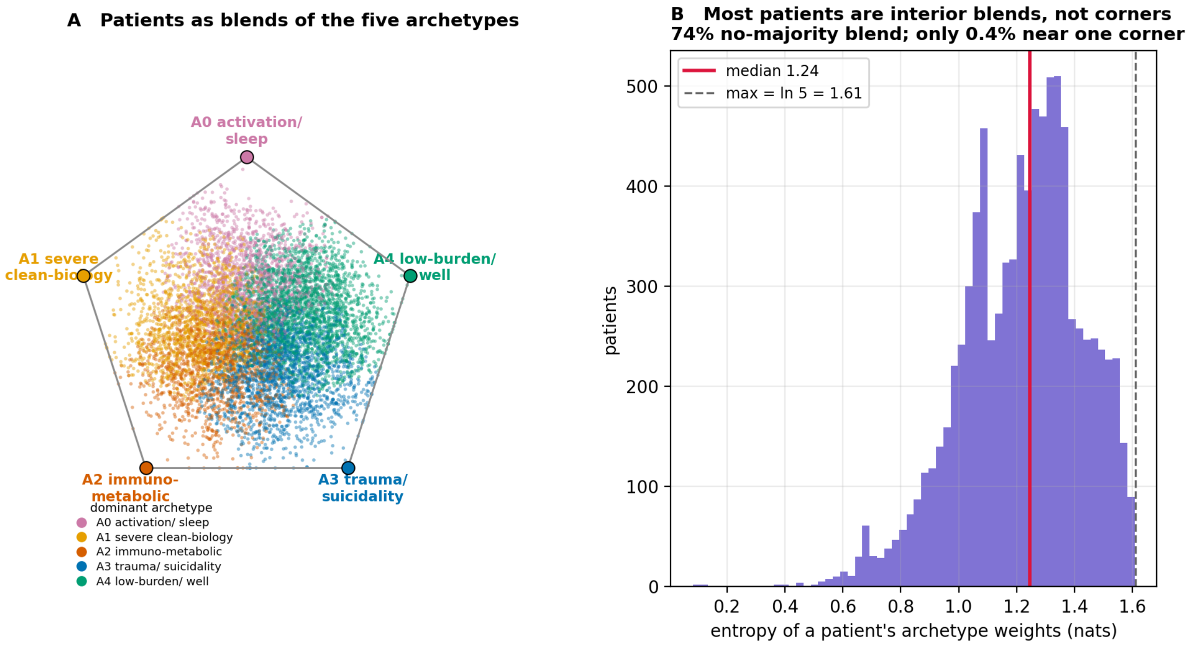

Figure 10: **The archetype simplex is a fog, not five clusters. (A)** Each of the 9,013 patients placed at the barycentre of the five archetype corners by its membership weights w_i, coloured by its *dominant* (largest-weight) archetype so the directional lean of the cloud toward each corner is visible. The colours form five contiguous territories that fan out toward their corners, yet the cloud fills the interior rather than the vertices: patients tilt toward an archetype without reaching it. **(B)** Distribution of the mixing entropy $H(w_i)$: median 1.24 of a maximum $\log 5 = 1.61$ nats, with 74% of patients having no majority archetype and only 0.4% near a single corner—the archetypes span the continuum, they do not partition it.

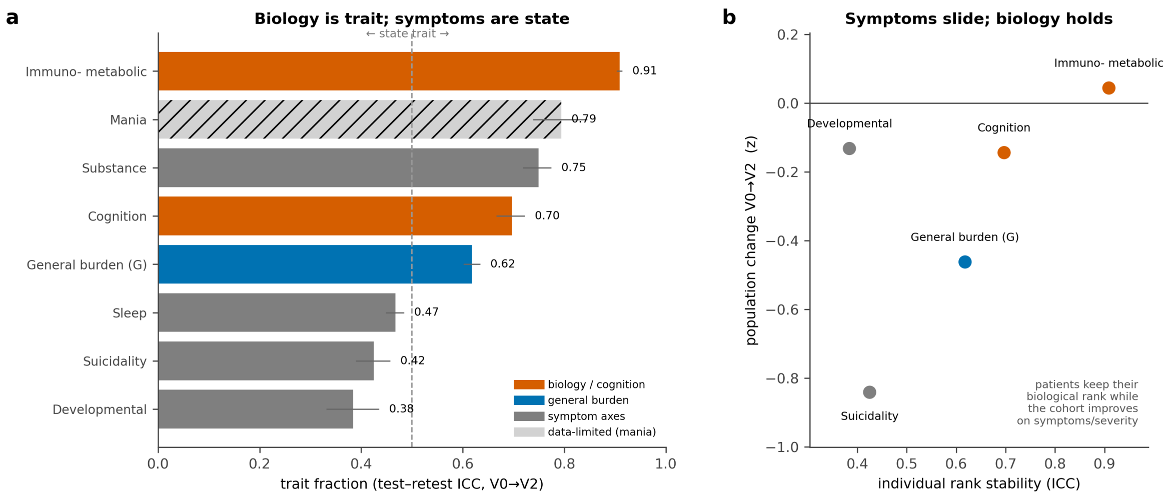

Figure 11: **Durable biology, moving symptoms.** (**a**) Trait fraction (test–retest intraclass correlation, with 94% credible intervals) over follow-up: the biological and cognitive axes are trait-like, the symptom axes state-like; mania is data-limited (hatched). (**b**) Individual rank stability versus population-level change: patients keep their biological rank (high intraclass correlation, near-zero population shift) while the cohort slides down the severity and symptom axes.

moved little over two years (median cosine similarity 0.81). *Clinically,* the reading is compact: *stratify on the durable biology; monitor the moving symptoms.*

A trait/state spectrum across the axes. Shown in full as a variance *thermometer* (Fig. 12) partitioning each axis into trait, state and measurement error, and ordered by error-corrected trait fraction the eight axes form a single durability spectrum from immunometabolic biology (ICC 0.91; 80% of its total variance is the stable between-patient trait) down to developmental risk (0.39), with cognition (0.70) and general burden (0.62) trait-dominant and the symptom axes mixed. Separating state from measurement error also shows that the symptom axes (sleep, suicidality, developmental risk) carry genuine within-patient change rather than mere noise, whereas mania's variance is dominated by measurement error—its two indicators leave it data-limited and its trait verdict is not relied upon. The biology that anchors the map is again its most durable dimension, exactly the property a stratification axis should have.

3.11 Archetype position predicts group-level two-year functioning (the map predicts)

The clinically actionable prognostic signal sits at the level of *position on the map*, not individual point prediction, and it is visible before any model-comparison statistic. Ranking patients by their dominant archetypal extreme, two-year functional remission forms a gradient that reproduces *within* every diagnosis: the immunometabolic corner (A2) is the worst- or near-worst-remitting archetype in bipolar disorder, schizophrenia and depression alike (bipolar 27% vs. 73% at the low-burden pole A4; schizophrenia 9% vs. 25%; depression 31% vs. 72%; Figure 13a)—so the gradient is a property of the map, not a cohort-composition artefact. Pooled across cohorts the same ranking spans 22%–63% (Figure 1), a secondary, marginal framing reported only alongside the within-cohort result. The archetype coordinate is also a far more compact descriptor of where a patient sits on the map than diagnosis: it accounts for a mean η^2 of 0.256 across the eight axes, against 0.026 for DSM-5 category and 0.018 for cohort—an ≈ 9.7-fold tighter summary (Figure 13b). This is compactness on the map's own axes, not a claim about outcome variance

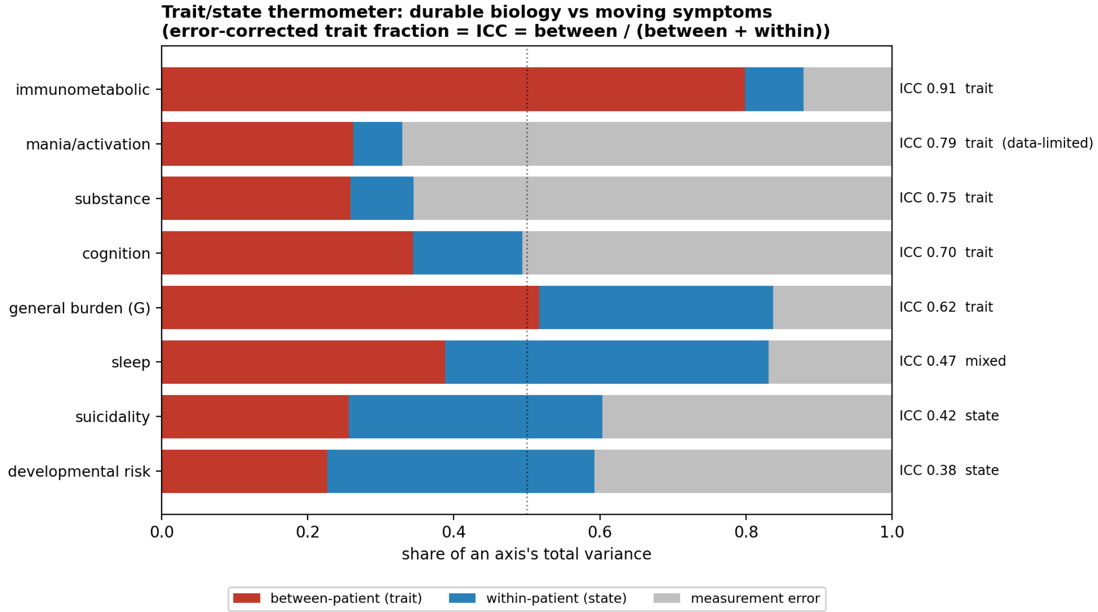

Figure 12: **Trait/state thermometer.** Each axis's variance across the three visits, partitioned into a stable between-patient (trait) part, a moving within-patient (state) part and measurement error, normalised to the axis total and ordered by the error-corrected trait fraction (ICC = between/(between+within)). The immunometabolic biology axis is the most durable (ICC 0.91), developmental risk the most state-like (0.39); mania is mostly measurement error (two indicators, data-limited). This variance view complements Fig. 11, separating genuine within-patient state from measurement noise.

explained; the outcome evidence is the remission gradient above, kept as a separate statement.

We then asked whether a baseline coordinate or phenotype predicts this two-year functioning outcome *incrementally beyond* diagnosis, current severity and the baseline value of the outcome, using an errors-in-variables Bayesian model that propagated each coordinate's measurement uncertainty. The map added predictive signal for *functioning* but not for severity (which was saturated by its own baseline autoregression). The signal was carried by the *phenotype*, not by any single laboratory axis: the five archetypal extremes added the most held-out predictive value (expected log predictive density gain +62.8 over the diagnosis-plus-severity-plus-baseline reference; Table 5), the seven specific axes added +38.1, while the isolated "durable biology" axes *alone* sat at the bar (+2.3, ambiguous). Biology still predicts functioning, but as a *corner of the phenotype* (the immunometabolic archetype) rather than as a standalone block—the immunometabolic axis itself carries a credible adverse direction (higher load → worse future functioning). No hard partition improved on the continuous and archetype encodings (every tessellation added ≈ +16 to +20, strictly dominated by the archetypes); the operative number of clusters is therefore *none*—the actionable object is a continuous position, not a category. The map was *co-informative* with DSM-5 rather than redundant with it (functioning: map +17.3, DSM-5 +29.0, both +62.6; Figure 13c), and *course-dependent*, with the signal concentrated in the episodic, open-course cohorts (bipolar disorder and depression) and weak when bipolar disorder was removed. At the *individual* level, the same map adds only +0.010 to remission-prediction AUC over a diagnosis-plus-severity reference—the map's prognostic value is one of stratification and durable measurement (test–retest ICC 0.91), not a replacement for individual clinical judgment.

This within-cohort gradient (Figure 13a) is a genuine within-diagnosis effect, not a cohort-composition artefact: the rank held inside every cohort (cohort-adjusted best-versus-worst odds ratio ≈ 6.3), standardizing to a common cohort mix moved it by only ≈ 4%, and the

Table 5: **Incremental predictive value for the two-year outcome (ΔELPD).** Held-out expected log predictive density gain of each baseline encoding when added to the reference model (diagnosis + baseline severity + baseline outcome autoregression), for functioning (EGF) and clinician severity (CGI-S). The archetype encoding is the strongest predictor of *functioning*; the same encoding gains little on *severity* (saturated by its own baseline); the continuous and archetype encodings dominate every hard tessellation (operative number of clusters = none).

Encoding added to the reference	ΔELPD functioning	ΔELPD severity
+ A=5 archetypes (all axes)	+**62.8**	+11.9
+ continuous coordinates (8 axes)	+38.1	+13.3
+ A=5 archetypes ($\perp$ G)	+33.5	+14.0
+ tessellation $K = 3\,/\,4\,/\,2$	+19.6 / 16.6 / 15.9	+5.6 / 4.4 / 4.9
+ durable (immunometabolic) axis	+2.3 (amb.)	−0.7 (amb.)
reference (autoregression)	0	0

corner-by-cohort interaction was non-significant. A2 also remained worse than the equally severe clean-biology corner A1 within diagnosis (22% vs 34% pooled)—the severity-is-not-biology dissociation, now translated into outcome. A wide spread in *group* remission rates alongside the small *individual* AUC gain noted above is the expected signature of groups that differ in average risk but overlap substantially at the patient level, layered on an already-strong autoregressive baseline—the map enriches and forecasts at the group level without sharply separating individuals. The functioning signal survived inverse-probability weighting for attrition (+54.4), restriction to either two of the three cohorts (+56.4, +61.8), and a permutation null (which correctly vanished). Finally, against the full set of raw indicators under a matched gradient-boosting classifier, the eight-dimension map was a *sufficient* representation for predicting deterioration (equal AUC) and *near-sufficient* for recovery (raw indicators +0.04 AUC), with 92–97% of the raw recovery signal already contained within the eight factors (Extended Data Fig. E5)—the compression is faithful, not lossy in any axis the map omits. *Clinically,* the atlas is useful for group-level enrichment and monitoring, not as an individual remission calculator.

3.12 A variational re-estimation reproduces the atlas at a fraction of the cost

Because the audit-grade NUTS fit is expensive, we asked whether the same measurement model can be recovered by a much cheaper estimator. Re-estimating the eight-factor map as a generalized linear latent-variable model fit by stochastic variational inference (Methods; Supplement G) reproduces the NUTS atlas on its two load-bearing objects at roughly 100× less cost (≈4.5 minutes on a CPU versus hours). The loading map is the same map: loading columns are congruent on all eight factors (Tucker φ 0.96–0.999; G 0.957) and the full item×factor loading scatter is $r = 0.993$ with no sign flips (Figure 14a). Patients land in the same place: per-patient coordinates correlate with the NUTS coordinates at $r \geq 0.90$ on all eight axes (≥ 0.99 for the G and immunometabolic axes; Figure 14b). The map is also a faithful generative model—9,013 synthetic patients drawn from it match the observed marginals (Kolmogorov–Smirnov median 0.04) and covariance (correlation SRMR 0.078, as for NUTS). The one quantity the variational engine does *not* reproduce is the inter-factor correlation Φ: its structure is correct (the biology $\perp G$ zeros are exact) but the off-diagonals are systematically attenuated by ∼21% under the mean-field posterior—a structural property of variational inference, not a tunable defect, which is why NUTS is retained for correlation magnitudes and all credible intervals (Supplement G). The variational engine is therefore a fast operational and generative stand-in for the map and coordinates; a fuller development is left to future work.

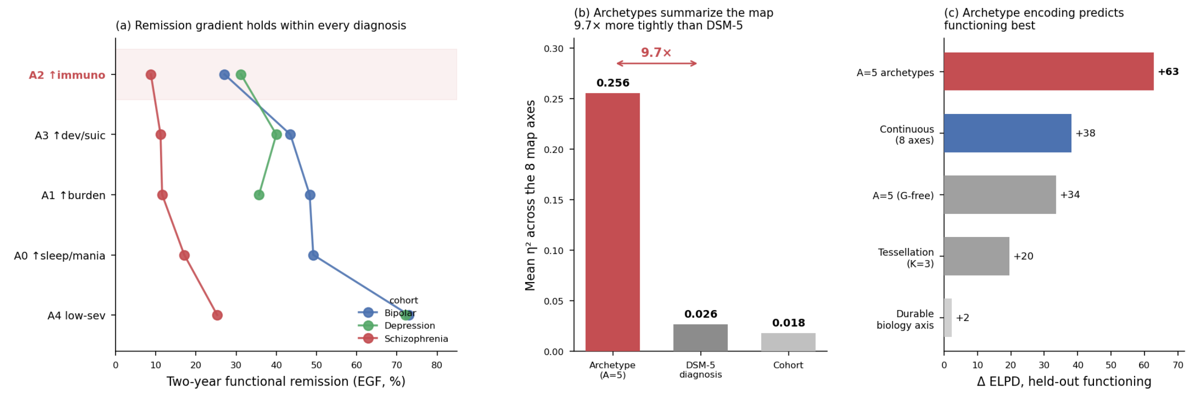

Figure 13: **The archetype coordinate carries outcome structure diagnosis misses.** (**a**) The within-cohort remission gradient: two-year functional-remission rate by baseline dominant archetype, shown separately within each cohort (cells with $n \geq 50$ patients). The immunometabolic pole (A2) is the lowest-remitting archetype in every diagnosis (bipolar 27% vs. 73% at the low-burden pole A4; schizophrenia 9% vs. 25%; depression 31% vs. 72%)—a within-diagnosis gradient, not a cohort-composition artefact. (**b**) The archetype assignment summarises the eight map axes 9.7× more compactly than DSM-5 (mean η^2 across the eight axes: archetype 0.256, DSM-5 diagnosis 0.026, cohort 0.018); this is compactness on the map's own axes, not outcome variance explained. (**c**) Held-out incremental predictive value (ΔELPD) for two-year functioning: the archetype encoding is the strongest single predictor, ahead of the continuous coordinates and every hard tessellation. The individual-level remission AUC gain over diagnosis is small (+0.010); the map's prognostic value is in stratification and durable measurement, not individual point prediction.

4 Discussion

Across three cohorts of severe mental illness, a single Bayesian measurement pipeline places clinical and biological measurements in one transdiagnostic coordinate system, holds that eight-dimension map fixed, and validates it for structure, temporal coherence and prognosis. The contribution is not a new estimator but the discipline of its integration, novel in combination on three counts. The measurement layer is *missingness-native*: one global observed-cell likelihood carries the modelled pool's ≈ 40% mean cell-missingness and two-cohort indicators, fit jointly rather than imputed or deleted. *Uncertainty is propagated end-to-end*: each patient's posterior coordinates and their covariance flow into structure-testing, temporal decomposition and errors-in-variables prognosis rather than collapsing to point scores at the measurement boundary. And the map is *fixed with diagnosis held out, then validated*—frozen before any validation stage touches it, which is what licenses the downstream results as properties of the measurement rather than artifacts fitted to an outcome. The immunometabolic dissociation that runs through the paper is the existence proof for that discipline: the same rigor that makes the map honest reveals structure a looser measurement would blur—one recurring empirical property in which immunometabolic load is the domain least entangled with overall functional severity. The cardiometabolic and inflammatory markers resolve as a single immunometabolic axis anchored by body composition (BMI loading ≈ 0.95, weight 0.88, waist circumference 0.84), with smaller metabolic, inflammatory and cardiovascular contributions (C-reactive protein ≈ 0.37); that axis sits nearly independent of G. Because the one inflammatory marker available here, CRP, is itself substantially adiposity-driven [43, 44], we read this as an adiposity-anchored cardiometabolic axis and do not interpret it as an inflammatory dimension independent of body composition—the IL-6-type signal implicated in mood pathology [45, 46] was not measured here. The axis is nonetheless more than body habitus: excluding the anthropometric indicators and refitting, a

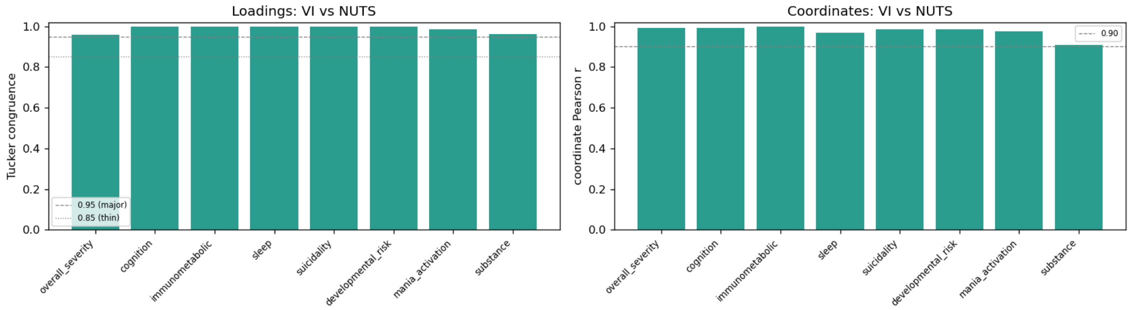

Figure 14: **A variational re-estimation reproduces the NUTS atlas at ∼100× less cost.** (**a**) Per-factor loading congruence (Tucker φ) between the variational GLLVM and the NUTS copula fit; all eight axes clear the major (0.95) and thin-factor (0.85) bars. (**b**) Per-patient coordinate agreement (Pearson r); all eight axes exceed 0.90. The full loading scatter ($r = 0.993$), the Φ comparison, and the model and inference details are in Supplement G.

coherent cardiometabolic–inflammatory axis survived, highly congruent with the full axis (Tucker $\varphi = 0.90$; Supplement A.7). If biological burden simply tracked how impaired a patient is, it would add little to a clinician's global impression. The data say the opposite—biological risk rides on an axis the severity picture does not see—and the five archetypal extremes make the consequence concrete. The immunometabolic corner and the severe, clean-biology corner carry comparable functional burden but opposite biological profiles, and they diverge in prognosis (the immunometabolic corner is the worst-prognosis pole and the well pole the best, two-year functional remission spanning 22% to 63% across the five corners). This is the property a biology-aware stratification can exploit: a durable, prognostically meaningful separation between patients who look equally ill but are biologically opposite.

This headline is corroborated, in direction, by a large and independent literature, which is why we state it as relative rather than absolute independence. Metabolic burden in severe mental illness is driven by medication, adiposity, age and chronicity more than by diagnosis or symptom severity, is present before chronic severity accrues, and is comparable across schizophrenia, bipolar disorder and depression [14–16, 47, 48]; inflammation marks a subgroup and specific symptoms rather than the severity continuum [49–51]; and the immuno-metabolic-depression programme has independently arrived at a separable biological dimension mapping onto atypical, energy-related symptoms [11–13, 52, 53]. We also state the boundary conditions plainly: case–control inflammatory elevations are real [54–56], a modest severity or duration dose–response for metabolic burden exists [57, 58], metabolic dysfunction does track cognition and functioning in mood disorders [59], and much of the inflammation–symptom association is carried by adiposity rather than inflammation *per se* [43, 60]. Our small immunometabolic–G couplings are best read as a quantification of a known qualitative pattern, sitting at its low end because G indexes *functional burden*—from which biology is most decoupled—rather than the case–control contrast of being ill. That they survive, and even fall slightly under, adjustment for medication, adiposity and site is the strongest available evidence that the dissociation is not a confound.

What is new is less the direction of this finding—which the literature above corroborates—than where biology sits in the model. The dimensional-psychopathology tradition builds its coordinate systems from symptom report alone, with no biological measurement inside the space [2], and normative modelling charts per-measure deviation from a reference rather than a shared low-dimensional map [8, 9]. The closest biological precedent identifies a separable immuno-metabolic profile *within* major depression, anchored on symptom dimensions with biology entered as a correlated *outcome* [13, 52], while the B-SNIP biotype programme resolves discrete brain-based groups rather than a continuum [17]. Placing routine biology as a co-equal latent axis *inside* a single diagnosis-blind map across bipolar disorder, schizophrenia and depression—and testing its decoupling, durability and prognosis in that same model—is the slot none of these occupy.

A second recurring result is structural. On these clinical–biological coordinates the patient space is a graded continuum, not a set of discrete biotypes, and it cuts almost entirely across DSM-5. This aligns with normative-modelling work reporting extensive individual heterogeneity in the same disorders [8–10], and it stands in apparent tension with the brain-based B-SNIP biotypes [17, 18]. The tension is more apparent than real: the two programmes interrogate different measurement spaces, B-SNIP itself found DSM diagnoses to be a severity continuum, and recent syntheses endorse both categorical and dimensional views [19]; clustering choices can also resurface apparent biotypes from continuous data [42]. We therefore claim no biotypes *in this eight-dimensional clinical–biological space*—a claim we make rigorous with a single-Gaussian falsification null (best-partition silhouette 0.140 against a structureless-Gaussian null of 0.137 ± 0.002, $z = 1.13$) rather than by the absence of an elbow—not that biotypes are absent from all spaces. The five archetypes are corners of this continuum, not clusters within it; they cut almost orthogonally across DSM-5 (adjusted Rand index 0.021) yet describe a patient's position on the eight-dimensional map about 9.7× more tightly than diagnosis does (mean η^2 0.256 versus 0.026). The corners span a severity gradient *and* a biological one, so the simplex separates on profile as well as severity. Anchoring G on functioning remains consistent with the impairment reading of the p factor [4, 5].

The remaining contributions are deliberately calibrated. The map persists over two years—durable biology, moving symptoms—which both validates the measurement longitudinally and motivates a clinical division of labour between traits to stratify on and states to monitor. The immunometabolic axis is the single most durable dimension (test–retest ICC 0.91, fully invariant across visits; it remains trait-like, ICC 0.76, even with the anthropometric indicators removed, so the durability is not merely body mass's stability—Supplement A.7), cognition is also trait-like, while severity and symptoms slide as the population improves: the durable part of the map is precisely its biological part.

This division is itself a design principle—*stratify on what stays put; monitor what moves*. Because the immunometabolic axis is both the durable trait and the dimension least entangled with functional burden, it is the stable axis on which to fix a patient's position—for instance at the immunometabolic corner (**A2**); general burden is durable only *by rank* (ICC 0.62), with patients largely keeping their relative position even as the cohort improves in absolute terms. The symptom axes, by contrast, slide—population shifts of −0.46 on severity and −0.84 on suicidality from baseline to two years, against an essentially static +0.04 on biology—and are therefore quantities to track over time rather than to fix a stratum on. We frame this as an interpretive design principle, read off the baseline structure and its forward-scored longitudinal coherence rather than from any validated protocol: what the durability buys is a principled separation of the axes worth stratifying on from those worth monitoring, not an individual treatment rule.

The map predicts future functioning incrementally beyond diagnosis and severity (archetypes ΔELPD +62.8 held-out, robust to inverse-probability weighting and permutation), but the gain is modest and group-level—the individual-binary lift is small (AUC +0.010)—it is co-informative with rather than superior to DSM-5, and it is carried by the biological *phenotype* (the immunometabolic archetype corner) rather than by any isolated laboratory axis. "Biology predicts functioning" is therefore a phenotype-level claim: the immunometabolic axis carries a credible adverse direction, but the predictive object is the fuller archetype representation in which it is embedded. That the eight-dimension map is nonetheless a *sufficient* representation of the full raw V0 indicator block (143 items, the representation benchmark's raw arm) for predicting deterioration, and near-sufficient for recovery (the residual is within-factor item-level compression, not a missing axis), shows the parsimony is faithful rather than lossy. We report these modest and null findings as deliberately as the positive ones, and are explicit about the gap between demonstrated scientific validity and undemonstrated individual-level clinical utility.

In sum, this atlas reframes a clinically actionable hypothesis with measured coordinates: not "which biotype is this patient?" but "where does this patient sit on continuous, partly biological axes that diagnosis and severity do not capture, and how durable is that position?" The immunometabolic axis a patient keeps over time forecasts the functional state they move toward—a group-level, transdiagnostic signal that complements diagnosis and identifies the immunometabolic corner as a concrete enrichment and monitoring target for prospective study. Separating a stable, prognostic biological burden from clinical severity is what makes mechanistic stratification in severe mental illness a measurable rather than a rhetorical goal.

Limitations

Three classes of limitation bound these claims. *Internal validity only:* the map is discovered and validated within FACE, with no external-cohort replication, and the invariance partials that exist are documented rather than hidden (G re-anchors in schizophrenia, which lacks the FAST functioning anchor; self-rated mania floors in depression). Because FACE recruits at French expert centres, the cohort likely differs from community, primary-care, first-episode and non-French samples in chronicity, comorbidity and treatment intensity; the continuum geometry, the archetype simplex and the prognostic gradient may shift in younger, first-episode or less-treated populations—a boundary external replication must map. *Outcomes are scale trajectories, not recorded events:* prognostic endpoints are state transitions defined from repeated functioning and severity scales, not registered relapses or hospitalizations—a deliberate de-scope from event-level prognosis. *Measurement coverage and horizon:* two axes are thin (mania has two indicators and is reported as data-limited; substance use is a two-cohort axis), education is 40.8% missing overall (and 74.5% in schizophrenia)—a structured missingness the observed-likelihood design handles but that limits that covariate—developmental risk scores partly as "state" because childhood-adversity reports carry recall noise rather than true change; attrition is informative (addressed by inverse-probability weighting), and the follow-up window is two years across three visits. Establishing individual-level clinical utility would require incident-event outcomes and external validation beyond this baseline observational cohort.

Measurement precision is not uniform across axes. A distinct limit is the resolution the current instrument bank affords each axis. Six of the eight axes are estimated with high reliability (0.85 and above), but two—mania/activation and substance use—cannot exceed a reliability of about 0.45 regardless of sample size (maxima 0.408 and 0.429). This is a property of the instrument bank, not of missing data: 95–96% of patients have at least one item on these axes, but mania rests on only two summary scores and substance use on four items, two of which are rare binary flags that carry almost no information. Because the measurement model is missingness-native and fully Bayesian, the consequence is graceful degradation rather than bias—an under-instrumented axis returns a wide posterior rather than a false-precise point score, so a barely-measured axis is never mistaken for a precisely located one. Two archetype poles (**A0**, **A4**) are partly defined by the mania axis; these survive nonetheless because an archetype coordinate is a group mean, and averaging over more than a thousand patients per pole resolves the between-pole mania contrast to within ≈ 0.02 standard-deviation units even though any single patient's mania score is uncertain. The model thus reports its own per-axis precision and names mania and substance use as the priorities for the next data-collection wave—specifically graded severity instruments (for example AUDIT- or DAST-style quantity–frequency measures) rather than additional binary screens, which would add coverage without adding information. This sharpens the external-validity boundary above: what a replication cohort should collect is not merely more patients but better-instrumented mania and substance measures.

Data availability

The FACE database per-patient cohort data are confidential and governed by FondaMental's data-access procedures, to which qualified researchers may apply. No per-patient data are reproduced in this article. Aggregate parameters, derived structure and all figure/table source values are available in the project's regenerated aggregate outputs.

Code availability

The full analysis pipeline (data harmonization, the Bayesian measurement engine, stratification, temporal and prognosis modules), the harmonized variable dictionary, and configuration files are available at https://github.com/AndriiKulakovskyi/face-trans-diagnostic-phenotypes.git. Every reported quantity is reproducible from the numbered pipeline; per-patient data are never required to regenerate the aggregate results, tables and figures.

Ethics

The FACE cohorts were approved by the relevant institutional review boards and all participants provided written informed consent. [**TODO:** Insert exact IRB/ethics committee names, approval numbers, and trial/cohort registration identifiers.]

Competing interests

None.

Acknowledgements

We thank the FACE–FondaMental Expert Centre network, participating clinicians and, above all, the participants. [**TODO:** List the FondaMental Advanced Centres of Expertise (FACE) collaborators and any consortium authorship to be acknowledged.]

Funding

[**TODO:** Insert funding sources and grant numbers (e.g. Fondation FondaMental; Investissements d'Avenir program ANR-. . . ; and any additional grants), and state the role of the funders. If none, state "This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors."]

Author contributions

[**TODO:** Insert author-contribution statement, e.g. in CRediT terms: Conceptualization; Methodology; Software; Formal analysis; Data curation; Writing—original draft; Writing—review & editing; Visualization; Supervision; Funding acquisition. Name which author performed each role.]

Trial and cohort registration

[**TODO:** Insert the registration identifier(s) for the FACE–FondaMental cohorts (e.g. ClinicalTrials.gov / other registry numbers), or state "Not applicable" for an observational cohort not requiring prospective registration.]

Corresponding author

[**TODO:** Insert corresponding author name, affiliation and contact e-mail.]

Supplementary materials — derivations, proofs, diagnostics and reproducible tables

A The measurement model: full derivation

This annex develops the global measurement model from first principles: the generative factor model and its defining local-independence assumption (A.2); the mixed-likelihood observation model and the Gaussian-copula marginal (A.3); the observed-data likelihood and why missingness is handled by marginalization rather than imputation (A.4); the soft-prior loadings, the regularized-horseshoe sparse-ESEM that lets a few evidence-backed cross-loadings emerge, and the rotation problem they resolve (A.5); the closed-form covariance identity that carries the headline estimand (A.6); the Woodbury evaluation and its exact equivalence with full-information maximum likelihood (A.8); and the staged estimation and in-engine confirmation (A.9). Propositions are proved; everything reported in the main text is an evaluation of the objects defined here.

A.1 Variable accounting and indicator selection

The harmonized common dictionary contains 225 variables; the fitted map carries 109 as indicators. A variable is admitted as a *factor indicator* only if it satisfies four criteria: (i) it indexes a *current* clinical or biological state—not a history, onset or outcome record; (ii) it is not a treatment-, demographic- or outcome-driven *confound*; (iii) it is not *redundant* with an already-admitted indicator; and (iv) it is present in at least two cohorts. Above that two-cohort floor, admission is deliberately *coverage-agnostic*: because the observed-cell likelihood (Prop. 1, Eq. (A37)) lets an indicator contribute only for the patients in whom it is measured, a marker observed in two of the three cohorts is a valid transdiagnostic indicator—it simply contributes no term for the third cohort, with no imputation. Coverage therefore does *not* decide admission: of the 109 fitted indicators only about two-thirds (66) are present in all three cohorts and about one-third (43) in two—the latter including much of the biology panel (measured in bipolar disorder and depression) and the substance axis (recorded in bipolar disorder and schizophrenia). The remaining 116 variables are set aside for one of six reasons (Table A6); a single-cohort variable is the only purely coverage-based exclusion, because it cannot be placed on a transdiagnostic axis.

"Set aside" is not "unused." The demographic covariates residualize every indicator before fitting; the validation labels (diagnosis, cohort, visit) are held out of fitting and used only to test measurement invariance and transdiagnosticity; and the redundant *numeric* variables form the raw arm of the representation benchmark (Supplement D, the 143-variable raw pool), where they are usable as raw predictors though not admitted as factor indicators. Only the treatment/outcome confounds and the single-cohort variables are excluded from the analysis entirely—the former because admitting them would inject medication or outcome signal into the latent axes (prolactin, for example, is a direct D_2-blockade marker, and natremia tracks lithium and carbamazepine), the precise contamination the diagnosis-held-out, confound-adjusted design is built to avoid.

A.2 Generative model and local independence

Index patients $i = 1, \dots, N$, indicators $j = 1, \dots, J$, specific factors $k = 1, \dots, K$ and one general factor G. A few latent causes generate the many measured quantities: each patient carries a factor vector $f_i = (G_i, D_{i1}, \dots, D_{iK})^\top \in \mathbb{R}^{K+1}$ with

$$f_i \sim \mathcal{N}_{K+1}(\mathbf{0}, \Phi), \qquad \operatorname{diag}(\Phi) = \mathbf{1}, \tag{A26}$$

the factor *correlation* matrix Φ carrying a unit diagonal to fix the latent scale. Each indicator is a linear reflection of the factors plus item-specific noise, through a single linear predictor

$$\eta_{ij} \;=\; \alpha_j \;+\; \lambda_{jG}\, G_i \;+\; \sum_{k=1}^{K} \lambda_{jk}\, D_{ik} \;+\; \beta_j^{\top} c_i, \tag{A27}$$

with item intercept α_j, general loading λ_{jG}, specific loadings λ_{jk} (row j of the loading matrix $\Lambda = [\,\lambda_G \mid \Lambda_D\,] \in \mathbb{R}^{J\times(K+1)}$), and item-level covariates c_i that adjust an indicator's *mean* but are never factor indicators. Stacking one patient's items, with the general column separated from the specific block,

$$x_i = \mu_i + \underbrace{\lambda_G G_i}_{\text{general}} + \underbrace{\Lambda_D D_i}_{\text{specifics}} + \varepsilon_i, \qquad \mu_i = \alpha + B c_i, \qquad \varepsilon_i \sim \mathcal{N}(\mathbf{0}, \Psi), \tag{A28}$$

with diagonal residual covariance $\Psi = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_J^2)$.

Local independence (the defining assumption). Given a patient's latent scores, the items are conditionally independent,

$$p(x_i \mid f_i, \Theta) = \prod_{j\in O_i} p(x_{ij} \mid f_i, \Theta), \tag{A29}$$

i.e. the factors are the only reason items covary. In the Gaussian linear model this is *identical* to "Ψ is diagonal": local independence and diagonal residual covariance are two faces of one assumption—that the factors explain away all shared covariance. (It is also why filling missing cells is forbidden: an invented value injects covariance the model would mistake for a factor.) Because f_i is unobserved and the likelihood must depend on Θ alone, we integrate it out and multiply across independent patients,

$$L(\Theta) = \prod_{i=1}^{N} p(x_i \mid \Theta) = \prod_{i=1}^{N} \int \Big[\prod_{j\in O_i} p(x_{ij} \mid f_i, \Theta) \Big]\, \mathcal{N}(f_i; \mathbf{0}, \Phi)\, \mathrm{d}f_i. \tag{A30}$$

This is the complete first-principles likelihood: a product over patients (independent units), a product over items (local independence), and an integral that removes the latent (the sum rule). Equation (A27) is a bifactor exploratory structural-equation model [20, 21]—bifactor because every indicator may load on G *and* on its home specific factor; exploratory because an indicator is *allowed* small cross-loadings on plausibly related factors, governed by a sparsity-inducing prior (A.5) that keeps the map mostly simple-structure yet lets a few evidence-backed cross-loadings emerge, rather than forbidding them outright as in confirmatory factor analysis. The fitted map has $K{+}1 = 8$ factors: the general factor G (overall burden) and seven specific axes—cognition, immunometabolic, sleep, mania/activation, suicidality, developmental-risk and substance.

A.3 Mixed-likelihood observation model and the Gaussian copula

Coercing variables onto a common scale would manufacture covariance and bias the very factors being estimated. Each indicator instead keeps the observation density appropriate to its type, all sharing the linear predictor (A27):

$$\begin{aligned}
\text{continuous / } z\text{-score:} &\quad X_{ij} \sim \mathcal{N}(\eta_{ij},\ \sigma_j^2), && \text{(A31)}\\
\text{skewed laboratory:} &\quad \log X_{ij} \sim \mathcal{N}(\eta_{ij},\ \sigma_j^2), && \text{(A32)}\\
\text{ordinal } (C_j \text{ levels}): &\quad \Pr(X_{ij} \le c) = \mathrm{logit}^{-1}(\tau_{jc} - \eta_{ij}), \quad \tau_{j1} < \cdots < \tau_{j,C_j-1}, && \text{(A33)}\\
\text{binary:} &\quad X_{ij} \sim \mathrm{Bernoulli}\Big(\mathrm{logit}^{-1}(\eta_{ij})\Big), && \text{(A34)}\\
\text{count:} &\quad X_{ij} \sim \mathrm{NegBinom}(\mu_{ij} = e^{\eta_{ij}},\ \phi_j)\,. && \text{(A35)}
\end{aligned}$$

The Gaussian-copula marginal (reported map). The reported map replaces the raw Gaussian likelihood on the continuous block with a semiparametric *Gaussian-copula* model: each continuous indicator j is mapped to a latent Gaussian score by its frozen empirical distribution function,

$$z_{ij} = \Phi_{\mathcal{N}}^{-1}\big(\hat{F}_j(X_{ij})\big), \tag{A36}$$

and the factor model of A.2 is fit on z_{ij}. This makes the continuous block invariant to monotone re-scaling and robust to non-normal margins (heavy tails, skew, bounded scales) while leaving the latent dependence structure—the loadings and factor correlations that the scientific claims live in—unchanged. Because the rank transform (A36) is monotone, it alters none of the identification arguments below. Fitting the same structure under the plain Gaussian likelihood recovers the same loadings and correlations to within seed variability; that the central findings hold under both is reported in the main text as a robustness result. Writing $F_j(\cdot)$ for indicator j's density and $O_i \subseteq \{1, \dots, J\}$ for its *observed* indicators, the observed-data log-likelihood sums over observed cells only,

$$\log p(X_{\text{obs}} \mid \Theta) = \sum_{i=1}^{N} \sum_{j \in O_i} \log F_j(X_{ij} \mid \eta_{ij}, \theta_j)\,, \tag{A37}$$

so a missing cell contributes no term and no completed $N \times J$ matrix is ever formed; deterministic skip-logic decodes structural zeros as *observed* values.

A.4 Missing data is the marginal property, not imputation

Proposition 1 (Marginalization)**.** *For patient i with observed continuous set $C_i \subseteq O_i$, the model-implied distribution of the observed sub-vector is obtained by* deleting *the unobserved coordinates from (μ, Σ):*

$$X_{i,C_i} \sim \mathcal{N}(\mu_{i,C_i},\ \Sigma_{C_i})\,, \qquad \Sigma_{C_i} = \Lambda_{C_i}\Phi\Lambda_{C_i}^{\top} + \Psi_{C_i}, \tag{A38}$$

with Λ_{C_i} the observed-row loadings and $\Psi_{C_i} = \mathrm{diag}(\sigma_j^2)_{j\in C_i}$.

Proof. A foundational property of the multivariate normal is that marginals are read off by deleting coordinates: if $x \sim \mathcal{N}(\mu, \Sigma)$ then any sub-vector x_C is $\mathcal{N}(\mu_C, \Sigma_{CC})$. Apply this to $x_i \sim \mathcal{N}(\mu_i, \Sigma)$ with $\Sigma = \Lambda\Phi\Lambda^{\top} + \Psi$ (Proposition 3); the submatrix Σ_{CC} retains exactly the observed rows/columns of Λ and the observed diagonal of Ψ, giving (A38). □

Hence "integrating out" the missing cells is not an extra step but the marginal property itself; summing $\log \mathcal{N}(\cdot)$ over each patient's observed sub-vector is precisely the full-information maximum-likelihood (FIML) objective for incomplete multivariate-normal data. This is the formal justification of the no-imputation invariant: any completed matrix would inject between-cell covariance that (A29) would misattribute to a latent factor.

A.5 Soft-prior loadings, sparse-ESEM cross-loadings and the rotation problem

Theory enters through the prior on each loading. Each (indicator, factor) cell is assigned a role $\rho(j,k)$—one of PRIMARY, plausible CROSS, UNLIKELY, or FORBIDDEN—and the prior on that loading is

$$\lambda_{jk} \mid \rho(j,k) \ \sim \begin{cases} \mathcal{N}_{+}(0.70,\ 0.25^2) & \text{PRIMARY (HOME CELL)}, \\ \text{regularized horseshoe (A40)} & \text{PLAUSIBLE CROSS}, \\ \mathcal{N}(0,\ 0.05^2) & \text{UNLIKELY}, \\ \delta_0 & \text{FORBIDDEN}, \end{cases} \tag{A39}$$

where $\mathcal{N}_+$ truncates to $(0, \infty)$ and δ_0 fixes the loading at zero. Remaining priors are weakly informative: $\alpha_j \sim \mathcal{N}(0, 1.5^2)$; $\sigma_j = 0.05 + \text{HalfNormal}(1)$ (the floor guards against a Heywood case); ordered-logit cutpoints $\tau_{j\cdot}$ ordered; the negative-binomial concentration half-normal; and $\Phi \sim \text{LKJ}(\eta{=}2)$, whose density $\propto \det(\Phi)^{\eta-1}$ gently favours weakly correlated factors [61]. Latent scores use a non-centred parameterization to defuse the hierarchical funnel.

The cross-loading prior: a regularized horseshoe (default-off, evidence-on, magnitude-capped). A measurement map must choose how each indicator relates to factors other than its home factor. Both naive choices fail here. *Hard-zero* (rigid CFA) fixes every off-home specific↔specific cell at exactly 0: clean and identified, but it forbids an instrument from loading weakly on a second axis even when the data support it. *Free / soft-Normal* (naive ESEM) frees every off-home cell at $\mathcal{N}(0, 0.05^2)$: this floods every factor's column with weak cross-loadings, and a *thin* factor—one defined by very few indicators (substance, four; mania, two)—loses its identity as the flood of spurious cross-loadings into its column overwhelms its handful of genuine home loadings, going multimodal and poisoning convergence. The map needs the middle ground: let an instrument carry a small loading on several axes when the data genuinely support it, without collapsing a low-instrument factor. The classic tool for "mostly zero, but let real signal through" is a sparsity-inducing prior, and the right one is the regularized ("Finnish") horseshoe [23, 24]. For each off-home specific↔specific cross-loading λ_j (the PLAUSIBLE CROSS cells),

$$\lambda_j = z_j \, \tau \, \tilde{\eta}_j, \qquad \tilde{\eta}_j = \sqrt{\frac{c^2 \, \eta_j^2}{c^2 + \tau^2 \eta_j^2}}, \qquad z_j \sim \mathcal{N}(0,1), \;\; \eta_j \sim \text{HalfCauchy}(1), \;\; \tau \sim \text{HalfNormal}(\tau_0), \tag{A40}$$

with $\tau_0 = 0.05$ and slab $c = 0.30$, fit non-centred (z separate from the scales). Three pieces, each doing one job. The *global shrinkage* τ pulls the whole set of cross-loadings toward 0—the prior's default belief is "no cross-loadings." The *local shrinkage* η_j has heavy (Cauchy) tails, so a single cross-loading the data insist on can take a large η_j and escape the global shrinkage—the sharp spike at 0 with heavy tails that defines the horseshoe. The *slab* c caps the escape ($\tilde{\eta}_j \to c/\tau$ for large η_j), so an escaped loading stays near $|\lambda| \approx c$ (cross-loadings remain *small*—they are not home loadings) and the funnel is tamed so the sampler mixes.

This prior is exactly matched to the scientific situation. Dilution happens when indicators from *other* factors acquire spurious cross-loadings *into* a thin factor's column; under the horseshoe those spurious cells get small local η and are crushed to ≈ 0 by the global τ, so the thin column stays defined only by its home indicators (which keep their separate, sign-anchored positive prior and are *not* horseshoed). A genuine weak cross-loading—clinically real but carried by few instruments—can still emerge through the heavy tails and be reported with a credible interval. The prior is therefore default-off, evidence-on, magnitude-capped: the honest way to handle a weak signal in a factor with very few indicators. A map whose credible cross-loadings are few, small and clinically sensible is positive evidence the structure is well-specified, not an artefact of rigid zeros.

The data-supported cross-loadings. A continuous sparse-ESEM validator freed all off-home specific↔specific cells simultaneously and let the horseshoe select: $\approx 83\%$ shrank to ≈ 0 (median 0.002), so the simple structure is *recovered from the data, not imposed*, while the heavy tails let a handful of small, well-converged, clinically sensible cells survive; the thin factors stayed intact (substance home $|\lambda|$ 0.585, mania 0.484). Folding the survivors into the full mixed map—which arbitrates them under the correlated factor block—keeps three credible cross-loadings, all childhood-trauma / sleep items loading on cognition, each with a 95% credible interval excluding zero:

cross-loading	median	95% CI	reading
CTQ-37 → cognition	−0.094	[−0.134, −0.053]	childhood adversity ↔ cognition
PSQI latency → cognition	+0.057	[+0.011, +0.102]	sleep ↔ cognition
PSQI daytime → cognition	−0.070	[−0.122, −0.020]	sleepiness ↔ cognition

The candidate mania cross-loadings widened to non-credible and were dropped: a mania↔sleep relation belongs in the factor correlation ($\phi_{\text{mania,sleep}} \approx 0.23$), not an item cross-loading on a two-indicator factor. The headline of this exercise is the validation: given total freedom the model reproduces known clinical cross-talk and nothing spurious—positive evidence the 8-factor map is well-specified.

Proposition 2 (Rotation invariance of the likelihood)**.** *For any invertible $M \in \mathbb{R}^{(K+1)\times(K+1)}$, the reparameterization $\Lambda^\star = \Lambda M$, $\Phi^\star = M^{-1}\Phi M^{-\top}$, $f^\star = M^{-1}f$ leaves the implied covariance, and hence the Gaussian likelihood, unchanged:*

$$\Lambda^\star\Phi^\star\Lambda^{\star\top} + \Psi = \Lambda M\left(M^{-1}\Phi M^{-\top}\right)M^\top\Lambda^\top + \Psi = \Lambda\Phi\Lambda^\top + \Psi = \Sigma. \tag{A41}$$

Proof. Immediate from $MM^{-1} = I$: the inner $M(M^{-1}\Phi M^{-\top})M^\top$ telescopes to Φ. Since x_i is determined in distribution by (μ_i, Σ) and Σ is invariant, so is the likelihood. □

A factor model is therefore not identified from the likelihood alone; confirmatory factor analysis breaks this by fixing loadings to exactly zero, whereas the soft-prior model breaks it with the prior. The likelihood is rotation-invariant but the prior is *not*: a rotation moves a loading the prior wanted near zero away from zero, lowering $p(\Lambda)$, so the posterior $p(\Lambda \mid X) \propto p(X \mid \Lambda)\,p(\Lambda)$ concentrates at the orientation where small loadings align with the soft-zero priors and home loadings are positive and substantial; the positive truncation removes the residual sign-flip.

Corollary 1 (Flat-prior confirmation)**.** *Under a flat loading prior (identification constraints only) the posterior mode equals the maximum-likelihood (FIML) solution. Hence a prior-free refit landing on the same loadings demonstrates that the structure is recovered from the data, not manufactured by the priors. Empirically this refit reproduced the map to Tucker congruence $\varphi = 1.00$.*

A.6 The covariance identity and the headline estimand

Proposition 3 (Bifactor covariance)**.** *Under* (A28) *with* $\operatorname{Var}(G_i) = 1$, $\operatorname{Cov}(D_i) = \Phi_D$, $\mathbb{E}[\varepsilon_i\varepsilon_i^\top] = \Psi$ *diagonal, and the bifactor orthogonality* $\mathbb{E}[G_iD_i^\top] = \mathbf{0}$ *together with* $(G_i, D_i) \perp \varepsilon_i$,

$$\Sigma = \operatorname{Cov}(x_i) = \underbrace{\lambda_G\lambda_G^\top}_{\textit{general: rank one}} + \underbrace{\Lambda_D\Phi_D\Lambda_D^\top}_{\textit{specific factors}} + \underbrace{\Psi}_{\textit{private noise}} = \Lambda\,\Phi\,\Lambda^\top + \Psi, \qquad \Phi = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \Phi_D \end{pmatrix}. \tag{A42}$$

Proof. Expand $\Sigma = \mathbb{E}[(x_i - \mu_i)(x_i - \mu_i)^\top]$ with $x_i - \mu_i = \lambda_G G_i + \Lambda_D D_i + \varepsilon_i$. The nine resulting terms include cross-terms $\lambda_G\,\mathbb{E}[G_iD_i^\top]\,\Lambda_D^\top$ and $\mathbb{E}[(\lambda_G G_i + \Lambda_D D_i)\varepsilon_i^\top]$; the first vanishes by $\mathbb{E}[G_iD_i^\top] = \mathbf{0}$ and the second by independence of the factors and the noise. The survivors are $\lambda_G\lambda_G^\top\operatorname{Var}(G_i)$, $\Lambda_D\Phi_D\Lambda_D^\top$ and Ψ, giving (A42); the zero off-diagonal block of Φ encodes $G \perp$ specifics. □

The general factor leaves a *rank-one* footprint: every pair of items covaries through G by exactly $\lambda_{jG}\lambda_{j'G}$—one shared source touching all items.

The headline lives in the off-diagonal block. The correlated-G model *unzips* the zero block, freeing $\phi_{Gd} = \mathrm{Corr}(G, D)$:

$$\Phi = \begin{pmatrix} 1 & \phi_{Gd}^\top \\ \phi_{Gd} & \Phi_D \end{pmatrix}, \qquad \Sigma = \lambda_G \lambda_G^\top + \underbrace{\lambda_G \phi_{Gd}^\top \Lambda_D^\top + \Lambda_D \phi_{Gd} \lambda_G^\top}_{\text{the terms orthogonality removed}} + \Lambda_D \Phi_D \Lambda_D^\top + \Psi. \tag{A43}$$

The vector ϕ_{Gd}—the correlation of each specific factor with general burden—is the headline estimand of the enterprise: "biology is the least severity-entangled domain" is the statement that the immunometabolic entry is near zero while cognition and sleep are far larger (0.39 and 0.42). The bifactor orthogonality is the *null*; the correlated-G arm measures the departure from it; and the bifactor direct loading $|\lambda_{jG}|$ and the correlation ϕ_{Gd} agree on the ordering—reporting both is what turns "biology is separable" from a constraint artefact into a finding.

The immunometabolic factor is anchored by body composition and graded across metabolic, inflammatory and cardiovascular markers. Table A7 lists all 37 primary loadings with 95% credible intervals, every one excluding zero: the three body-size indicators (BMI 0.95, weight 0.88, waist 0.84) are the strongest, while the metabolic, inflammatory (CRP 0.37) and cardiovascular markers contribute smaller but credible loadings. The axis is therefore adiposity-anchored rather than inflammation-led; a sensitivity refit excluding the three anthropometric indicators, reported below, tests whether a coherent metabolic–inflammatory–cardiovascular axis survives once the body-size anchors are removed.

A.7 Sensitivity: the axis without its anthropometric anchors

To test whether the immunometabolic factor is merely a body-habitus axis, we refit the full eight-factor map after *excluding* the three anthropometric indicators (BMI, weight, waist circumference), leaving the metabolic, inflammatory and cardiovascular markers to define the axis on their own (balanced $N = 2{,}000$, otherwise identical specification; 0 divergences, $\hat{R} = 1.09$, so magnitudes are read as approximate). The axis did not collapse. All 34 remaining indicators loaded with 95% credible intervals excluding zero, and the column re-anchored on blood pressure (systolic and diastolic, standing and supine, 0.63–0.67), heart rate (0.44–0.48), the total/HDL cholesterol ratio (0.44), triglycerides (0.40) and inflammatory markers (white-cell count 0.36, C-reactive protein 0.32, neutrophils 0.28)—a coherent cardiometabolic–inflammatory syndrome rather than a residue. Against the canonical axis the loading column was highly congruent on the shared items (Tucker $\varphi = 0.90$), and the factor remained nearly uncorrelated with G and the symptom axes (specific–specific $|\phi| \leq 0.02$, apart from the known immunometabolic–substance coupling ≈ 0.20). The body-size indicators are therefore the strongest *anchors* of the axis, not its content: removing them redistributes the loading onto the cardiovascular and metabolic markers without dissolving the dimension. Because the one inflammatory marker available, CRP, is itself substantially adiposity-driven [43–45], we name the axis cardiometabolic and adiposity-anchored and do not claim an inflammatory dimension independent of body composition.

The axis is also durable without its anthropometric anchors. Scored forward onto the 12- and 24-month visits under the fixed excluded model (the temporal protocol of C), the immunometabolic coordinate remained trait-like: a measurement-error test–retest intraclass correlation of 0.76 (94% CI $[0.74, 0.78]$), with an essentially static population trajectory (shift ≈ 0). This is below the full-axis 0.91—body mass is indeed the most stable single component—but still comfortably in the trait range, so the 0.91 durability is not merely the test–retest stability of body size but a property of the broader cardiometabolic–inflammatory constellation.

A.8 Woodbury evaluation and equivalence with FIML

Evaluating (A38) naively costs $\mathcal{O}(|C_i|^3)$ per patient. The Woodbury identity and matrix-determinant lemma reduce it to the $(K+1)$-dimensional factor space.

Lemma 1 (Woodbury reduction). *With Ψ_{C_i} diagonal and Λ_{C_i} the observed-row loadings,*

$$\Sigma_{C_i}^{-1} = \Psi_{C_i}^{-1} - \Psi_{C_i}^{-1}\Lambda_{C_i}\big(\Phi^{-1} + \Lambda_{C_i}^{\top}\Psi_{C_i}^{-1}\Lambda_{C_i}\big)^{-1}\Lambda_{C_i}^{\top}\Psi_{C_i}^{-1}, \tag{A44}$$

$$\log|\Sigma_{C_i}| = \log\Big|\Phi^{-1} + \Lambda_{C_i}^{\top}\Psi_{C_i}^{-1}\Lambda_{C_i}\Big| + \log|\Phi| + \sum_{j\in C_i}\log\sigma_j^2. \tag{A45}$$

The inner solve is $\mathcal{O}\big((K+1)^3\big)$, independent of $|C_i|$; patients sharing a missingness pattern share the inner $(K+1)\times(K+1)$ factorization, computed once per pattern. This removes the per-patient latent funnel for the continuous block and makes full-sample ($N = 9{,}013$) estimation tractable on a workstation; the non-Gaussian block retains explicit per-patient latents.

Proposition 4 (Bayes/FIML equivalence). *The marginalized Bayesian observed-data model and FIML optimize the* same *observed-data likelihood and differ only by the priors* (A39)*. In particular (Corollary 1) the flat-prior MAP equals the FIML estimate, so a standalone FIML estimator adds no independent evidence and confirmation is done in-engine.*

Proof. By Proposition 1 the per-patient marginal density is the incomplete-data multivariate-normal density on C_i, whose log summed over patients is the FIML objective. The posterior is this likelihood times the prior; with a flat prior the log-posterior equals the log-likelihood up to a constant, so its mode is the FIML maximizer. □

Factor scores. Each patient's coordinates are the expected a posteriori (EAP) factor scores from their observed cells; for an all-continuous pattern,

$$\hat{f}_i = \mathbb{E}[f_i \mid X_{i,C_i}] = \Phi\Lambda_{C_i}^{\top}\Sigma_{C_i}^{-1}(X_{i,C_i} - \mu_{i,C_i}), \quad S_i = \Phi - \Phi\Lambda_{C_i}^{\top}\Sigma_{C_i}^{-1}\Lambda_{C_i}\Phi, \tag{A46}$$

obtained by MCMC when non-Gaussian cells are present. The per-patient posterior covariance S_i is exported and propagated downstream (it is the noise term in the stratification mixture, Supplement B): a patient with one observed indicator for a factor is *not* treated as equally characterized as one with six.

A.9 Staged estimation and in-engine confirmation

The global posterior was reached by continuation (homotopy): a well-conditioned sub-model was fit and then deformed toward the full target one source of difficulty at a time. A clean hard-zero backbone was built first (continuous core at full N: G, cognition, immunometabolic, sleep), then extended to developmental-risk and mania/activation, each converged posterior warm-starting the next; the full 8-factor mixed map adds the suicidality and substance factors. The data-supported cross-loadings of A.5 are then folded onto this hard-zero solution—warm-starting the relaxed fit from the hard-zero basin is identified here, with the thin factors held in place. *Only the global fit is interpreted*; the earlier stages are convergence/identification checkpoints. Sampling used the No-U-Turn sampler [27] via PyMC [28] with the NumPyro/JAX backend [29]; a stage advanced only on $\hat{R} \leq 1.01$, effective sample size ≥ 400, zero divergences, no Heywood cases and acceptable posterior-predictive behaviour. The full-N continuous backbone met this strict gate; the joint 8-factor (mixed) map is reported at the largest sample that mixes ($\hat{R} \leq 1.03$, zero divergences) with a cross-seed stability guard (Tucker congruence $\varphi = 0.99$). Confirmation used (i) the flat-prior refit (Corollary 1); (ii) posterior-predictive checks (continuous Bayesian standardized root-mean-square residual ≈ 0.07; 21/22 non-Gaussian indicators within the 90% predictive interval, the lone exception a 90.8%-zero suicide-attempt count item—an item-level, not factor-level, misfit); and (iii) model comparison by WAIC and PSIS-LOO [30, 32], which decisively preferred the bifactor over unidimensional and correlated-factor alternatives. Classical fit indices (CFI/TLI, RMSEA) are not reported—they are weak under heavy missingness and non-normal indicators and add no independent evidence here.

B Stratification: structure testing, archetypes and soft regions

The stratification operates on the fitted coordinates $\{x_i, S_i\}_{i=1}^{N}$ with $x_i = \hat{f}_i \in \mathbb{R}^D$ ($D = 8$; the eight axes are overall severity G, cognition, immunometabolic, sleep, mania/activation, suicidality, developmental risk and substance) and known posterior covariance S_i from (A46); uncertainty is carried throughout rather than discarded. All $N = 9{,}013$ patients are scored on all eight axes (latent z-scale, 100% finite), with the immunometabolic, cognition and developmental coordinates tight, and mania, substance and suicidality wide.

B.1 Structure-discovery gate and the single-Gaussian falsification null

Before any clustering we test the cloud's shape with complementary statistics, each computed over posterior draws so measurement noise cannot manufacture structure. *Cluster tendency* uses the Hopkins statistic $H = \sum_i u_i^D / (\sum_i u_i^D + \sum_i w_i^D)$, where w_i is the nearest-neighbour distance of a sampled real point and u_i that of a uniform point over the data support ($H \approx \frac{1}{2}$ indicates no clustering); *modality* uses Hartigan's dip statistic [62] on PC1 and each axis; *model selection* uses the gap statistic [63] and the Gaussian-mixture BIC profile; *density* uses HDBSCAN [64] (fraction of points assigned to noise); *topology* a Mapper graph [65]; and *separation* the best silhouette over K. A continuum verdict does *not* require $H \to \frac{1}{2}$: Hopkins measures departure from *uniformity*, not from separability, and an elongated or heavy-tailed single Gaussian—exactly the symptom-axis geometry we observe—scores high on it. The operative criterion is instead a unimodal dip, the gap and BIC failing to prefer $K > 1$, HDBSCAN near-all-noise, a single Mapper component, and—the decisive separability test—a best-partition silhouette indistinguishable from the single-Gaussian null. Empirically: $H = 0.80$ (real), against a dispersion null of 0.756 ± 0.004, i.e. $z = 8.71$ versus a *uniform* reference—a large, significant departure from uniformity that we disclose openly rather than omit, because it does *not* contradict the continuum verdict (see below); PC1 dip $p \approx 1.0$; silhouette peak 0.146 (< 0.15, no separation); gap-statistic $k_{\text{opt}} = 1$; HDBSCAN 0 clusters (100% noise); Mapper a single connected component. Per-axis, six of the eight dip tests are unimodal ($p > 0.99$); only mania/activation and suicidality are multimodal ($p = 0$)—the symptom axes carry internal structure, but the joint space does not separate.

The high Hopkins value therefore does *not* indicate discrete clusters. Cluster *tendency* (departure from uniformity) and cluster *separability* (departure from a single, matched Gaussian) are distinct properties, and only the latter bears on whether the map contains discrete biotypes. Against the uniform reference used by Hopkins, $z = 8.71$ is unsurprising for any elongated, correlated cloud of clinical coordinates and carries no information about separability. The operative separability test is the silhouette against the single-Gaussian falsification null detailed just below: real 0.140 versus null 0.137 ± 0.002 ($z = 1.13$, n.s.)—not reliably better separated than a structureless Gaussian blob. Cluster tendency and cluster separability are thus dissociated in these data: the cloud is non-uniform (as any clinical population is) but not separable into discrete regions, which is precisely the signature of a graded continuum rather than a set of biotypes.

Because all of these are absolute statistics, we anchor them with a *falsification null*. We draw synthetic data from a single multivariate Gaussian with the empirical mean and covariance of the real coordinates—a structureless cloud by construction—and run the identical pipeline. If the real cloud were truly clustered, its best partition would separate points *better* than the null's.

$$z_{\text{sil}} = \frac{s^\star_{\text{real}} - \mathbb{E}[s^\star_{\text{null}}]}{\text{sd}[s^\star_{\text{null}}]} = 1.13, \tag{B47}$$

i.e. the real data's best partition (silhouette 0.140) is *not* reliably better separated than a structureless Gaussian blob (0.137 ± 0.002; $z = 1.13$, n.s.); the mixture-optimal number of components collapsed to one in all 20 posterior draws. This turns "no elbow" into a calibrated negative result.

B.2 Archetypal analysis (the continuum's shape)

Each patient is represented as a convex combination of A extreme phenotypes, each itself a convex combination of patients [38]:

$$\min_{W,B} \sum_{i=1}^{N} \Big\| x_i - \sum_{a=1}^{A} w_{ia}\, z_a \Big\|^2 \quad \text{s.t.} \quad z_a = \sum_{j=1}^{N} b_{ja}\, x_j, \quad w_{ia}, b_{ja} \geq 0, \quad \textstyle\sum_a w_{ia} = 1, \ \sum_j b_{ja} = 1, \tag{B48}$$

i.e. $X \approx WZ$ with $Z = BX$ and both W and B row-stochastic, which forces the archetypes z_a onto the convex hull of the cloud. The simplex weights $w_i \in \Delta^{A-1}$ are the soft archetype membership. Uncertainty is propagated by refitting (B48) across posterior draws and projecting each draw onto the fixed archetypes (simplex-constrained least squares). The number of corners A is set by *cross-seed stability* (largest A whose minimum cross-seed Tucker congruence ≥ 0.8), not by parsimony: $A = 2$ (0.999), $A = 4$ (0.997) and $A = 5$ (0.979, explained variance 0.601) are stable, with a clean stability cliff at $A = 6$ (0.979$\to$0.436) and below ($A = 7, 8$ at 0.20–0.55). The operational rule selected $A = 5$ (Table 3). **A0** *activation/sleep* is defined by elevated sleep (+2.8) and mania (+1.7) over a below-average severity floor; it is the manic/sleep-disturbed corner and is bipolar-enriched ($N = 1{,}448$). **A1** *severe, clean-biology* pairs high overall severity (+1.9) with markedly *low* immunometabolic load (−1.9) and low developmental risk ($N = 1{,}584$)—patients who are functionally ill but biologically quiet. **A2** *immunometabolic burden* is the biology corner: the highest immunometabolic coordinate in the space (+3.5) carried alongside high severity (+2.4) and elevated suicidality (+1.2); a secondary developmental-risk elevation (+1.2) also loads here, but the canonical developmental corner is **A3** ($N = 1{,}426$). **A3** *trauma/suicidality* is anchored on developmental risk (+2.8) and suicidality (+1.4) with immunometabolic load running the other way (−2.1); it is the childhood-adversity corner and the most populous symptom corner ($N = 2{,}004$). **A4** *low-burden/well* is the healthy-reference extreme—low sleep (−2.7), low severity (−2.0) and low mania—and the single largest corner ($N = 2{,}551$).

The clinically important geometry. Two features make the simplex substantive rather than decorative. First, **A1** and **A2** carry comparable functional burden (both high on G) but *opposite* biology—low versus highest immunometabolic load—so the corners render the biology–G dissociation as a discrete contrast: two extremes that look comparably ill but are biologically opposite. Second, the symptom space resolves into two distinct corners rather than one severity gradient—**A0** (activation/sleep) and **A3** (trauma/suicidality)—while **A4** marks the low-burden pole that closes the hull. Crucially, the biology corner **A2** is a *direction of maximal phenotype*, not a partition boundary: immunometabolic load varies continuously across the cloud with no density gap (Hopkins $\to \frac{1}{2}$, no preferred mixture K), so **A2** is an archetypal corner of the continuum rather than a cluster carved out of it. This is why the archetypes and the continuous coordinates, not any tessellation, are the primary object for biology, and why the dominant-archetype labels enter only as summaries while the continuous simplex weight vector remains the inferential object.

B.3 Measurement-error mixture and the nested K-family

Treating x_i as a noisy reading of a latent position, $x_i \mid \theta_i \sim \mathcal{N}(\theta_i, S_i)$, with θ_i from a K-component mixture yields the extreme-deconvolution likelihood [39]

$$x_i \ \sim \ \sum_{k=1}^{K} \pi_k\, \mathcal{N}(m_k,\ \Sigma_k + S_i)\,, \qquad \pi \sim \text{Dirichlet}\big(\tfrac{\alpha}{K}\mathbf{1}\big)\,, \tag{B49}$$

in which *every* observation carries its own additive, known covariance S_i, so imprecise coordinates spread their mass over components and prior-dominated axes self-down-weight. The soft regions are the posterior responsibilities $r_{ik} = \pi_k \mathcal{N}(x_i \mid m_k, \Sigma_k + S_i) / \sum_l \pi_l \mathcal{N}(x_i \mid m_l, \Sigma_l + S_i)$. On a continuum the mixture BIC is essentially flat (185,006–185,557 across $K = 2$–4, $< 0.4\%$; the minimum at $K = 4$ but with the lowest stability), so no K is privileged; we export a nested family ($K = 2, 3, 4$) as labelling conventions and *defer the operative granularity to incremental predictive validity* (Supplement D), which returns "no hard K". Every K splits first on psychiatric symptom burden—mania ($\eta^2 = 0.224$) and suicidality ($0.225 \to 0.476 \to 0.523$ across $K = 2, 3, 4$)—while finer K progressively picks up severity (G: $0.050 \to 0.108 \to 0.203$) and the immunometabolic gradient ($0.027 \to 0.057$) that $K = 2$ discards; biology has no density gap and so never forms a region boundary at any K. A coarsest-K default ($K = 2$) is carried only as a hand-off contract.

B.4 Validation gates

The representation is tested on four gates, run on the $K = 2$ contract default (the gates hold across the confident-stable family). *Not-just-severity*: variance explained per axis $\eta_d^2 = 1 - \sum_k \sum_{i \in \mathcal{C}_k} (x_{id} - \bar{x}_{kd})^2 / \sum_i (x_{id} - \bar{x}_d)^2$ peaks on the specific axes (mean $\eta^2_{\text{spec}} = 0.077$, driven by mania 0.224 and suicidality 0.225) above G (0.050). *Transdiagnosticity*: the adjusted Rand index $\text{ARI} = (\text{RI} - \mathbb{E}[\text{RI}]) / (\max \text{RI} - \mathbb{E}[\text{RI}])$ against cohort and the seven DSM-5 subtypes is ≈ 0 (0.011 and 0.006). *Stability and not-a-missingness-artefact*: cross-seed ARI 0.991, and a classifier predicting region membership from each patient's per-axis coverage pattern underperforms the majority baseline (accuracy 0.611 vs 0.683, lift -0.072). *Tighter than DSM-5*: a free mixture beats one constrained to the DSM-5 partition by information criterion (185,557 at $K = 2$ vs 188,200 for DSM-5 at $K = 7$; mean η^2 0.074 vs 0.026, roughly $3\times$ the explained variance at lower BIC).

These four gates characterise the $K = 2$ soft *tessellation*; the load-bearing $A = 5$ *archetype* simplex is validated separately and more strongly. Read against the held-out labels, the archetype assignment is essentially orthogonal to DSM-5 (ARI 0.021; cohort 0.017) yet explains $\eta^2 = 0.256$ of the eight-axis coordinate variance against 0.026 for the DSM-5 grouping ($\approx 9.7\times$, at lower BIC). The "not-just-severity" reading—driven by mania and suicidality above G—is specific to the coarse $K = 2$ split; the archetypes instead separate on severity ($\eta_G^2 = 0.47$) *and* biological/symptom profile (immunometabolic 0.39, sleep 0.38, developmental 0.35), so equally severe corners (A1, A2) remain biologically opposite. Per-cohort composition is in Table 4.

Embeddings (PCA, Figure 8; UMAP [66]) are used for visualization only, never as clustering inputs.

C Temporal coherence: forward scoring, invariance and trait/state

C.1 Forward scoring under the fixed measurement model

Follow-up visits were scored on the *fixed* measurement model: the baseline ($V0$) parameters $(\hat{\Lambda}, \hat{\Phi}, \hat{\Psi})$, the frozen rank-copula maps $\hat{F}_j$ from (A36), and the frozen $V0$ covariate residualization are held constant, and a follow-up record is projected, never re-discovered. Concretely, a V_t score is obtained by (i) orienting and copula-transforming each gaussianized cell onto the $V0$ z-scale via $\hat{F}_j$; (ii) applying the frozen-$V0$ covariate residualization (age-spline, sex, education, site) with the visit's covariates; and (iii) projecting onto the fixed $\hat{\Lambda}, \hat{\Phi}, \hat{\sigma}$ using the EAP map (A46) for the continuous block and the corresponding full-N projection for the explicit block. As a validity check, scoring $V0$ through this path reproduces the baseline coordinates at Pearson $r = 0.99$–1.00 across the continuous axes, so $V0$, $V1$ and $V2$ live on a common scale (panel of 16,241 visit-records: $V0$ 9,013, $V1$ 4,270, $V2$ 2,958).

C.2 Longitudinal measurement invariance

Re-fitting the simple-structure backbone per visit (scale-invariant by design) and computing the Tucker congruence of the primary loadings against $V0$, every backbone axis was invariant at both follow-ups ($\varphi \geq 0.95$, 4/4; minimum congruence φ: sleep 0.996, cognition 0.995, overall severity 0.991, immunometabolic 0.987; Extended Data Fig. E4). The merged immunometabolic biology axis is measured consistently over time ($\varphi = 0.987$, fully invariant), so coordinate change on it reads as patient change rather than instrument drift. The explicit axes (suicidality, developmental risk, substance) are not part of the continuous backbone test, and mania is data-limited (two indicators).

C.3 Trait/state decomposition

For each axis we fit a measurement-error random-intercept model across visits. Writing x_{it} for patient i's coordinate at visit t, with known baseline measurement variance s_i^2 (the relevant diagonal entry of S_i),

$$x_{it} = \underbrace{\gamma_t}_{\text{visit fixed effect}} + \underbrace{u_i}_{\text{trait}} + \underbrace{e_{it}}_{\text{state}} + \underbrace{m_{it}}_{\text{measurement}}, \quad u_i \sim \mathcal{N}(0, \tau^2),\ e_{it} \sim \mathcal{N}(0, \omega^2),\ m_{it} \sim \mathcal{N}(0, s_i^2), \tag{C50}$$

so that plugging the known s_i^2 prevents a low-reliability axis from masquerading as state, and the visit fixed effects γ_t prevent a shared population trajectory from being miscounted as individual instability. The trait fraction is the intraclass correlation

$$\text{ICC} = \frac{\tau^2}{\tau^2 + \omega^2}, \tag{C51}$$

estimated as a posterior with 94% credible interval. The immunometabolic biology axis is the single most durable axis (ICC 0.91 $[0.90, 0.91]$), with cognition (0.70 $[0.67, 0.72]$) trait beside it; the moving symptom axes are state or mixed (substance 0.49, sleep 0.47, suicidality 0.43, developmental risk 0.39 $[0.33, 0.44]$); general burden is trait by rank (0.62 $[0.60, 0.63]$). The population shift $\gamma_{V2} - \gamma_{V0}$ was strongly negative on severity (-0.46) and suicidality (-0.84) and near zero on biology ($+0.04$), the formal statement of "the cohort improves on symptoms while individual biological rank is held." A complementary geometric route confirmed that the archetype weight vector moved little across the $A = 5$ simplex (median cosine similarity 0.81 over $V0 \to V2$ completers, $n = 2{,}958$), with severity sliding more than biology (spine-moves-not-biology $0.234 > $ biology-moves-not-spine 0.163, consistent with the ICC ranks). The error-corrected ICC carries the trait/state headline; the geometric route is read as persistence of archetype identity in the weights rather than as an independent confirmation of the variance ranks (the per-axis Spearman of reliable-change rate against ICC is near zero, $\rho = 0.07$, $p = 0.87$, with the merged biology axis, the $A = 5$ argmax churn, and the orthogonal/thin substance and data-limited mania axes muddying an eight-point correlation). Developmental risk scores as state partly because childhood-adversity reports carry recall noise rather than true change; substance now carries no inter-factor correlation (pinned orthogonal) and rests on two binary indicators, so less of its variance reads as durable trait; mania is data-limited (two indicators) and its trait/state verdict is not relied upon.

D Prognosis: errors-in-variables incremental validity

D.1 Errors-in-variables Bayesian GLM and the nested ladder

The map coordinates are estimated, not observed, so a naive regression on $\hat{f}_i$ would suffer attenuation. We use an errors-in-variables (EIV) Bayesian generalized linear model that treats the latent coordinate as a parameter with a measurement model attached. For outcome y_i (functioning EGF or severity CGI-S at two years, on its native scale) and design W_i,

$$\theta_{id} \sim \mathcal{N}(\hat{f}_{id},\, s_{id}^2), \qquad g(\mathbb{E}[y_i]) = \beta_0 + \beta_W^\top W_i + \beta_\theta^\top \theta_i, \tag{D52}$$

where s_{id} is the per-patient posterior standard deviation from (A46), so the regression "knows" how precisely each coordinate is measured. We compare a fixed ladder of designs W: a reference R_{3y} = {age, sex, baseline outcome, baseline severity}; then +DSM-5 subtype, +map (continuous coordinates, the seven specific axes, the five archetypes, or a tessellation), or +both. The estimand is incremental: does the map add *beyond* diagnosis, current severity and the baseline value of the outcome.

D.2 Incremental value, operative K, and discrimination

Incremental value is the held-out expected log predictive density, estimated by Pareto-smoothed importance-sampling leave-one-out [30]:

$$\Delta\mathrm{ELPD} = \widehat{\mathrm{elpd}}_{\mathrm{loo}}(\mathrm{model}) - \widehat{\mathrm{elpd}}_{\mathrm{loo}}(\mathrm{R}_{3y}), \tag{D53}$$

declared *predictive* when it exceeds twice its standard error. For functioning (N = 2,114): archetypes +62.8, seven specific axes +38.1, every tessellation $\approx$ +16 to +20, the isolated durable-biology axes +2.3 (ambiguous). The continuous and archetype encodings strictly dominate any hard partition, so the operative-K selector returns *none*—the actionable object is a continuous position. For severity the increments are small and ambiguous (autoregression-saturated). Head-to-head, map and DSM-5 are co-informative (functioning: +map 17.3, +DSM-5 29.0, +both 62.6). For binary endpoints we also report the change in AUC under five-fold cross-validation; functional-remission AUC rose only 0.745 → 0.755 (+0.010), the signature of a strong group-level signal that does not translate into a large individual gain. The two-year functional-remission gradient across the five archetypes (22% at the immunometabolic biology corner to 63% at the low-burden pole) was a within-diagnosis effect: the rank held inside each cohort (biology-vs-low-burden cohort-adjusted odds ratio $\approx$ 6.3; within each cohort BP 27% → 73%, DR 31% → 72%, SZ 9% → 25%), direct standardization to a common cohort mix moved it only $\approx$ 4%, and the corner-by-cohort interaction was non-significant ($p = 0.79$).

D.3 Robustness

Stressing the predictive winner (the archetypes) on functioning: the signal survived inverse-probability-of-attrition weighting (+54.4, a 13% attenuation), dropping DR or SZ (+56.4, +61.8), and vanished under a permutation null (−2.4, correctly); it weakened sharply when bipolar disorder was removed (+7.1), so the increment is carried by the episodic, open-course cohorts. The isolated durable-axis block does *not* clear the bar on its own (+2.3, ambiguous), even though the immunometabolic axis itself has a credible adverse direction (higher immunometabolic loading forecasts worse functioning, coefficient −0.053, 95% interval $[-0.088, -0.019]$); the robust predictive object is the fuller five-archetype phenotype, where the biology lives as the worst-prognosis corner.

D.4 Representation sufficiency benchmark

Because the eight-factor map is a roughly twentyfold compression of the 143 raw indicators, the honest question is sufficiency, not supremacy. Under one fixed regularized gradient-boosting classifier with identical cross-validation folds, we compared three feature sets—reference (DSM-5 + severity + baseline GAF), reference + map (coordinates, uncertainty and the five archetypes), and reference + the raw indicators—predicting recovery (impaired→GAF$\geq$ 71) and deterioration

(GAF drop ≥ 10), pooled across V1 and V2. The map was *sufficient* for deterioration (AUC tie; raw$-$map +0.009 at V1, +0.005 at V2) and *near-sufficient* for recovery (raw +0.04 AUC; +0.040 at V1, +0.039 at V2); a TreeSHAP attribution showed that 92% (V1) to 97% (V2) of the raw recovery signal already lives *within* the eight modelled factors (top drivers: CRP and BMI loading on immunometabolic, verbal memory and processing speed on cognition, nicotine dependence on substance, sleep-rhythm items on sleep), so the residual is within-factor compression—the only off-map carriers are the depression/anxiety window items—not a missing axis (Extended Data Fig. E5). The map also transported across held-out cohorts at least as well as the raw indicators for deterioration.

E Extended Data figures and the claims ledger

E.1 Extended Data figures

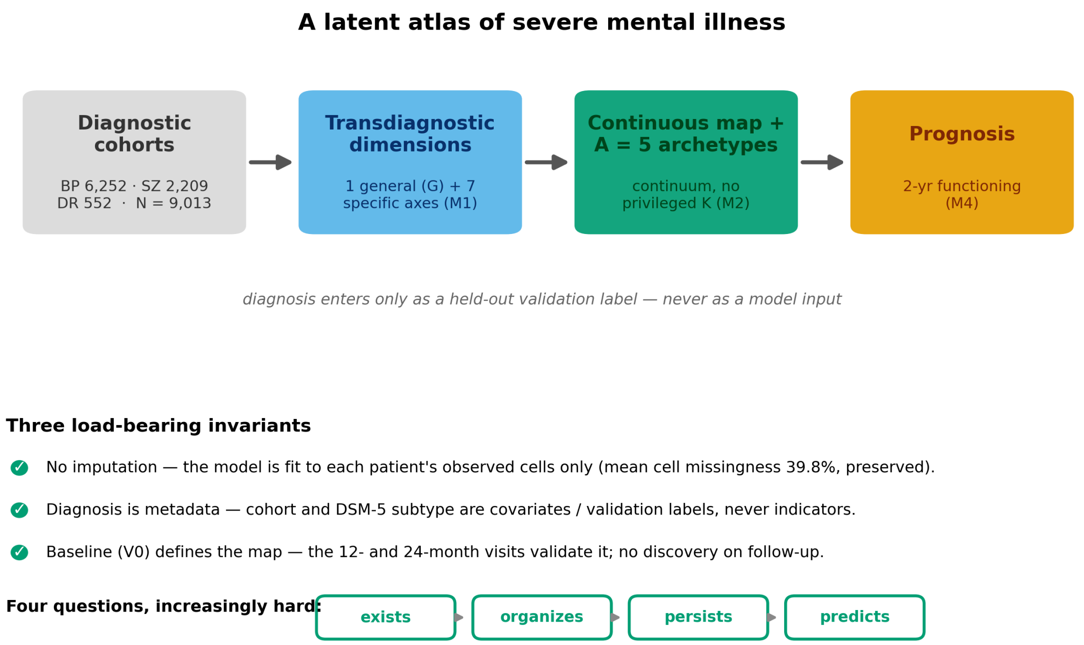

Figure E1: **Study design, invariants and the four questions.** The harmonized three-cohort FACE baseline (bipolar disorder, schizophrenia, depression; $N = 9{,}013$) is reorganized around diagnosis-agnostic axes by one global, missingness-aware Bayesian bifactor / exploratory structural-equation model fit to each patient's observed cells only, under three core modelling principles (no imputation; diagnosis as metadata; baseline defines the map and follow-up validates it). The fitted map is then interrogated through four dependent, increasingly demanding questions—exists, organizes, persists, predicts—with positive and null results reported alike.

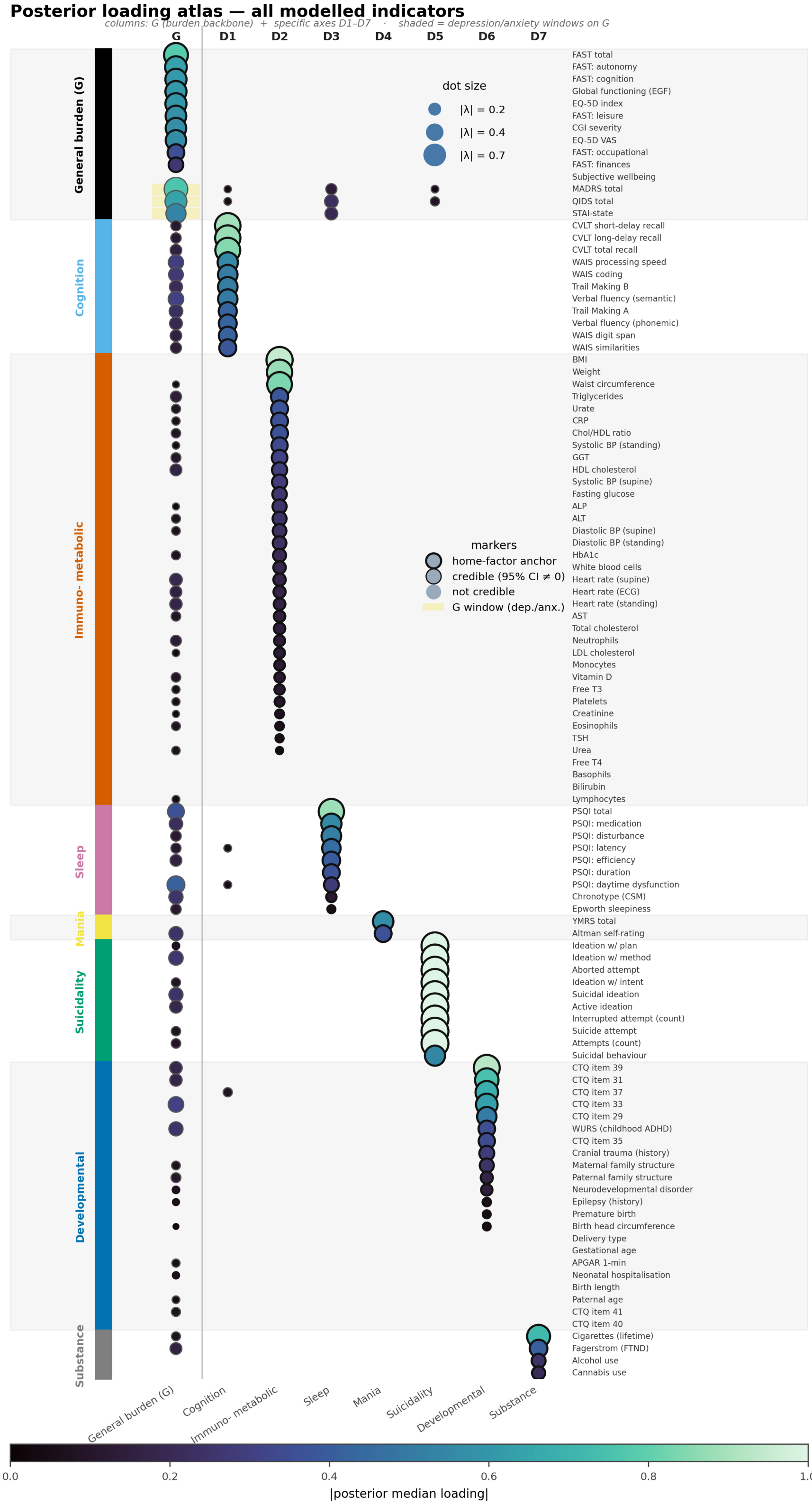

Figure E2: **The complete posterior loading atlas.** As Figure 5a but showing *every* modelled indicator (all 109 items: 88 continuous and 21 explicit) grouped by home factor, with no per-block cap. Dot size and colour encode the absolute posterior median loading; outlined dots have a 95% credible interval excluding zero (heavier ring = home-factor anchor). Three data-supported cross-loadings render as outlined dots in the cognition column—childhood adversity (CTQ-37, −0.094) and two poor-sleep items (PSQI latency +0.057, PSQI daytime dysfunction −0.070), each with a 95% credible interval excluding zero. The diagonal block structure—each clinical block loading cleanly on its own specific axis, with a sparse general-burden (G) column—is the full anatomy of the measurement model summarised in the main-text Figure 5.

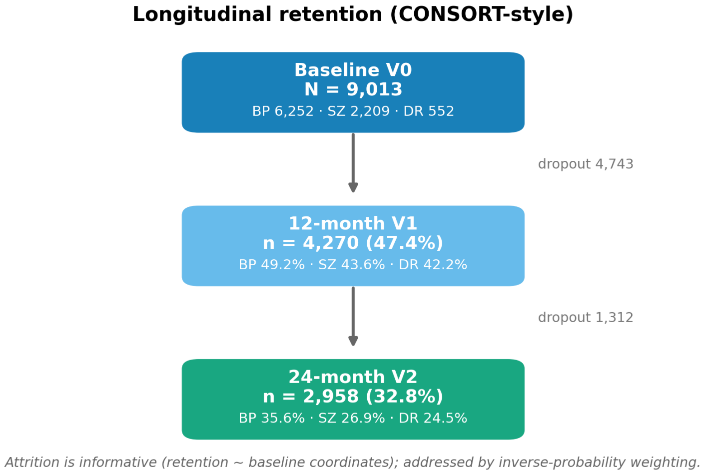

Figure E3: **Longitudinal retention.** Patients assessed at baseline and retained at the 12- and 24-month follow-up visits, overall and by cohort. Attrition is informative (retention depends on baseline coordinates) and is addressed throughout by inverse-probability weighting.

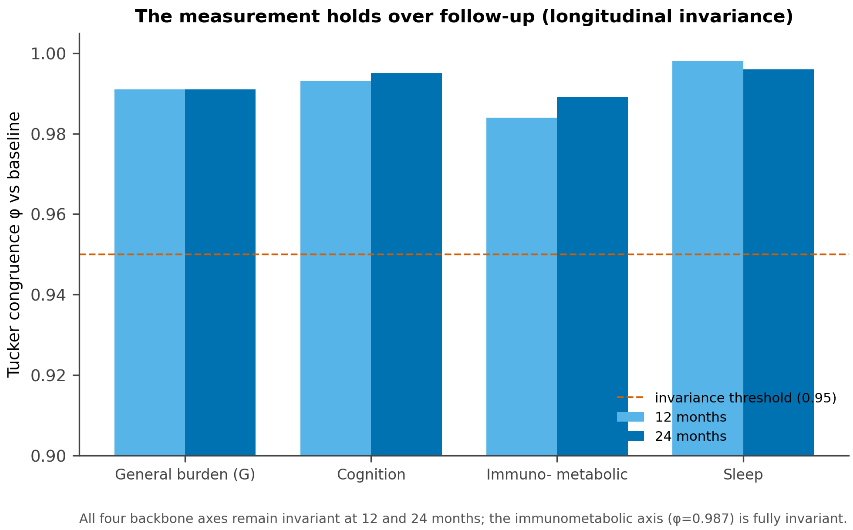

Figure E4: **Longitudinal measurement invariance.** Tucker congruence of each backbone axis against baseline at 12 and 24 months. All four backbone axes (G, cognition, immunometabolic, sleep) remain invariant ($\varphi \geq 0.95$); the merged immunometabolic biology axis is fully invariant ($\varphi = 0.987$).

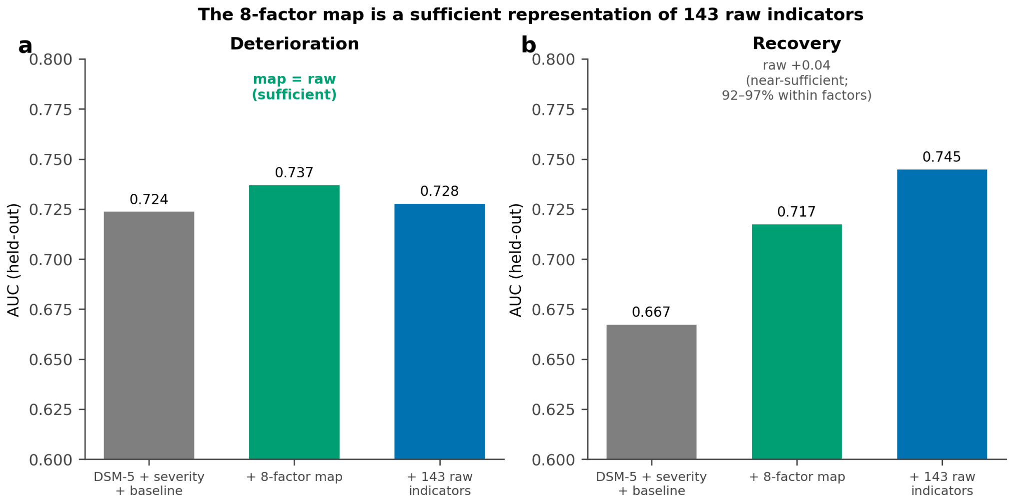

Figure E5: **The eight-factor map is a sufficient representation of the 143 raw indicators.** Held-out AUC for a matched gradient-boosting classifier predicting two-year deterioration and recovery, using the reference (DSM-5 + severity + baseline), the reference + the eight-factor map, or the reference + all 143 raw indicators. The map equals the raw indicators for deterioration (sufficient) and trails by ≈ 0.04 AUC for recovery (near-sufficient); $\geq 92\%$ of the raw recovery signal lies within the eight factors.

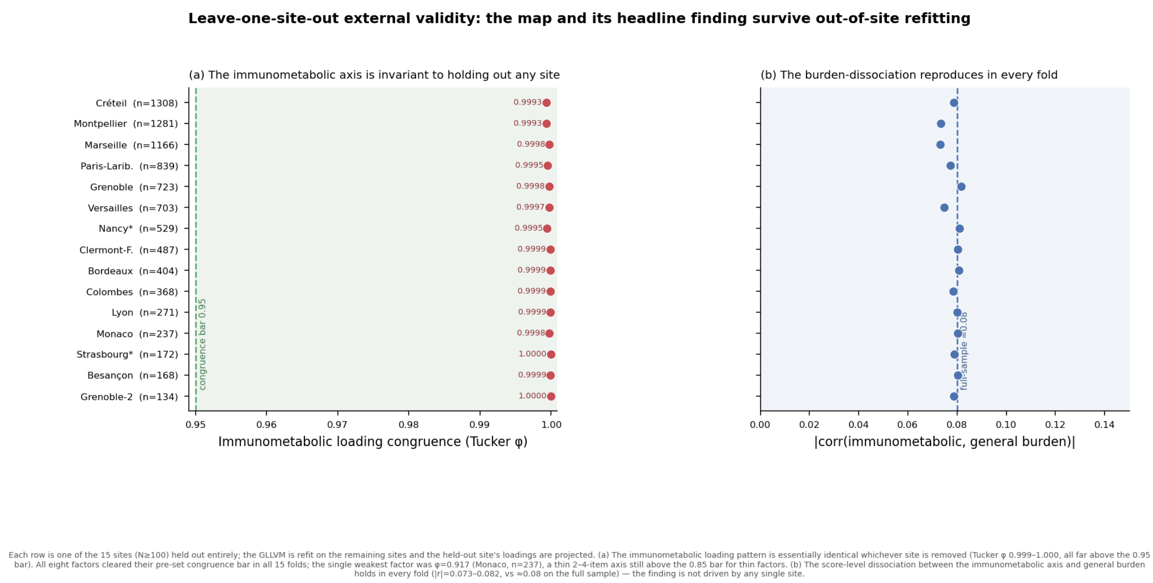

Figure E6: **Leave-one-site-out external validity: the map and its headline finding survive out-of-site refitting.** Each row is one of the 15 sites ($N \geq 100$) held out entirely; the GLLVM is refit on the remaining sites and the held-out site's loadings are projected. **(a)** The immunometabolic loading pattern is essentially identical whichever site is removed (Tucker $\varphi = 0.999$–1.000, all far above the 0.95 congruence bar). All eight factors cleared their pre-set congruence bar in all 15 folds; the single weakest factor was $\varphi = 0.917$ (Monaco, $n = 237$), a thin two-to-four-item axis still above the 0.85 bar for thin factors. **(b)** The score-level dissociation between the immunometabolic axis and general burden holds in every fold ($|r| = 0.073$–0.082, versus ≈ 0.08 on the full sample)—the finding is not driven by any single site.

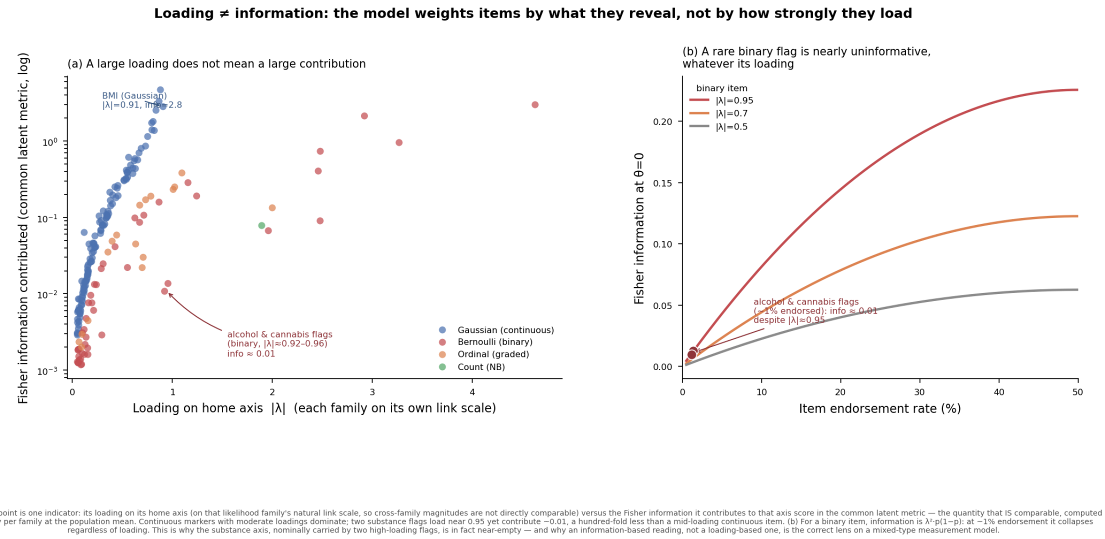

Figure E7: **Loading $\neq$ information: the model weighs items by what they reveal, not by how strongly they load. (a)** Every indicator's loading on its home axis (on that likelihood family's own link scale, so cross-family magnitudes are not directly comparable) against the exact Fisher information it contributes to that axis score at the population mean—the quantity that *is* comparable across families. Continuous markers with moderate loadings dominate; the two substance-use flags (alcohol, cannabis) load ≈ 0.92–0.96 yet contribute only ≈ 0.01–0.013, roughly a hundred-fold less than a mid-loading continuous marker such as body-mass index (specific-factor loading 0.908, information 2.8; the table-reported general loading is ≈ 0.95). **(b)** The mechanism: for a binary item, Fisher information is $\lambda^2 p(1-p)$, which collapses toward zero as endorsement approaches the extremes whatever the loading—so a rare binary flag is nearly uninformative regardless of how strongly it loads. This is why the substance-use axis, nominally carried by two high-loading flags, is in fact near-empty of information, and why an information-based reading of the map, not a loading-based one, is the correct lens on a mixed-type measurement model.

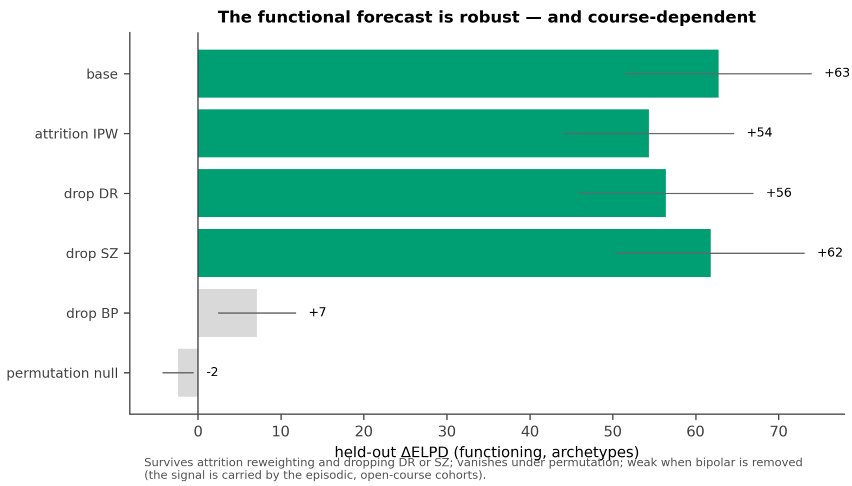

Figure E8: **The functional forecast is robust and course-dependent.** Held-out incremental value of the archetype phenotype for two-year functioning under attrition reweighting, cohort hold-outs and a permutation null. The signal survives attrition and dropping either DR or SZ, vanishes under permutation, and weakens when bipolar disorder is removed—it is carried by the episodic, open-course cohorts.

Table A6: **Variable accounting: the 225 harmonized common variables by role.** A variable is fitted as a factor indicator only if it is a current clinical/biological state, not a confound, not redundant with an admitted indicator, and present in ≥ 2 cohorts; above the two-cohort floor admission is coverage-agnostic (the observed-cell likelihood handles partial coverage without imputation, so two-cohort markers are used where measured). The six set-aside roles are mutually exclusive.

Role	n	Reason	Representative variables
Factor indicator (fitted)	109	current state; not confound/redundant; ≥ 2 cohorts (66 three-cohort, 43 two-cohort)	FAST, EGF, CVLT, WAIS, PSQI, BMI, lipids, CRP, leukocytes, YMRS, ISF, CTQ
Identifier	2	record linkage only	`usubjid`, `fondacode`
Covariate (residualization)	5	adjusted out of each indicator; not a phenotype	age, sex, education, education level, recruitment site
Validation / metadata label	3	held out of fitting; used for invariance and transdiagnosticity	DSM-5 subtype, visit, visit number
Treatment / outcome confound	16	medication- or outcome-driven; would contaminate the latent axes	prolactin, oxcarbazepine level, QTc/RR, sodium/potassium/calcium, CGI-improvement, hypnotic use, menopausal/hormonal status
Redundant / instrument sub-item	46	captured by an already-admitted indicator	electrolyte & haematology panel (albumin, haemoglobin, RBC, MCV, iron), C-SSRS items, FAST/PSQI sub-scores, MARS adherence, PRISM stigma
History / onset / service-use	22	antecedent or consequence, not a current state	medical & family-history flags, age at onset, hospitalisations, work leave, occupational status
Not usable	22	single-cohort, too sparse, or incomparable	clozapine flag (SZ-only), MDQ screen, comorbidity flags, un-normed neuropsych scores

Table A7: **All immunometabolic primary loadings with 95% credible intervals** (canonical Gaussian-copula 8-factor fit, $N = 9{,}013$), grouped by biological block and ordered by $|\lambda|$ within block. All 37 intervals exclude zero. The axis is anchored by body composition (BMI, weight, waist), with graded metabolic, inflammatory and cardiovascular contributions; the only available inflammatory marker, CRP, loads at 0.37 and is itself substantially adiposity-driven.

Block	Indicator	λ	95% CI
Anthropometric (body composition)			
	Body-mass index	0.946	[0.930, 0.963]
	Weight	0.875	[0.860, 0.891]
	Waist circumference	0.838	[0.821, 0.855]
Metabolic (lipids, glycaemia, hepatic/renal)			
	Triglycerides	0.382	[0.361, 0.404]
	Urate (uric acid)	0.368	[0.347, 0.389]
	Total/HDL cholesterol ratio	0.365	[0.337, 0.393]
	Gamma-glutamyl transferase	0.310	[0.285, 0.336]
	HDL cholesterol	0.291	[0.269, 0.314]
	Fasting glucose	0.261	[0.237, 0.285]
	Alkaline phosphatase	0.246	[0.215, 0.277]
	Alanine aminotransferase	0.241	[0.212, 0.270]
	Glycated haemoglobin (HbA_{1c})	0.205	[0.170, 0.238]
	Aspartate aminotransferase	0.150	[0.121, 0.180]
	Total cholesterol	0.147	[0.123, 0.172]
	LDL cholesterol	0.128	[0.103, 0.154]
	Vitamin D	0.118	[0.089, 0.147]
	Creatinine	0.072	[0.046, 0.098]
	Urea	0.034	[0.008, 0.063]
	Bilirubin	0.002	[0.000, 0.008]
Inflammatory / haematologic			
	C-reactive protein	0.366	[0.342, 0.390]
	White-cell count	0.184	[0.158, 0.209]
	Neutrophils	0.135	[0.111, 0.160]
	Monocytes	0.125	[0.095, 0.155]
	Platelets	0.102	[0.078, 0.124]
	Eosinophils	0.068	[0.043, 0.093]
	Basophils	0.020	[0.002, 0.045]
	Lymphocytes	0.001	[0.000, 0.004]
Cardiovascular / autonomic / endocrine			
	Systolic BP (standing)	0.337	[0.309, 0.366]
	Systolic BP (supine)	0.288	[0.267, 0.309]
	Diastolic BP (supine)	0.230	[0.208, 0.251]
	Diastolic BP (standing)	0.229	[0.202, 0.256]
	Heart rate (supine)	0.181	[0.155, 0.207]
	Mean heart rate (ECG)	0.173	[0.144, 0.201]
	Heart rate (standing)	0.157	[0.129, 0.185]
	Free triiodothyronine (T3)	0.112	[0.086, 0.139]
	Thyroid-stimulating hormone	0.055	[0.025, 0.085]
	Free thyroxine (T4)	0.021	[0.002, 0.047]

E.2 The claims ledger

Table E8 states, side by side, what the study claims and what it deliberately does not—the spine of the Limitations and of any response to reviewers.

Table E8: **What we claim, and what we do not.**

We claim	**We do not claim**
A validated eight-dimension transdiagnostic measurement map (internal validity).	Natural biotypes or subtypes; that no biotype exists in any measurement space.
Immunometabolic load *largely independent* of a general functional-burden axis (quantified, with boundary conditions).	Strict statistical orthogonality, or independence from *being ill* (case–control elevations).
A graded continuum, resolved by five stable archetypes, that is descriptively tighter than DSM-5.	That the regions are clinically superior to DSM-5 beyond the modest prognostic increment shown.
Temporal coherence: durable biology, moving symptoms.	That developmental risk is genuinely "state" (it partly reflects recall noise).
A modest, *group-level* incremental prognosis for *functioning*, carried by the biological phenotype, course-dependent.	A large individual-risk gain; prediction of *recorded* relapse/hospitalization events; a deployable individual rule.
Convergent validity with the immunometabolic literature.	External or out-of-sample validation (none yet).

F Glossary of metrics: meaning and values

This glossary lists, in plain language, every quantitative metric used in the paper, how to read it, and the value obtained in this study. It is intended as a reading aid for clinical readers; formal definitions and provenance are in the cited sections. Throughout, a *credible interval* is the Bayesian analogue of a confidence interval (the range that contains the true value with the stated probability), and "$\hat{f}_i$" denotes a patient's estimated position on a latent axis.

Metric	**What it measures**	**How to read it**	**Value in this study**
A. Did the model fit honestly? **(model convergence and goodness of fit)**			
$\hat{R}$ (Gelman–Rubin)	Whether the Bayesian sampler converged—independent chains agree on the answer	≈ 1.00 ideal; <1.05 acceptable; >1.1 suspect	Global 8-factor map 1.03; continuous backbone 1.01; anthropometry-excluded arm 1.09 (its magnitudes read as approximate)
Effective sample size (ESS)	How many independent posterior samples support an estimate	Higher is better; a few hundred is comfortable	1,514 (backbone); 38 in the excluded arm (hence approximate)
Divergences	Sampler failures that flag an untrustworthy posterior region	0 is ideal	0 in every reported fit

continued on next page

Metric	What it measures	How to read it	Value in this study
Bayesian SRMR	Average gap between the observed and model-implied indicator correlations (absolute fit)	$<$0.08 is good fit	0.074 (continuous block)
Posterior-predictive coverage	Share of indicators whose real data fall inside the model's 90% predictive band	Higher is better	21/22 non-Gaussian indicators
WAIC / PSIS-LOO (ΔELPD)	Out-of-sample fit used to compare model *shapes*	Larger ELPD = better fit; we treat it as *secondary* evidence (it favours bifactor models by form)	Bifactor preferred over correlated-factor ($\Delta \approx$ 2,720) and one-factor ($\Delta \approx$ 53,000)
Tucker congruence (φ)	How identically two fits reproduce the *same* factor	1.0 = identical; $>$0.95 excellent; $>$0.85 good	Prior-free refit 1.00; resampling $\geq$ 0.958; longitudinal 0.987–0.996; anthropometry-excluded axis 0.90
B. What are the axes? **(factor structure)**			
Factor loading (λ)	Strength with which one indicator reflects a latent axis (a standardized weight)	$\lvert\lambda\rvert>$0.3 substantial; sign gives direction	e.g. BMI$\to$immunometabolic 0.95, CRP 0.37, FAST$\to G$ 0.78 (all 37 immunometabolic loadings in Table A7)
Factor correlation (ϕ, Φ)	How strongly two axes move together	-1 to $+1$; near 0 = independent	Specific axes mean $\lvert\phi\rvert \approx 0.08$; under a free general factor $\phi(G, \text{cognition})$=0.39, $\phi(G, \text{sleep})$=0.42, $\phi(G, \text{immunometabolic}) \approx$ 0.10
95% credible interval	Bayesian uncertainty range for a loading or effect	Excluding 0 = reliably non-zero	All 37 immunometabolic loadings exclude 0
Coordinate uncertainty (S_i, reliability tier)	How precisely a patient's position on each axis is known, given how many indicators they had	Larger SD = less reliable; carried into every downstream analysis (no patient treated as exact)	Each patient has a per-axis posterior SD and tier (well / partial / prior-dominated)
C. Continuum or distinct biotypes? **(stratification, M2)**			
Silhouette	How cleanly patients separate into clusters	-1 to 1; $>$0.5 strong clusters; $\sim$0.14 = none	Best partition 0.140
Falsification-null z	Real-data clustering vs a structureless Gaussian of identical spread, in standard deviations	$\sim$0 = no real clustering; $>$2 would signal clusters	z = 1.13 (not significant; null silhouette 0.137 $\pm$ 0.002)

continued on next page

Metric	What it measures	How to read it	Value in this study
Hopkins statistic	Cluster *tendency* of the cloud vs. a *uniform* reference	0.5 = uniformly random; → 1 = clustered	0.79–0.81 (real); null 0.756 ± 0.004 (z = 8.71): high tendency vs. uniform, but *not* vs. a matched Gaussian—see the structure test (Annex B)
Dip test (p)	Whether an axis is multimodal (clumpy) vs smooth	p>0.05 = unimodal (smooth)	Unimodal on 6/8 axes
Gap statistic (k_{opt})	Data-driven optimal number of clusters	1 = no clusters	1
HDBSCAN	Density-based clustering (returns clusters or labels points as noise)	0 clusters / all-noise = a continuum	0 clusters
Adjusted Rand Index (ARI)	Agreement between the map's partition and DSM-5 diagnosis	0 = chance; 1 = identical	0.021 archetypes; 0.006 soft tessellation (both ≈ 0, transdiagnostic)
Information criterion (BIC)	Which description fits the coordinate cloud more tightly—a free continuum or one constrained to DSM-5	Lower = better (more parsimonious) description	185,600 (free) vs 188,200 (DSM-constrained): the map describes the space ≈3× more tightly at lower BIC
Archetype count (A)	Number of extreme "corner" phenotypes that reproduce across runs	The largest cross-seed-stable A	A = 5 (cross-seed φ = 0.979; collapses to 0.436 at A = 6)
Normalized entropy of weights	How *blended* a patient is between corners	0 = a pure corner; 1 = an even mix (continuum)	Median 0.65 (patients sit between corners)
D. Does the map hold over time? **(temporal coherence, M3)**			
Intraclass correlation (ICC, trait fraction)	Share of an axis that is a stable *trait* vs moving *state*, with measurement error removed	>0.6 trait; <0.4 state	Immunometabolic 0.91 (0.76 without the anthropometric items); cognition 0.70; severity 0.62 (trait by rank); sleep 0.47; suicidality 0.43; developmental 0.39
Population shift (slide)	Average 2-year movement of the whole cohort on an axis	Near 0 = population-stable; negative = improving	Biology +0.04 (static); severity −0.46; suicidality −0.84
Longitudinal invariance (φ)	Whether the instrument measures the same construct at follow-up	>0.95 = invariant	0.987–0.996 across the four backbone axes
Archetype persistence (cosine)	How stable a patient's archetype mix is over 2 years	1 = unchanged	Median 0.81
V0 reproduction (Pearson r)	Check that re-scoring reproduces baseline coordinates	∼1 = faithful rescoring	0.99
E. Does it forecast outcome? **(prognosis, M4)**			

continued on next page

Metric	What it measures	How to read it	Value in this study
Held-out ΔELPD	Improvement in 2-year outcome prediction from adding the map, on unseen patients	Larger = more predictive (group level)	Archetypes +62.8; the map adds +17.3 beyond DSM-5's +29.0 (co-informative); attrition-weighted +54.4
ΔAUC	Gain in telling apart, at the *individual* level, who will reach an outcome	0 = none; 0.5 = perfect-from-chance; here small	Functional remission +0.010
Functional-remission rate	Share reaching 2-year functional remission, by archetype corner	The clinical readout of the gradient	22% (immunometabolic corner) to 63% (low-burden pole)
Odds ratio (best vs worst corner)	Relative odds of remission across the corners	>1 favours the better pole	≈6.3 (cohort-adjusted)
Permutation null	Sanity check: the signal should vanish when outcome labels are shuffled	Vanishing = the signal is real	Vanished, as required
Representation sufficiency (ΔAUC vs raw)	Whether the 8 axes capture what the 143 raw indicators do for prognosis	Small gap = a sufficient summary	Tie for deterioration; raw +0.04 for recovery (92–97% captured)

G A variational GLLVM re-estimation: model, inference, and comparison to NUTS

The atlas of the main text is estimated by Hamiltonian Monte Carlo (the No-U-Turn sampler, NUTS), which is audit-grade but costs hours of multi-chain sampling. Here we re-estimate the *same* measurement model by stochastic variational inference (SVI), giving a ~100× faster operational engine that we calibrate against—rather than substitute for—the NUTS fit. We frame this as a generalized linear latent-variable model (GLLVM) [33] fit variationally [34, 35]; a fuller development of the variational engine (amortized scoring, richer posteriors, calibrated uncertainty) is left to future work. This section describes the model, the inference, the conceptual differences from NUTS, and the head-to-head comparison.

G.1 Generative model

The generative model is the atlas of Annex A, unchanged. For patient i with factor vector $f_i = (G_i, D_{i1}, \ldots, D_{iK})^\top$ and item j,

$$f_i \sim \mathcal{N}(\mathbf{0}, \Phi), \quad \Phi \text{ bifactor-faithful } (G \text{ and the pinned substance axis orthogonal}), \tag{G54}$$

$$\eta_{ij} = \alpha_j + \lambda_j^\top f_i, \quad \text{linear decoder; home loadings positive (softplus), forbidden cells exact zero,} \tag{G55}$$

$$x_{ij} \sim F_j(\eta_{ij}, \theta_j), \quad \text{per-item likelihood,} \tag{G56}$$

with the same ontology prior matrix as the folded operational map (immunometabolic merge; the three earned cross-loadings freed; everything else hard-zero). F_j is Gaussian on the rank-INT (Gaussian-copula) scale for continuous and high-cardinality items, and native Bernoulli-logit / ordered-logistic / negative-binomial for binary, low-cardinality ordinal and count items—the same tiered likelihood as the main fit. The log-likelihood is evaluated over *observed cells only* (a missingness mask removes absent cells; no imputation). One implementation difference from the NUTS engine: rather than marginalizing the continuous block analytically (the Woodbury identity of Annex A), the variational engine retains an explicit per-patient latent for *every* factor and attaches every likelihood directly to f_i, so it carries the full ≈ 142-indicator block with no continuous/explicit split.

G.2 Variational inference

The exact posterior $p(\{f_i\}, \Theta \mid X_{\text{obs}})$ is intractable, so SVI replaces it with a tractable family and *optimizes* rather than samples. Each patient receives an independent variational posterior over its coordinate,

$$q_i(f_i) = \mathcal{N}(\mu_i, \Sigma_i), \qquad \Sigma_i = \text{diag}(s_i^2) \text{ (mean-field default)}, \tag{G57}$$

with optional low-rank-plus-diagonal $\Sigma_i = \text{diag}(s_i^2) + U_i U_i^\top$ or full $K \times K$ covariance. Inference minimizes the negative evidence lower bound (ELBO),

$$\mathcal{J} \;=\; \underbrace{\mathbb{E}_q[-\log p(X_{\text{obs}} \mid f, \Theta)]}_{\text{observed-cell reconstruction}} \;+\; \sum_{i=1}^{N} \text{KL}\big[\, q_i(f_i) \,\|\, \mathcal{N}(\mathbf{0}, \Phi) \,\big] \;+\; \Omega(\Lambda) + \Omega(\Phi), \tag{G58}$$

where the reparameterization trick [37] makes the latent draws differentiable, the KL term regularizes each coordinate toward the bifactor prior (G54), and $\Omega(\cdot)$ are the ontology penalties—the soft loading prior written as a per-cell $\frac{1}{2}\sum_{\text{free}}((\lambda - m)/v)^2$ with (m, v) read from the prior matrix, plus a weak off-diagonal penalty on Φ. The objective is optimized by minibatch AdamW (per-patient terms scaled by N/B; global penalties added once), with the small $K \times K$ Cholesky/KL on CPU in double precision. The loadings, factor correlations, residual scales and cutpoints $(\Lambda, \Phi, \sigma, \text{cutpoints})$ are *maximum-a-posteriori* point estimates; loading credible bands are obtained from a seed ensemble, with NUTS retained as the uncertainty authority. A two-rung warm start (continuous backbone $\to$ full mixed map) protects the thin factors from ELBO-flat collapse. The full fit takes $\approx$4.5 minutes on a workstation CPU (PyTorch [36]), against the NUTS fit's hours.

G.3 NUTS versus variational inference

The two engines optimize the same model but differ in what they compute and what they cost (Table G10). NUTS draws asymptotically exact samples from the full joint posterior, delivering calibrated uncertainty for every quantity; SVI returns the best approximation *within a chosen family*, which is fast but biased by the family. The load-bearing consequence here is specific and predictable: a mean-field (diagonal) q_i cannot represent posterior covariance *between* factors, so to fit the data it shrinks the prior inter-factor correlations the likelihood would otherwise support. The result is faithful factor loadings and patient coordinates—which are governed by the posterior *means*—but systematically attenuated inter-factor correlations Φ and underestimated coordinate standard deviations, both governed by the posterior *covariance*. This is why we keep NUTS as the authority for Φ magnitudes and for any credible interval, and use the variational engine for the map, the coordinates, fast reruns and synthetic-cohort generation.

G.4 Comparison results

On the eight-factor operational map, the variational engine reproduces the NUTS fit on its two load-bearing objects—the measurement map and the patient coordinates—and matches it as a generative model.

Table G10: NUTS versus variational inference for the FACE-ATLAS measurement model.

	NUTS (main fit)	**Variational GLLVM**
Inference	MCMC: asymptotically exact samples from the full joint posterior	Optimization: best approximation within a chosen posterior family
Latent treatment	Continuous block marginalized (Woodbury); latents only for discrete items	Explicit per-patient latent for every factor; all ≈ 142 items
Global parameters	Full posterior (loadings, Φ, residuals) with credible intervals	MAP point estimates; loading bands from a seed ensemble
Coordinate uncertainty	Calibrated posterior SD per axis	Relative reliability signal (mean-field underestimates SD)
Inter-factor Φ	Posterior magnitudes with uncertainty	Structure correct; off-diagonals attenuated (mean-field bias)
Cost	Hours (multi-chain)	$\approx$4.5 min CPU ($\sim$100$\times$ faster)

The measurement map. Loading columns are congruent on all eight factors (Tucker φ: G 0.957, every specific axis 0.96–0.999; Table G11), and the full item$\times$factor loading scatter is $r = 0.993$ over 208 cells with no sign flips (Figure G9). G is the hardest axis—it carries every item's small bifactor loading—and is the only one near the major-axis bar; a richer posterior raises it to 0.974. *It is the same map.*

The patient coordinates. Per-patient coordinates correlate with the NUTS coordinates at $r \geq 0.90$ on all eight axes (Table G12: G 0.990, immunometabolic 0.998, cognition 0.992, down to substance 0.908, the orthogonal, instrument-poor axis). *Patients land in the same place.*

Table G11: Loading congruence (Tucker φ) vs NUTS.

Factor	cells	Tucker φ
G (overall severity)	106	0.957
cognition	16	0.998
immunometabolic	37	0.998
sleep	12	0.999
suicidality	12	0.998
developmental risk	21	0.998
mania/activation	2	0.984
substance	2	0.963

Table G12: Coordinate agreement (Pearson r) vs NUTS.

Factor	n (well)	r
G (overall severity)	8641	0.990
cognition	6451	0.992
immunometabolic	8825	0.998
sleep	7522	0.968
suicidality	8285	0.984
developmental risk	8767	0.986
mania/activation	6152	0.976
substance	4249	0.908

Generative fidelity. Drawing 9,013 synthetic patients from the fitted map and comparing to the observed data, the marginals match (Kolmogorov–Smirnov median 0.04; per-variable mean fidelity $r = 1.000$, SD fidelity $r = 0.998$) and so does the joint covariance (correlation SRMR 0.078, matching the NUTS fit's ≈ 0.07)—the eight factors reproduce the data's correlation structure, not merely its margins.

The one gap: inter-factor Φ. The correlation structure is reproduced—G and substance rows are exactly zero in both engines, and the sleep/suicidality/developmental-risk/mania block is positively correlated in both (Figure G10)—but the off-diagonals are systematically attenuated: correlated-block mean $|\phi|$ is 0.083 (variational) versus 0.105 (NUTS), a $\sim$21% shrinkage concentrated in the small negative cognition correlations (the substantive immunometabolic$\leftrightarrow$cognition coupling is preserved). Climbing the full posterior-richness ladder—mean-field $\rightarrow$ low-rank $\rightarrow$ full $K \times K$ covariance—reduces the shrinkage monotonically (21% $\rightarrow$ 20% $\rightarrow$ 18%) but does not

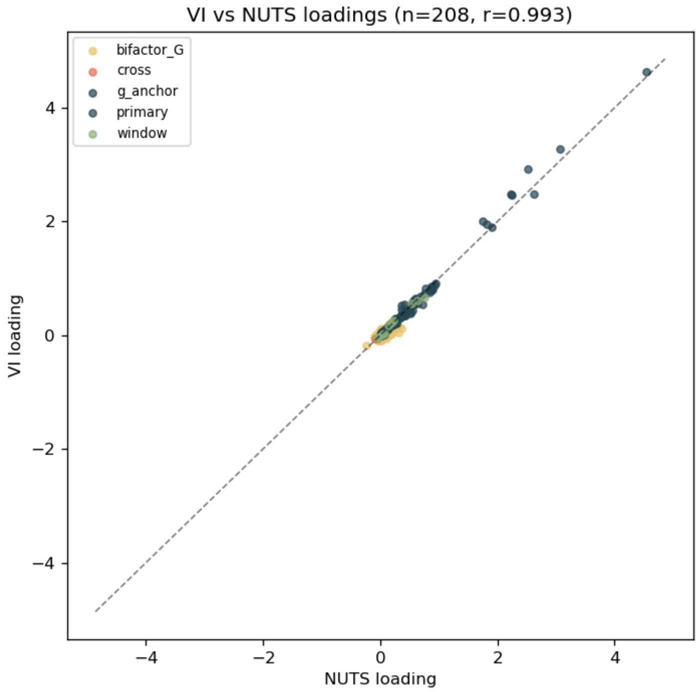

Figure G9: **The variational map is the NUTS map.** Item×factor loadings from the variational GLLVM against the NUTS copula fit ($n = 208$ cells, $r = 0.993$), coloured by cell type. Large home and G-anchor loadings align on the diagonal; cross, window and small bifactor-G cells cluster tightly near zero.

close it, establishing the residual as a *structural bias of variational inference on the inter-factor correlation hyperparameter* rather than a tunable defect (an exploratory arm removing the Φ penalty and loosening gradient clipping left Φ unchanged). This is precisely the quantity for which NUTS is retained as the authority; the variational Φ should be read as directionally correct but not as a magnitude.

Summary. The variational GLLVM is a calibrated, $\sim$100×-faster stand-in for the NUTS measurement map and patient coordinates (loadings Tucker 8/8, scatter $r = 0.993$; coordinates $r \geq 0.90$ on 8/8 axes) and a faithful generative model of the cohort, with the inter-factor correlation Φ as the one structurally attenuated quantity for which NUTS remains the authority. It is suited to fast reruns, sensitivity sweeps, synthetic-cohort generation, and as an independent-estimator robustness check on the map; developing it into a fully calibrated Bayesian engine is future work.

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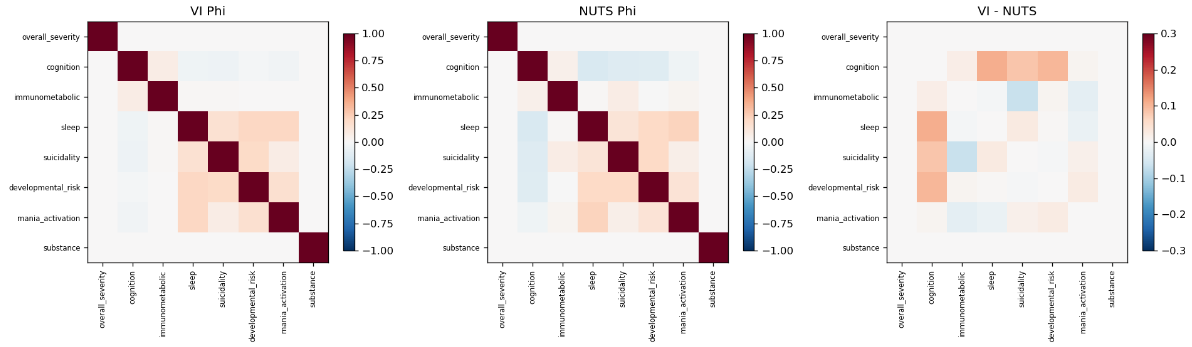

Figure G10: **Inter-factor correlation Φ: structure reproduced, magnitudes attenuated.** Variational Φ (left), NUTS Φ (middle), and their difference (right). The block-diagonal orthogonality (G, substance) is exact in both and the correlated specific block is positive in both; the difference panel localizes the ∼21% attenuation to the small cognition cross-correlations.

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